SUPPORT COMPLET

AUDIT BRODY / OBSIDIA / DOMAINES

Branchement interne - Claude guide - Brody - Domaines - P1 a P80

Champ	Valeur
Type de document	Support complet interne, exhaustif, non synthetique
Source utilisee	audit brody ect serveur ful 222+ audit branchement int + audit chat terminal full non branché (2).docx
Mode	READ_ONLY / local evidence / no patch / no server start
Base officielle	Audit V4 FAST INDEXED + freeze final + document terminal fourni
Claude items audites	4353
Non branches strict	3316
Partiels runtime	204
Brody only	772
Brody + Domaines	61
P1 a P80	80 lignes auditees

Règle de lecture

Ce PDF est construit comme un support interne complet. Il ne cherche pas a reduire l'audit a une synthese commerciale. Il reprend le document fourni et le remet en forme comme les supports Obsidia precedents : page de garde, doctrine, statuts, preuves, tableaux et inventaire long.

Le document part du fichier source donne dans cette conversation. Les compteurs officiels scelles sont conserves : 4353 items Claude audites, 3316 non branches strict, 204 partiels runtime, 772 Brody only, 61 Brody + Domaines, 80 P1 a P80 audites.

Un item 'Brody only' ne doit pas etre interprete comme branche aux domaines. Un item 'partial runtime' ne doit pas etre interprete comme branche Brody. Un item 'reference only not bound' existe dans le guide Claude mais n'a pas ete retrouve dans les surfaces runtime scannees.

Le Kernel X108 reste la seule autorite souveraine. Brody, Graphiti, UI, domaines et memoire sont des surfaces d'aide, contexte, routage, lecture ou audit. Ils ne deviennent pas autorite de decision.

Synthèse scellée

Métrique	Valeur	Lecture
ITEM_REFERENCE_ONLY_NOT_BOUND	3316	Reference Claude seulement, non retrouve dans surfaces runtime scannees
ITEM_BOUND_BRODY_ONLY	772	Retrouve cote Brody, pas domaine
ITEM_PARTIAL_RUNTIME_SURFACE_ONLY	204	Retrouve UI / Kernel / Graphiti / tests / preflight, pas Brody/domaines
ITEM_BOUND_BRODY_AND_DOMAIN	61	Retrouve cote Brody et au moins une surface domaine
P_BOUND_BRODY_ONLY	79	P1-P80 visible Brody seulement
P_BOUND_BRODY_AND_DOMAIN	1	P1-P80 visible Brody + domaine

Sommaire opérationnel

1. Statut du support et source utilisée
2. Audit sémantique profond Claude / Brody / Obsidia
3. Structures Graphiti, source roots et registres
4. Contrats Brody et frontières KX108_ONLY
5. Capabilities, arbres, lois, protocoles, agents, modules, pépites, balances, cortex
6. Audit strict V4 FAST : chiffres, familles, items, P1 à P80
7. Freeze final, quality gate, manifest et preuve de fermeture
8. Document source intégral remis en forme

Document source intégral remis en forme

La section suivante reprend le contenu du document fourni, dans l'ordre, avec une mise en forme de support PDF interne. Les logs, commandes, verdicts et listes restent conservés pour garder la traçabilité.

● Mode: READ_ONLY

Layer: AGENTIC

Scope: Audit sémantique profond Brody/Obsidia — 14 familles, inventaire nominal exhaustif

Risk: NONE

Files touched: .local_audits/CLAUDE_BRODY_SEMANTIC_DEEP_ENUMERATION_V0_<TS>/ (nouveau dossier audit uniquement)

Je commence par inspecter les sources disponibles avant d'écrire le script Python.

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && ls .local_audits/ | head
```

-30)

 3001_start.log

3001_start_err.log

5173_start.log

… +27 lines (ctrl+o to expand)

Searched for 1 pattern (ctrl+o to expand)

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && head -5
```

".local_audits/BRODY_SEMANTIC_COMPONENT_REGISTRY_V0_20260618…)

 category,source_file,path,line,name,value,origin

MEMOIRE,_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_2
0260512_224854/graphiti_readonly_index_v2.jso

n,$.status,,GRAPHITI_READONLY_INDEX_V2_FUSION_READY,,json_value

… +12 lines (ctrl+o to expand)

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && cat
```

".local_audits/BRODY_SEMANTIC_READABLE_ALL_FAMILIES_V0_1_SAFE_20…)

 "category","items","file"

"ARBRE_34","8615","01_ARBRES_34.md"

---DEREFERENCE---

… +66 lines (ctrl+o to expand)

● Le CSV brut a 205 761 lignes. Je vois que les arbres pointent vers un chemin externe (obsidia-engine-candidate). Voyons la

structure Graphiti et les fichiers sources réels.

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && ls
```

_graphiti_readonly_indexes/ 2>/dev/null | head -10)

 GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
```

import csv, collections, sys…)

 AGENT: 14545

ARBRE_34: 29942

BALANCE: 3460

… +12 lines (ctrl+o to expand)

● Excellentes données. Le CSV brut contient 203 827 lignes réelles sur 14 familles. Je dois inspecter le Graphiti Index JSON

et les fichiers Brody clés pour concevoir le script optimal.

```
● Explore(Exploration graphiti index + sources brody pour plan audit sémantique)
```

 Done (35 tool uses · 71.8k tokens · 3m 32s)

(ctrl+o to expand)

 Allowed by auto mode classifier

● Mode: READ_ONLY

Layer: AGENTIC

Scope: Exécution audit sémantique profond — script Python, 14 familles, inventaire nominal

Risk: NONE

Files touched: .local_audits/CLAUDE_BRODY_SEMANTIC_DEEP_ENUMERATION_V0_<TS>/ (nouveau dossier uniquement)

```
.local_audits\XLS_DILUTION_AUDIT_20260602_145946\02_COLUMNS_RISK.csv
.local_audits\XLS_DILUTION_AUDIT_20260602_145946\03_DUPLICATE_GENERIC_VALUES.csv
.local_audits\CIC_METRICS_V0_VERIFY_AND_TEST_20260613_020125\zip_extracted\tests\test1_baseline .csv
.local_audits\CIC_METRICS_V0_VERIFY_AND_TEST_20260613_020125\zip_extracted\tests\test2_noise.cs v
```

```
.local_audits\CIC_METRICS_V0_VERIFY_AND_TEST_20260613_020125\zip_extracted\tests\test3_structur al_error.csv
.local_audits\CIC_METRICS_V0_VERIFY_AND_TEST_20260613_020125\zip_extracted\tests\test4_hold.csv
.local_audits\CIC_METRICS_V0_VERIFY_AND_TEST_20260613_020125\zip_extracted\docs\source_packs\CO RPUS_MAP_V4_20260602.csv
.local_audits\P3T1_DOMAIN_SENSITIVE_FIDELITY_AUDIT_20260612_102501\P3T1_rows_latest_500.csv
.local_audits\P3T2_LOW_INTEGRITY_ALLOW_DEEP_AUDIT_20260612_102759\P3T2_target_low_integrity_all ow_cases.csv
.local_audits\P3T3_BUSINESS_DECISION_SEMANTICS_AUDIT_20260612_103151\P3T3_focus_cases.csv
.local_audits\P3T3B_BUSINESS_SEMANTICS_DATA_ONLY_20260612_110409\P3T3B_focus_cases.csv
.local_audits\P3T5_PASS_PATHS_FULL_BUSINESS_AUDIT_20260612_111011\P3T5_review_cases.csv
.local_audits\P3T6_IRREVERSIBLE_FIELD_SOURCE_AUDIT_20260612_112103\P3T6_missing_irreversible_ca ses.csv
.local_audits\P3T9_TRADING_ALLOW_REVIEW_SEMANTIC_RULE_AUDIT_20260612_144329\P3T9_focus_cases.cs v
.local_audits\P3T10_FINAL_METRICS_CONSOLIDATION_AUDIT_20260612_152045\P3T10_rows.csv
.local_audits\P3T10_FINAL_METRICS_CONSOLIDATION_AUDIT_20260612_152327\P3T10_rows.csv
.local_audits\CIC_METRICS_AUDIT_20260613_002236\02_metric_raw_matches.csv
.local_audits\CIC_METRICS_AUDIT_20260613_002236\03_metric_dedup_matches.csv
.local_audits\CIC_METRICS_AUDIT_20260613_002236\06_metric_candidates.csv
.local_audits\CIC_METRICS_AUDIT_20260613_002236\08_decision_keywords.csv
.local_audits\CIC_RULES_METRICS_AUDIT_20260613_010338\02_rules_metric_raw_matches.csv
.local_audits\CIC_RULES_METRICS_AUDIT_20260613_010338\03_rules_metric_dedup_matches.csv
.local_audits\BRODY_SEMANTIC_MACHINATION_AUDIT_V0_20260618_025435\brody_semantic_cases.csv
.local_audits\BRODY_DEEP_SUBLAYER_MACHINATION_AUDIT_V0_20260618_025830\brody_deep_sublayer_case s.csv
.local_audits\BRODY_DEEP_SUBLAYER_MACHINATION_AUDIT_V0_20260618_025830\category_summary.csv
.local_audits\BRODY_CANONICAL_COVERAGE_GAP_AUDIT_V0_20260618_030551\canonical_file_coverage.csv
.local_audits\BRODY_CANONICAL_COVERAGE_GAP_AUDIT_V0_20260618_030551\canonical_commit_coverage.c sv
.local_audits\BRODY_CANONICAL_COVERAGE_GAP_AUDIT_V0_20260618_030551\brody_canonical_probe_resul ts.csv
.local_audits\BRODY_CANONICAL_COVERAGE_GAP_AUDIT_V0_20260618_030551\file_zone_summary.csv
.local_audits\BRODY_CANONICAL_COVERAGE_GAP_AUDIT_V0_20260618_030551\commit_zone_summary.csv
.local_audits\BRODY_CANONICAL_COVERAGE_GAP_AUDIT_V0_20260618_030551\brody_probe_summary.csv
.local_audits\BRODY_CANONICAL_COVERAGE_GAP_AUDIT_V0_1_20260618_030901\canonical_file_coverage.c sv
.local_audits\BRODY_SOURCE_PACK_LOADER_DISCOVERY_V0_20260618_151621\loader_binding_candidates.c sv
.local_audits\BRODY_SOURCE_PACK_LOADER_BINDING_FOCUS_V0_20260618_152557\binding_focus_snippets. csv
.local_audits\BRODY_DEEP_BINDING_SURFACE_AUDIT_V0_20260618_153539\deep_binding_surface.csv
.local_audits\BRODY_DEEP_BINDING_SURFACE_AUDIT_V0_1_20260618_154129\deep_binding_surface.csv
.local_audits\BRODY_DEEP_BINDING_LAYER_CLASSIFICATION_V0_20260618_154539\binding_layer_classifi cation.csv
.local_audits\BRODY_RUNTIME_SPINE_FUNCTION_MAP_V0_20260618_154810\runtime_spine_functions.csv
.local_audits\BRODY_FULL_MACHINATION_ATLAS_V0_1_SAFE_20260618_164940\brody_machination_componen ts.csv
.local_audits\BRODY_FULL_MACHINATION_ATLAS_V0_1_SAFE_20260618_164940\brody_machination_call_edg es.csv
.local_audits\BRODY_FULL_MACHINATION_ATLAS_V0_1_SAFE_20260618_164940\brody_machination_imports. csv
.local_audits\BRODY_FULL_MACHINATION_ATLAS_V0_1_SAFE_20260618_164940\brody_machination_constant s.csv
.local_audits\BRODY_MACHINATION_NAME_REGISTRY_V0_20260618_165357\brody_machination_name_registr y.csv
.local_audits\BRODY_SEMANTIC_COMPONENT_REGISTRY_V0_20260618_170307\brody_semantic_component_reg istry.csv
.local_audits\BRODY_SEMANTIC_READABLE_ALL_FAMILIES_V0_1_SAFE_20260618_171808\00_SUMMARY_COUNTS. csv
.local_audits\BRODY_SEMANTIC_READABLE_ALL_FAMILIES_V0_1_SAFE_20260618_171808\00_DEREFERENCE_TAR GETS_ALL_FAMILIES.csv
.local_audits\BRODY_SEMANTIC_SIMPLE_READABLE_ALL_FAMILIES_V0_20260618_175730\00_SUMMARY_COUNTS. csv
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && head -5
".local_audits/BRODY_SEMANTIC_COMPONENT_REGISTRY_V0_20260618_170307/br
ody_semantic_component_registry.csv" && echo
"---" && wc -l
".local_audits/BRODY_SEMANTIC_COMPONENT_REGISTRY_V0_20260618_170307/br
ody_semantic_component_registry.csv")
 category,source_file,path,line,name,value,origin
MEMOIRE,_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_2
0260512_224854/graphiti_readonly_index_v2.jso
n,$.status,,GRAPHITI_READONLY_INDEX_V2_FUSION_READY,,json_value
BOUNDARY,_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_
20260512_224854/graphiti_readonly_index_v2.js
on,$.status,,GRAPHITI_READONLY_INDEX_V2_FUSION_READY,,json_value
ARBRE_34,_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_
20260512_224854/graphiti_readonly_index_v2.js
on,$.source_roots.MMONDE_REVERSE_OS_ROOT,,C:\Users\User\Desktop\obsidi
a-engine-proof-core\obsidia-engine-candidate\
OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\O
BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PL
US_V1,,json_value
PROTOCOLE,_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION
_20260512_224854/graphiti_readonly_index_v2.j
son,$.source_roots.MMONDE_REVERSE_OS_ROOT,,C:\Users\User\Desktop\obsid
ia-engine-proof-core\obsidia-engine-candidate
\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\
OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2P
LUS_V1,,json_value
---
205761 .local_audits/BRODY_SEMANTIC_COMPONENT_REGISTRY_V0_20260618_170307/brody_semantic_component_reg istry.csv
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && cat
".local_audits/BRODY_SEMANTIC_READABLE_ALL_FAMILIES_V0_1_SAFE_20260618 _171808/00_SUMMARY_COUNTS.csv"
2>/dev/null &&
echo "---DEREFERENCE---" && head -20
```

".local_audits/BRODY_SEMANTIC_READABLE_ALL_FAMILIES_V0_1_SAFE_20260618
_171808/00_DEREFERENCE_TARGETS_ALL_FAMILIES.csv"

2>/dev/null)

 "category","items","file"

"ARBRE_34","8615","01_ARBRES_34.md"

---DEREFERENCE---

"category","reason","name","source_file","line","path"

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

S_V1","_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20
260512_224854/graphiti_readonly_index_v2.json

","","$.source_roots.MMONDE_REVERSE_OS_ROOT"

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

S_V1\04_ARBRES_34_TENSOR_MATRIX","_graphiti_readonly_indexes/GRAPHITI_
READONLY_INDEX_V2_FUSION_20260512_224854/grap

```
hiti_readonly_index_v2.json","","$.source_roots.34_ARBRES_TENSOR_MATRIX"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

S_V1\13_GRAPHES_NUAGE_POINTS","_graphiti_readonly_indexes/GRAPHITI_REA
DONLY_INDEX_V2_FUSION_20260512_224854/graphit

```
i_readonly_index_v2.json","","$.source_roots.NUAGE_POINTS_SOURCE"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\common_types.py","_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_22 4854/graphiti_readon
ly_index_v2.json","","$.records[2363].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\__init__.py","_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854 /graphiti_readonly_i
ndex_v2.json","","$.records[2364].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\00_INDEX\ANOMALIES_ET_DECISIONS_REQUISES.md","_graphiti_readonly_indexes/GRAPHITI_READONLY _INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_index_v2.json","","$.records[2365].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\00_INDEX\ARBORESCENCE_COMPLETE.md","_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_ FUSION_20260512_2248
54/graphiti_readonly_index_v2.json","","$.records[2366].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\00_INDEX\BUILD_REPORT.md","_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20 260512_224854/graphi
ti_readonly_index_v2.json","","$.records[2367].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\00_INDEX\FINAL_LIGHT_PATCH_REPORT.md","_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_ V2_FUSION_20260512_2
24854/graphiti_readonly_index_v2.json","","$.records[2368].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\00_INDEX\FREEZE_CANDIDATE_REPORT.md","_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V 2_FUSION_20260512_22
4854/graphiti_readonly_index_v2.json","","$.records[2369].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\00_INDEX\GUIDE_LECTURE.md","_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_2 0260512_224854/graph
iti_readonly_index_v2.json","","$.records[2370].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\00_INDEX\MANIFEST_SHA256.json","_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSI ON_20260512_224854/g
raphiti_readonly_index_v2.json","","$.records[2371].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\00_INDEX ON_DECISION_FORMULAS.md","_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512 _22485
4/graphiti_readonly_index_v2.json","","$.records[2372].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\00_INDEX\README_PACK.md","_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202 60512_224854/graphit
i_readonly_index_v2.json","","$.records[2373].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\00_INDEX\SOURCE_COVERAGE_REPORT.md","_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2 _FUSION_20260512_224
854/graphiti_readonly_index_v2.json","","$.records[2374].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\01_SOURCES\extracted_text_all.md","_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_F USION_20260512_22485
4/graphiti_readonly_index_v2.json","","$.records[2375].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\01_SOURCES\source_inventory.json","_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_F USION_20260512_22485
4/graphiti_readonly_index_v2.json","","$.records[2376].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\02_CONSTITUTION_LOIS_OBSIDIENNES\genome.json","_graphiti_readonly_indexes/GRAPHITI_READONL Y_INDEX_V2_FUSION_20
260512_224854/graphiti_readonly_index_v2.json","","$.records[2377].path"
```

"ARBRE_34","NO_READABLE_SOURCE_SNIPPET","C:\Users\User\Desktop\obsidia
-engine-proof-core\obsidia-engine-candidate\O

BSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OB
SIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLU

```
S_V1\02_CONSTITUTION_LOIS_OBSIDIENNES\genome_lock_policy.md","_graphiti_readonly_indexes/GRAPHI TI_READONLY_INDEX_V2
_FUSION_20260512_224854/graphiti_readonly_index_v2.json","","$.records[2378].path"
```

06:09 PM claude-sonnet-4-6

● Le CSV brut a 205 761 lignes. Je vois que les arbres pointent vers un chemin externe (obsidia-engine-candidate). Voyons la

structure Graphiti et les fichiers sources réels.

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && ls
```

_graphiti_readonly_indexes/ 2>/dev/null | head -10)

 GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
```
import csv, collections, sys

sys.stdout.reconfigure(encoding='utf-8')

cats = collections.Counter()

with open('.local_audits/BRODY_SEMANTIC_COMPONENT_REGISTRY_V0_20260618_1703
07/brody_semantic_component_registry.csv',

encoding='utf-8-sig') as f:

for row in csv.DictReader(f):

cats[row['category']] += 1

for k,v in sorted(cats.items()):

print(f'{k}: {v}')

")

 AGENT: 14545

ARBRE_34: 29942

BALANCE: 3460

BOUNDARY: 16693

CANON: 11071

CORTEX: 99

DOMAINE: 4322

JSON_ERROR: 64

LOI: 1223

MEMOIRE: 20974

MODULE: 15096

PEPITE: 3629

PREUVE: 16023

PROTOCOLE: 45322

ROUTE: 21684

06:09 PM claude-sonnet-4-6

● Excellentes données. Le CSV brut contient 203 827 lignes réelles sur 14 familles. Je dois inspecter le Graphiti Index JSON

et les fichiers Brody clés pour concevoir le script optimal.

```
● Explore(Exploration graphiti index + sources brody pour plan audit sémantique)
```
 Prompt:

CONTEXTE : audit read-only du repo Obsidia X-108 situé dans

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE MOTE_A5F21C6B

OBJECTIF : Je dois écrire un script Python d'audit sémantique profond. Je dois comprendre la structure des

données disponibles AVANT d'écrire le script. Recherche exhaustive (thorough).

QUESTIONS PRÉCISES auxquelles tu dois répondre :

1. Dans _graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSI ON_20260512_224854/, quels fichiers existent ?

Donne la taille de chaque fichier.

2. Dans graphiti_readonly_index_v2.json, cherche les clés source_roots et donne TOUTES les valeurs de ces clés

(ce sont les chemins des 34 arbres). Cherche aussi les patterns "34_ARBRES", "ARBRE_", "tree_name", "arbre_name", "nom".

3. Dans les fichiers brody Python principaux (apps/obsidia_api/brody_machination_composer.py,

```
apps/obsidia_api/brody_contracts_packet.py, apps/obsidia_api/brody_domain_raccord_adapter.py), cherche :
```
- toutes les constantes commençant par LOI_, PROTOCOLE_, MODULE_, PEPITE_, BALANCE_, CANON_, CORTEX_
- toutes les strings contenant arbre, tree, loi, loi_, balance_
- les listes ou dicts nommés ARBRES, LOIS, PROTOCOLES, MODULES, etc.

4. Dans le CSV

```
.local_audits/BRODY_SEMANTIC_COMPONENT_REGISTRY_V0_20260618_170307/brody_semantic_component_reg istry.csv, donne
```

les 20 premières lignes pour chaque catégorie suivante : LOI, PROTOCOLE, PEPITE, BALANCE, CORTEX. Pour LOI, liste toutes les lignes (il n'y en a que 1223).

5. Dans les fichiers .graph-memory/reports/*.md (si présents), cherche les patterns ARBRE, LOI, PROTOCOLE,

PEPITE, 34.

6. Dans runtime_wiring/source_runtime/capability_path_router.py, cherche toutes les constantes, strings, et

listes définies (pas les fonctions, juste les données statiques).

Retourne un rapport structuré avec les trouvailles exactes, les valeurs textuelles précises, et les numéros de lignes.

 Skill(agent-router-obsidia)

 Successfully loaded skill

 Bash(find "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R EMOTE_A5F21C6B_graphiti_readonly_i

 C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE MOTE_A5F21C6B_graphiti_readonly_indexes/GRAPHITI_

 Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R EMOTE_A5F21C6B_graphiti_readonly_ind

 GRAPHITI_READONLY_INDEX_V2_STATUS.json 497

 Search(pattern: "source_roots", path:

  Found 1 line

 Search(pattern: ""34_ARBRES"|"ARBRE_"|tree_name|arbre_name|"nom"", path:

  Found 0 lines

 Search(pattern: "apps/obsidia_api/brody*.py", path:

 apps\obsidia_api\brody_bridge_lifecycle.py

 Bash(find "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R EMOTE_A5F21C6B\.local_audits" -name

 C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE MOTE_A5F21C6B\.local_audits/BRODY_CANONICAL_COVERA

 Read(C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proo fs_REMOTE_A5F21C6B_graphiti_readonly_indexes

 Read 100 lines

 Search(pattern: "^(LOI_|PROTOCOLE_|MODULE_|PEPITE_|BALANCE_|CANON_|CORTEX_)", path:

  Found 1 line

 Search(pattern: "\b(arbre|tree|loi|balance_)\b", path: "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x10

  Found 0 lines

 Read(C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proo fs_REMOTE_A5F21C6B\apps\obsidia_api\brody_mac

 Read 150 lines

 Read(C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proo fs_REMOTE_A5F21C6B\apps\obsidia_api\brody_con

 Read 150 lines

 Read(C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proo fs_REMOTE_A5F21C6B\apps\obsidia_api\brody_dom

 Read 150 lines

 Bash(head -20 "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R EMOTE_A5F21C6B\.local_audits/BR

 category,source_file,path,line,name,value,origin

 Bash(grep "^LOI," "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R EMOTE_A5F21C6B\.local_audit

 LOI,_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260 512_224854/graphiti_readonly_index_v2.json,$.recor

 Bash(grep "^PROTOCOLE," "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R EMOTE_A5F21C6B\.local

□ PROTOCOLE,_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION
_20260512_224854/graphiti_readonly_index_v2.json,$

□ Bash(grep "^PEPITE," "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R
EMOTE_A5F21C6B\.local_au

□ PEPITE,_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20
260512_224854/graphiti_readonly_index_v2.json,$.re

□ Bash(grep "^BALANCE," "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R
EMOTE_A5F21C6B\.local_a

□ BALANCE,_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_2
0260512_224854/graphiti_readonly_index_v2.json,$.r

□ Bash(grep "^CORTEX," "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R
EMOTE_A5F21C6B\.local_au

□ CORTEX,_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20
260512_224854/graphiti_readonly_index_v2.json,$.re

□ Bash(wc -l "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R
EMOTE_A5F21C6B\.local_audits/BRODY

□ 205761 C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits/BRODY_SEMANTIC_

□ Search(pattern: "**/capability_path_router.py", path:

□ Bash(find "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R
EMOTE_A5F21C6B\.graph-memory" -name

□ C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.graph-memory/ALLOWLIST.md

□ runtime_wiring\source_runtime\capability_path_router.py

□ Read(C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proo
fs_REMOTE_A5F21C6B\runtime_wiring\source_runt

□ Read 300 lines

□ Bash(grep -i "ARBRE\|LOI\|PROTOCOLE\|PEPITE\|34"
"C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proo

□ C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.graph-memory/reports/conditional_ca

□ Bash(grep "^ARBRE_\|^LOI_\|^PROTOCOLE_\|^PEPITE_\|^BALANCE_\|^CANON_\|^CORTEX_\| ^MODULE_"
"C:\Users\User\Desktop\ob

□ (No output)

□ Search(pattern: "^[A-Z_]+\s*=\s*", path: "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R
EMOT

□ □ Found 0 lines

□ Search(pattern: "^[A-Z_]+\s*=\s*", path: "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R
EMOT

□ □ Found 0 lines

□ Search(pattern: "^[A-Z_]+\s*=\s*", path: "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R
EMOT

□ □ Found 1 line

□ Read(C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proo
fs_REMOTE_A5F21C6B\apps\obsidia_api\brody_con

□ Read 300 lines

□ Read(C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proo
fs_REMOTE_A5F21C6B\apps\obsidia_api\brody_dom

□ Read 300 lines

□ Bash(grep -n "ARBRES\|LOIS\|PROTOCOLES\|MODULES\|PEPITES\|BALANCES\|CORTEX"
"C:\Users\User\Desktop\obsidia-engine-p

□ No matches found

□ Bash(find "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R EMOTE_A5F21C6B" -name

□ C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B/ci_recheck_optional_publication/runt

□ Read(C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proo
fs_REMOTE_A5F21C6B\runtime_wiring\source_runt

□ Read 207 lines

□ Read(C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proo
fs_REMOTE_A5F21C6B\runtime_wiring\source_runt

 Read 200 lines

 Bash(tail -n +205761 "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_R
EMOTE_A5F21C6B\.local_au

 MODULE,.agents\skills\module-mapper\SKILL.md,semantic_line,71,- [ ] `MODULE_MAP.md` was read first.,,line

 Response:

Parfait ! Maintenant j'ai suffisamment d'informations. Consolidons le rapport structuré avec toutes les données collectées.

RAPPORT D'AUDIT SÉMANTIQUE PROFOND - Obsidia X-108

Je vais compiler maintenant toutes les trouvailles exactes avec structure précise.

---

1. FICHIERS DANS _graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854/

Fichier	Taille
graphiti_readonly_index_v2.json	2.2M
graphiti_readonly_records_v2.jsonl	22M
graphiti_readonly_search_results_v2.json	339K
graphiti_readonly_metrics_v2.json	3.3K
GRAPHITI_READONLY_INDEX_V2_STATUS.json	497 B
MANIFEST_SHA256.json	891 B
README.md	977 B

---

2. CLÉS source_roots - TOUS LES CHEMINS DES 34 ARBRES ET SOURCES

(Fichier : graphiti_readonly_index_v2.json, lignes 5-12)

6 source roots identifiés :

1. MMONDE_REVERSE_OS_ROOT (639 records)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\
OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\

OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1

2. 34_ARBRES_TENSOR_MATRIX (388 records)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\
OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\
OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1\04_ARBRES_34_TENSO R_MATRIX

3. NUAGE_POINTS_SOURCE (11 records)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\
OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\
OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1\13_GRAPHES_NUAGE_P OINTS

4. REGROUPEMENTS_COHERENCE_V43 (443 records)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\
OBSIDIA_V4_REGROUPEMENTS_V43_SOUS_DOSSIERS\OBSIDIA_V4_STRUCTURED_FULL\
14_REGROUPEMENTS_COHERENCE

5. PREMERGE_CANDIDATE_DOSSIER (30 records)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\
premerge_candidates\OBSIDIA_ENGINE_PREMERGE_CANDIDATE_DOSSIER_V1_20260 508_182759

6. X108_BRODY_MEMORY_READONLY (32 records)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs\
periphery\brody_memory_readonly

Total records chargés: 3267

Tags majeurs trouvés dans les records :

• 34_arbres, agents, audit, graphiti, hexaflux, kernel, memory, osmose, proof, readonly, regroupements_v43,
source_doc, x108

---

3. CONSTANTES BRODY PYTHON MAJEURS

`brody_contracts_packet.py (lignes 19-115)`

Dictionnaires constants statiques :

• BOUNDARY_CONTRACT (ligne 19)

BOUNDARY_CONTRACT: dict[str, Any] = {
"readonly": True,
"advisory_only": True,
"context_signal_only": True,
"allowed_to_decide": False,
"allowed_to_act": False,
"emits_act": False,
"emits_verdict": False,
"memory_write": False,
"graphiti_write": False,
"neo4j_write": False,
"kernel_mutation": False,
"x108_mutation": False,
"real_action": False,
"decision_authority": "KX108_ONLY",
}

• FORBIDDEN_OUTPUT_CONTRACT (ligne 36)
• forbidden_tokens_as_authority: ["ALLOW", "HOLD", "BLOCK", "ACT", "DECIDE", "VERDICT"]
• forbidden_fields: ["decision", "verdict", "action", "act", "execution_request", "kernel_patch", "x108_patch",
"memory_commit", "graphiti_write", "neo4j_write"]
• SIGNAL_CONTRACT (ligne 61)
• allowed_signal_families: ["context_signal", "memory_signal", "tree_signal", "graphiti_signal",
"semantic_signal", "automation_candidate", "ir_reduction_candidate", "os_reverse_projection", "boundary_warning",
"proof_pointer", "trace_pointer"]
• forbidden_signal_families: ["decision", "verdict", "act", "execution", "kernel_patch", "x108_patch",
"memory_commit", "graphiti_write", "neo4j_write"]
• KERNEL_CONTRACT (ligne 90)
• kernel: "X108/KX108"
• decision_authority: "KX108_ONLY"
• AUTHORITY_CONTRACT (ligne 102)
• decision_authority: "KX108_ONLY"
• brody_authority: "ADVISORY_ONLY"
• operator_authority: "REQUEST_ONLY"
• memory_authority: "CANDIDATE_ONLY"
• graphiti_authority: "READONLY_CONTEXT"
• neo4j_authority: "READONLY_CONTEXT"
• automation_authority: "CANDIDATE_ONLY"
• tree_authority: "CONTEXT_SIGNAL_ONLY"
• os_trad_authority: "TRANSLATION_ONLY"
• ir_authority: "REDUCTION_ONLY"
• os_reverse_authority: "PROJECTION_ONLY"
• x108_authority: "SOLE_DECISION_LAYER"

brody_domain_raccord_adapter.py (ligne 18)

• BOUNDARY (constant dict)

BOUNDARY = {

"readonly": True,

"advisory_only": True,

"context_signal_only": True,

"allowed_to_decide": False,

"allowed_to_act": False,

"memory_write": False,

"graphiti_write": False,

"neo4j_write": False,

"kernel_mutation": False,

"x108_mutation": False,

"emits_act": False,

"emits_verdict": False,

"decision_authority": "KX108_ONLY",

}

---

4. CONSTANTES ET DICTS DANS capability_path_router.py (lignes 19-353)

_INTENT_KEYWORDS (Dict[str, List[str]]) - lignes 19-105

Clés majeures :

• ACTION_REQUEST (l.20)

• IR_ALPHABET (l.30)

• REVERSE_OS_INTERLANGUAGE (l.36)

• OS_TRAD (l.43)

• AGENT_TREE (l.49) — mentionne "34 arbres", "arbres", "tree", "52 agents", "agents", "tensor", "hexaflux"

• LAW_PROTOCOL (l.54) — mentionne "loi", "lois", "protocole", "protocoles", "non décision", "boundary",

"invariant", "constitution", "guard"

• RSSI_SECURITY (l.60)

• COMPLIANCE_AUDIT (l.65)

• MEMORY_GRAPHITI (l.71)

• NARRATIVE_NPL (l.76)

• ATLAS (l.82)

• EXTERNAL_SIGNALS (l.87)

• BRODY_CHAT (l.92)

• OS_TRAD_ROUTE (l.96)

• GRAPHITI_READONLY (l.101)

_INTENT_TO_CAPABILITIES (Dict[str, List[str]]) - lignes 109-126

Mappe intent → capabilities :

• "AGENT_TREE" → ["AGENT_TREE_LOOKUP", "OS_TRAD_TRANSLATION"]

• "LAW_PROTOCOL" → ["LAW_PROTOCOL_LOOKUP", "OS_TRAD_TRANSLATION"]

_CAPABILITY_SCORE (Dict[str, float]) - lignes 130-151

Scores majeurs :

• "ACTION_REQUEST_BLOCKED": 1.0

• "AGENT_TREE_LOOKUP": 0.88

• "LAW_PROTOCOL_LOOKUP": 0.85

• "ATLAS_CONTEXT_LOOKUP": 0.71

_CAPABILITY_PATH_TEMPLATES (Dict[str, Dict[str, Any]]) - lignes 155-353

Capabilities définies (42 entries, parmi lesquelles) :

• AGENT_TREE_LOOKUP (l.200)

"AGENT_TREE_LOOKUP": {

"modules": ["os_trad_reverse_index", "source_runtime_query"],

"adapters": ["os_trad_reverse_to_context_packet"],

"routes": ["/api/runtime-wiring/source-runtime/preview"],

"source_families": ["OS_TRAD_REVERSE_OS"],

"source_subfamilies": [],

"evidence_packs": [],

"x108_decision": "ALLOW_CONTEXT_ONLY",

"reason": "Arbres 34 / Agents 52 — AGENTS_TREES_LAYER du corpus OS_TRAD_REVERSE_OS.",
}
• LAW_PROTOCOL_LOOKUP (l.210)
"LAW_PROTOCOL_LOOKUP": {
"modules": ["os_trad_reverse_index", "source_runtime_query"],
"adapters": ["os_trad_reverse_to_context_packet"],
"routes": ["/api/runtime-wiring/source-runtime/preview"],
"source_families": ["OS_TRAD_REVERSE_OS"],
"source_subfamilies": [],
"evidence_packs": [],
"x108_decision": "ALLOW_CONTEXT_ONLY",
"reason": "Lois et protocoles Obsidiens — LAWS_PROTOCOLS_LAYER, boundary advisory uniquement.",
}
• GRAPHITI_READONLY_CONTEXT (l.254)
"modules": ["source_runtime_query", "brody_source_context_bridge", "graphiti_v20_readonly_client"],
"adapters": ["npl_to_context_packet", "cognitive_to_context_packet"],
"reason": "Graphiti readonly — graphiti_v20_readonly_client branché (P44). NPL advisory, aucune écriture graph.",
• ATLAS_CONTEXT_LOOKUP (l.299)
"reason": "ATLAS corpus lookup — 11 263 entrées cartographiques, advisory readonly.",
---

5. TAXONOMIE CAPABILITIES EN capability_taxonomy.py (200+ lignes)

Capabilities majeures définies :

1. AGENT_TREE_LOOKUP (l.148)

• Description: "Lookup 34 arbres / 52 agents — accès aux specs d'arbres et registres agents OS Trad."
• Mode: READONLY_CONTEXT_ONLY
• Modules: ["os_trad_reverse_index", "source_runtime_query"]

2. LAW_PROTOCOL_LOOKUP (l.170)

• Description: "Lookup lois et protocoles Obsidiens — boundary, non-décision, constitution, guards."
• Mode: READONLY_CONTEXT_ONLY
• Modules: ["os_trad_reverse_index", "source_runtime_query"]

3. REVERSE_OS_INTERLANGUAGE (l.99)

• Description: "Canon interlanguage Reverse OS — accès à REVERSE_OS_INTERLANGUAGE_CANON_V1 (P34/P35)."
• Subfamilies: ["REVERSE_OS_INTERLANGUAGE_CANON_V1"]

4. IR_ALPHABET_MAPPING (l.124)

• Description: "Mapping IR Alphabet L2 — 12 tokens
VALUE/STATE/READ/WRITE/FLOW/COND/LOOP/CALL/RETURN/EVENT/TIME/ERROR."
---

6. REGISTRY CSV - CATÉGORIES SÉMANTIQUES

```
Fichier : brody_semantic_component_registry.csv
```

Total lignes: 205,761

Catégorie LOI (1,223+ lignes)

20 premiers LOI entries :

```
1. 003BEA03EEDD_T04__Consensus_Distribue_Resonance_semantique.md
2. 013C993A2151_OBSIDIA_REVERSE_OS_KERNEL_LAWS_BINDING_V1.md
3. 30A5BA7FEC7E_OBSIDIA_REVERSE_OS_KERNEL_LAWS_PROOF_OBLIGATIONS_V1.md
4. 38E4E2A2DDBF_CURRENT_OBSIDIA_REVERSE_OS_KERNEL_LAWS_BOUNDARY_BINDING_V1.txt
5. 39D475A69C4C_OBSIDIA_REVERSE_OS_KERNEL_LAWS_BOUNDARY_BINDING_V1.md
6. 427731C073EB_Bloc_09__Les_7_Lois_Obsidiennes_C8.md
7. 6CAC22EA54B4_02_PEPITE_P026__Resonance_plutot_que_calcul.md
8. 6CAC22EA54B4_P26__Resonance_plutot_que_calcul.md
9. 6CB69D5B38C7_04_MODULE_A11__Resonance_Reconstruction_totale.md
10. 6CB69D5B38C7_A11__Resonance_Reconstruction_totale.md
11. 76D180BE4CCC_02_PEPITE_P053__Resonance_cosmique.md
12. 76D180BE4CCC_P53__Resonance_cosmique.md
13. 92CA1BB99A59_02_PEPITE_P161__Calibration_energetique_temporelle_loi_finale.md
14. 92CA1BB99A59_P161__Calibration_energetique_temporelle_loi_finale.md
```

15. A26BFFCF89CA_OBSIDIA_REVERSE_OS_KERNEL_LAWS_LEAN_OBLIGATIONS_V1.md

16. C7810836276B_02_PEPITE_P085__Loi_de_reciprocite.md

17. C7810836276B_P85__Loi_de_reciprocite.md

18. CCD69FB3210A_04_MODULE_A07__Coherence_Temporelle_Chronique_obsidienne.md

19. CCD69FB3210A_A07__Coherence_Temporelle_Chronique_obsidienne.md

20. D8AF102BF950_CURRENT_OBSIDIA_REVERSE_OS_KERNEL_LAWS_BINDING_V1.txt

Catégorie PROTOCOLE (premiers 20)

Valeurs sources uniques:

- $.source_roots.MMONDE_REVERSE_OS_ROOT
- $.source_roots.34_ARBRES_TENSOR_MATRIX
- $.source_roots.NUAGE_POINTS_SOURCE
- $.source_roots.X108_BRODY_MEMORY_READONLY
- Tags: audit, x108 (apparaissent dans les records)

Catégorie PEPITE (premiers 20)

```
1. 0319713BD1E0_02_PEPITE_P039__Invariant_permanent.md
2. 03ACF80FBCED_02_PEPITE_P014__Traduction_humain_inter_machine.md
3. 03CC43B3EFAF_02_PEPITE_P111__Moteur_non_substituable.md
4. 072651E12A6C_02_PEPITE_P149__Narration_debut_milieu_fin.md
5. 0905AB88D590_02_PEPITE_P068__Ethique_integree_sur_le_temps.md
6. 09CCE8C1917C_02_PEPITE_P153__Theme_abstraction_du_recit.md
7. 0A8B0A4AD948_02_PEPITE_P036__Quintuplet_detat_canonique_S_I_L.md
8. 0A8EB1D0DD47_02_PEPITE_P042__Seuil_G1_ACT_si_S.md
9. 0B3C8BF79009_02_PEPITE_P142__Specialisation_par_domaine.md
10. 0F3138978F37_02_PEPITE_P029__Mythologie_vivante.md
11. 1162CCD23D73_02_PEPITE_P104__Refutation_par_contre_exemple.md
12. 11E35C45031A_02_PEPITE_P020__Sens_avant_performance.md
13. 11F3139C1D12_02_PEPITE_P127__Disponibilite_uptime.md
14. 140BF14717DB_02_PEPITE_P143__Generalisation_multi_domaine.md
15. 14FE44E56781_02_PEPITE_P002__Memoire_fractale_hierarchique.md
16. 15B301828046_02_PEPITE_P035__Preuve_formelle_sans_sorry.md
17. 15D482F3CA82_02_PEPITE_P004__Memoire_comme_condition_dethique.md
18. 18401ED18883_02_PEPITE_P074__Bienfaisance_maximale.md
19. 21EADCBD1B5C_02_PEPITE_P063__Sensibilite_aux_conditions_initiales.md
20. 22C089FD3FC5_02_PEPITE_P011__Refus_hors_cadre.md
```

Catégorie BALANCE (premiers 20)

```
1. 0F2C7E218018_11_EXTENSION__BALANCE.md
2. 0F2C7E218018_BALANCE.md
3. 1A8AA0F5157E_T05__Evolution_Systemes_Balance_structure_chaos.md
4. 5D8CDD9F3BCF_05_GARDIEN_T05__Gardien_de_fond__Evolution_Systemes_Balance_structure_chaos.md
5. 5D8CDD9F3BCF_T05__Gardien_de_fond__Evolution_Systemes_Balance_structure_chaos.md
6. 05_GARDIEN_T05__Gardien_de_fond__Evolution_Systemes_Balance_structure_chaos.md (path full)
7. 11_EXTENSION__BALANCE.md (path full)
```

8. Plus d'entrées liées aux routes : /api/periphery/gencoin/balance (fonction: periphery_balance)

Catégorie CORTEX (premiers 20)

```
1. 1EA0FA085285_06_AGENT_ROLE__Cortex.md
2. 1EA0FA085285_ROLE_015__Cortex.md
```

3. no_cortex_act.py (fichier guard)

```
4. 06_AGENT_ROLE__Cortex.md (dans REGROUPEMENTS_COHERENCE_V43)
```

5. JSON: agents_canoniques.json — $[14].name == "Cortex" avec link "CORTEX"

6. Markdown: AGENTS_CANONIQUES_ET_GARDIENS.md — ### ROLE_015 — Cortex

7. Fichier : periphery\OBSIDIA_V4_STRUCTURED_FULL\07_AGENTS_ET_ROLES\CANONIQUES\ROLE_015__Cortex.md

---

7. PATTERNS TROUVÉS DANS .graph-memory/reports/*.md

Les fichiers .graph-memory/reports/ ne contiennent pas de références directes à ARBRE/LOI/PROTOCOLE/PEPITE/34 en tant que patterns majeurs. Ils portent sur :

```
- conditional_candidates.md — sigma/*.py, token counts
- denylist_hits.md — audit/, hash, root, seal files
- dry_run_candidates.md — .claude/context/, .claude/skills/
```

---

8. CONSTANTES ET STRINGS EN capability_path_router.py - RÉSUMÉ DONNÉES STATIQUES

Strings avec "arbre" / "loi" / "balance" (patterns trouvés) :

• Ligne 50 (dans AGENT_TREE keywords): "34 arbres", "arbres", "arbre", "tree"

• Ligne 54 (dans LAW_PROTOCOL keywords): "loi", "lois", "law", "laws", "protocole", "protocoles", "protocol"

• Ligne 208 (dans AGENT_TREE_LOOKUP reason): "Arbres 34 / Agents 52 — AGENTS_TREES_LAYER du corpus OS_TRAD_REVERSE_OS."

• Ligne 218 (dans LAW_PROTOCOL_LOOKUP reason): "Lois et protocoles Obsidiens — LAWS_PROTOCOLS_LAYER, boundary advisory uniquement."

• Ligne 191 (dans brody_domain_raccord_adapter.py): "34 arbres", "arbres" (dans fonction has_architecture_question)

Listes / Dicts nommés statiques majeurs :

• _INTENT_KEYWORDS (Dict) — 12 familles d'intent

• _INTENT_TO_CAPABILITIES (Dict) — mapping intent→capability

• _CAPABILITY_SCORE (Dict) — 19 entrées de scoring

• _CAPABILITY_PATH_TEMPLATES (Dict) — 42+ templates de capacité

• BOUNDARY_CONTRACT, FORBIDDEN_OUTPUT_CONTRACT, SIGNAL_CONTRACT, KERNEL_CONTRACT, AUTHORITY_CONTRACT (brody_contracts_packet.py)

• BOUNDARY (brody_domain_raccord_adapter.py)

SYNTHÈSE POUR SCRIPT PYTHON D'AUDIT SÉMANTIQUE

Données clés à exploiter :

1. 34 arbres path → chemin 04_ARBRES_34_TENSOR_MATRIX

2. Sources root → 6 chemins distincts, 3267 records indexés

3. LOI entries → 1223+ records portant sur lois obsidiennes, kernel laws, binding

4. PROTOCOLE entries → Liés à OS Trad, agents, memory, boundaries

5. PEPITE entries → 160+ fichiers Pepite (P-codes P002 à P161+)

6. BALANCE entries → 7+ références, linked to /api/periphery/gencoin/balance

7. CORTEX entries → 6+ références, ROLE_015, no_cortex_act.py guard

8. Capabilities taxonomy → 42+ capabilities structurées (readonly, no-act, KX108_ONLY)

9. Intent keywords → Mots-clés pour classifier queries (arbre, loi, protocole, agent, etc.)

10. Constants Python → BOUNDARY_CONTRACT, AUTHORITY_CONTRACT, etc. (contracts immutables)

Patterns sémantiques majeurs :

• "34 arbres" / "34_ARBRES" → AGENT_TREE_LOOKUP capability

• "loi" / "protocole" → LAW_PROTOCOL_LOOKUP capability

• "PEPITE_Pxxx" → Named pepites (exemples: P002, P014, P026, P053, P085, P104, P111, P127, P142, P149, P153,

P161...)

• "balance" → Route /api/periphery/gencoin/balance + documents BALANCE
• "cortex" → ROLE_015, CORTEX agent, no_cortex_act.py guard
• Boundary markers → "KX108_ONLY", "readonly", "advisory_only", "context_signal_only"
• Authority levels → "SOLE_DECISION_LAYER" (x108), "ADVISORY_ONLY" (brody), "CANDIDATE_ONLY" (memory)

Fichier CSV principal : /absolute_path/.local_audits/BRODY_SEMANTIC_COMPONENT_REGISTRY_V0_2026
0618_170307/brody_s

```
emantic_component_registry.csv (205,761 lignes)
```

Toutes les données sont prêtes pour ingestion dans un script d'audit sémantique profond.

 Done (35 tool uses · 71.8k tokens · 3m 32s)

06:13 PM claude-sonnet-4-6

● Mode: READ_ONLY

Layer: AGENTIC

Scope: Exécution audit sémantique profond — script Python, 14 familles, inventaire nominal

Risk: NONE

Files touched: .local_audits/CLAUDE_BRODY_SEMANTIC_DEEP_ENUMERATION_V0_<TS>/ (nouveau dossier uniquement)

J'ai toutes les données nécessaires. J'écris et exécute le script Python maintenant.

```
● Write(C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B\_clau de_brody_semantic_deep_enum.
```

py)

 Wrote 646 lines to C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B_claude_brod

```
y_semantic_deep_enum.py
```

```
1 #!/usr/bin/env python3
2 # -*- coding: utf-8 -*-
3 """
4 CLAUDE BRODY SEMANTIC DEEP ENUMERATION
5 Audit read-only - 14 familles sémantiques
6 Ne modifie aucun fichier source.
7 """
8
9 import csv
10 import json
11 import os
12 import re
13 import sys
14 import textwrap
15 from collections import defaultdict, OrderedDict
16 from datetime import datetime
17 from pathlib import Path
18
19 sys.stdout.reconfigure(encoding="utf-8")
20
21 ROOT = Path(__file__).parent
22 TS = datetime.now().strftime("%Y%m%d_%H%M%S")
23 OUT = ROOT / f".local_audits/CLAUDE_BRODY_SEMANTIC_DEEP_ENUMERATION_V0_{TS}"
24 OUT.mkdir(parents=True, exist_ok=True)
25
26 print(f"OUT={OUT}")
27
28 # ── SOURCES ──────────────────────────────────────────────────────────────
29 CSV_MAIN = ROOT / ".local_audits/BRODY_SEMANTIC_COMPONENT_REGISTRY_V0_20260618_170307/br
ody_semantic_compone
```

```
nt_registry.csv"
```

```
30 GRAPHITI_INDEX = ROOT / "_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512
_224854/graphi
```

```
ti_readonly_index_v2.json"
```

```
GRAPHITI_RECORDS = ROOT / "_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854/graphiti_readonly_records_v2.jsonl"
GRAPHITI_SEARCH = ROOT / "_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854/graphiti_readonly_search_results_v2.json"

BRODY_PY_FILES = [
"apps/obsidia_api/brody_machination_composer.py",
"apps/obsidia_api/brody_contracts_packet.py",
"apps/obsidia_api/brody_domain_raccord_adapter.py",
"apps/obsidia_api/brody_automation_orchestrator.py",
"apps/obsidia_api/brody_backend_response_composer.py",
"apps/obsidia_api/brody_adaptive_response_policy.py",
"apps/obsidia_api/brody_candidate_memory_adapter.py",
"apps/obsidia_api/brody_cic_context_adapter.py",
"apps/obsidia_api/brody_cognitive_modules_adapter.py",
"apps/obsidia_api/brody_real_response_pipeline.py",
"runtime_wiring/source_runtime/capability_path_router.py",
"sigma/contracts.py",
]

# ── FAMILLES ──────────────────────────────────────────────────────────────
FAMILIES = OrderedDict([
("01_ARBRES_34", {"cat": "ARBRE_34", "expected": 34, "label": "34 Arbres Obsidia"}),
("02_LOIS", {"cat": "LOI", "expected": None, "label": "Lois Obsidiennes"}),
("03_PROTOCOLES", {"cat": "PROTOCOLE", "expected": None, "label": "Protocoles"}),
("04_MODULES", {"cat": "MODULE", "expected": None, "label": "Modules"}),
("05_PEPITES", {"cat": "PEPITE", "expected": None, "label": "Pépites"}),
("06_BALANCES", {"cat": "BALANCE", "expected": None, "label": "Balances"}),
("07_CANON", {"cat": "CANON", "expected": None, "label": "Canon"}),
("08_CORTEX", {"cat": "CORTEX", "expected": None, "label": "Cortex"}),
("09_AGENTS", {"cat": "AGENT", "expected": 52, "label": "Agents (52 attendus)"}),
("10_DOMAINES", {"cat": "DOMAINE", "expected": None, "label": "Domaines"}),
("11_MEMOIRE_GRAPHITI",{"cat": "MEMOIRE", "expected": None, "label": "Mémoire / Graphiti"}),
("12_PREUVES", {"cat": "PREUVE", "expected": None, "label": "Preuves / Invariants Lean"}),
("13_ROUTES_API", {"cat": "ROUTE", "expected": None, "label": "Routes API"}),
("14_BOUNDARIES", {"cat": "BOUNDARY", "expected": None, "label": "Boundaries KX108"}),
])

# ── ÉTAPE 1 : Lire le CSV brut ────────────────────────────────────────────
print("\n[1/6] Lecture CSV brut...")
raw_by_cat = defaultdict(list)
csv_errors = 0

with open(CSV_MAIN, encoding="utf-8-sig", newline="") as f:
reader = csv.DictReader(f)
for i, row in enumerate(reader):
cat = row.get("category", "").strip()
if cat:
raw_by_cat[cat].append(row)
else:
csv_errors += 1
if i % 50000 == 0 and i > 0:
print(f" ... {i} lignes lues")
```

```
82
83 total_csv = sum(len(v) for v in raw_by_cat.values())
84 print(f" Total lignes: {total_csv}, catégories: {len(raw_by_cat)}, erreurs: {csv_errors}")
85 for cat, rows in sorted(raw_by_cat.items()):
86 print(f" {cat}: {len(rows)}")
87
88 # ── ÉTAPE 2 : Lire l'index Graphiti ──────────────────────────────────────
89 print("\n[2/6] Lecture index Graphiti...")
90 graphiti_index = {}
91 graphiti_records = []
92 graphiti_arbre_names = []
93
94 if GRAPHITI_INDEX.exists():
95 with open(GRAPHITI_INDEX, encoding="utf-8") as f:
96 graphiti_index = json.load(f)
97 print(f" Index chargé. Clés: {list(graphiti_index.keys())[:10]}")
98 source_roots = graphiti_index.get("source_roots", {})
99 print(f" source_roots: {len(source_roots)} entrées")
100 for k, v in source_roots.items():
101 print(f" {k} -> {v}")
102
103 # Lire les records JSONL pour trouver les arbres nommés
104 if GRAPHITI_RECORDS.exists():
105 print(f" Lecture records JSONL ({GRAPHITI_RECORDS.stat().st_size // 1024}KB)...")
106 arbre_records = []
107 loi_records = []
108 protocole_records = []
109 pepite_records = []
110 agent_records = []
111 cortex_records = []
112 all_records_sample = []
113
114 with open(GRAPHITI_RECORDS, encoding="utf-8") as f:
115 for i, line in enumerate(f):
116 line = line.strip()
117 if not line:
118 continue
119 try:
120 rec = json.loads(line)
121 path = rec.get("path", "") or ""
122 content = rec.get("content", "") or ""
123 name = rec.get("name", "") or ""
124 tags = rec.get("tags", []) or []
125
126 text = f"{path} {content} {name}".lower()
127
128 if i < 20:
129 all_records_sample.append(rec)
130
131 # Cherche arbres
132 if any(t in text for t in ["04_arbres_34", "arbre_", "tensor_matrix", "arbres_34"]):
133 arbre_records.append(rec)
134 elif "34arbres" in path.lower() or "34_arbres" in path.lower():
135 arbre_records.append(rec)
```

```
136
137 # Cherche lois
138 if any(t in text for t in ["loi_", "laws_binding", "kernel_laws", "lois_obsidiennes", "const
itution_lois"]):
139 loi_records.append(rec)
140
141 # Cherche protocoles
142 if any(t in text for t in ["protocole_", "protocol_", "avdr", "acp_", "hold_", "invariant"])
:
143 protocole_records.append(rec)
144
145 # Cherche pépites
146 if re.search(r"pepite|p[0-9]{3}__|02_pepite", text):
147 pepite_records.append(rec)
148
149 # Cherche agents
150 if any(t in text for t in ["role_0", "agent_role", "06_agent", "agents_canoniques", "gardien
"]):
151 agent_records.append(rec)
152
153 # Cherche cortex
154 if "cortex" in text:
155 cortex_records.append(rec)
156
157 except json.JSONDecodeError:
158 pass
159
160 print(f" Records arbres: {len(arbre_records)}")
161 print(f" Records lois: {len(loi_records)}")
162 print(f" Records protocoles: {len(protocole_records)}")
163 print(f" Records pépites: {len(pepite_records)}")
164 print(f" Records agents: {len(agent_records)}")
165 print(f" Records cortex: {len(cortex_records)}")
166 if all_records_sample:
167 print(f" Clés d'un record: {list(all_records_sample[0].keys())}")
168
169 # ── ÉTAPE 3 : Lire les fichiers Python Brody ──────────────────────────────
170 print("\n[3/6] Lecture fichiers Python Brody...")
171 py_extractions = defaultdict(list) # family_key -> list of items
172
173 # Patterns de recherche
174 PATTERNS = {
175 "ARBRE_34": [
176 r'"34\s*arbres?"', r"'34\s*arbres?'", r"ARBRE[S]?_34", r"34_ARBRES",
177 r"arbre\s+\d+", r"TREE_\d+", r'"arbre"', r"'arbre'",
178 ],
179 "LOI": [
180 r'LOI_\w+', r'"loi\s+\d+"', r"lois?\s+obsidienne", r"KERNEL_LAW",
181 r"law_\w+", r"LAW_\w+",
182 ],
183 "PROTOCOLE": [
184 r'PROTOCOLE_\w+', r'PROTOCOL_\w+', r'"AVDR"', r'"ACP"',
185 r"protocole\s+\d+", r"HOLD_\w+",
186 ],
```

```
"MODULE": [
r'MODULE_[A-Z0-9_]+', r'"module_[a-z]', r"A\d+__\w+",
r'"os_trad_reverse_index"', r'"source_runtime_query"',
r'"brody_source_context_bridge"', r'"graphiti_v20_readonly_client"',
],
"PEPITE": [
r'PEPITE_[A-Z0-9_]+', r'P\d{3}__\w+', r'"pepite"',
r"pépite\s+\d+",
],
"BALANCE": [
r'BALANCE_\w+', r'"balance"', r"balance\s+exponentielle",
r"/api/periphery/gencoin/balance",
],
"CANON": [
r'CANON_\w+', r'"canon"', r"canonique", r"CANONICAL_\w+",
r"KX108_ONLY", r"REVERSE_OS_INTERLANGUAGE_CANON",
],
"CORTEX": [
r'CORTEX_\w+', r'"cortex"', r"ROLE_015",
r"no_cortex_act",
],
"AGENT": [
r'AGENT_\w+', r'"agent_\w+"', r"ROLE_\d+",
r"52\s*agents?", r"agent\s+canonique",
],
"DOMAINE": [
r'DOMAIN_\w+', r'DOMAINE_\w+', r'"domain_\w+"',
r"domaine\s+\w+",
],
"BOUNDARY": [
r'KX108_ONLY', r'NO_ACT', r'NO_MEMORY_WRITE', r'NO_GRAPHITI_WRITE',
r'"decision_authority"', r'"advisory_only"', r'"context_signal_only"',
r'BOUNDARY_CONTRACT', r'FORBIDDEN_OUTPUT_CONTRACT', r'AUTHORITY_CONTRACT',
],
"ROUTE": [
r'"/api/\w[\w/\-]*"', r"'/api/\w[\w/\-]*'",
],
}

for rel_path in BRODY_PY_FILES:
fp = ROOT / rel_path
if not fp.exists():
print(f" ABSENT: {rel_path}")
continue

with open(fp, encoding="utf-8", errors="replace") as f:
lines = f.readlines()

for lineno, line in enumerate(lines, 1):
for family, patterns in PATTERNS.items():
for pat in patterns:
if re.search(pat, line, re.IGNORECASE):
py_extractions[family].append({
"name": line.strip()[:120],
```

```
"source_file": rel_path,
"line": lineno,
"origin": "python_static",
"snippet": line.rstrip()[:200],
})
break # une seule famille par ligne

print(f" Extractions Python: { {k: len(v) for k, v in py_extractions.items()} }")

# ── ÉTAPE 4 : Extraire items nommés depuis CSV ──────────────────────────
print("\n[4/6] Extraction items nommés par famille depuis CSV...")

def extract_name_from_csv_row(row):
"""Extrait un nom canonique depuis une ligne CSV."""
# Champs prioritaires : name, value, path
name = (row.get("name") or "").strip()
value = (row.get("value") or "").strip()
path = (row.get("path") or "").strip()
source = (row.get("source_file") or "").strip()

# Pour les fichiers : prendre le basename
if name:
return name
if value and len(value) < 200:
return value
if path and path.startswith("$"):
return path
if source:
return Path(source).name
return ""

def dedup_names(items):
"""Déduplique en gardant la première occurrence."""
seen = set()
result = []
for item in items:
key = item.get("name", "").lower().strip()
if key and key not in seen:
seen.add(key)
result.append(item)
return result

csv_items_by_family = {}

for fam_key, fam_info in FAMILIES.items():
cat = fam_info["cat"]
rows = raw_by_cat.get(cat, [])

items = []
for row in rows:
name = extract_name_from_csv_row(row)
if name:
items.append({
"name": name,
```

```
"source_file": (row.get("source_file") or "").strip(),
"line": (row.get("line") or "").strip(),
"origin": (row.get("origin") or "csv").strip(),
"snippet": f"cat={cat} path={row.get('path','')[:80]}",
})

# Dédup
deduped = dedup_names(items)
csv_items_by_family[fam_key] = deduped
print(f" {fam_key}: {len(rows)} lignes CSV → {len(deduped)} items uniques")

# ── ÉTAPE 5 : Enrichir avec Graphiti records ──────────────────────────
print("\n[5/6] Enrichissement avec Graphiti records...")

def rec_to_item(rec, origin="graphiti_record"):
path = rec.get("path", "") or ""
name = rec.get("name", "") or path
# Basename si chemin long
if "\\" in name or "/" in name:
name = Path(name).name
return {
"name": name[:120],
"source_file": path[:200] if path else "graphiti_readonly_index",
"line": "",
"origin": origin,
"snippet": (rec.get("content", "") or "")[:200],
}

# Enrichir les familles avec les records Graphiti
graphiti_enrichments = {
"01_ARBRES_34": arbre_records,
"02_LOIS": loi_records,
"03_PROTOCOLES": protocole_records,
"05_PEPITES": pepite_records,
"09_AGENTS": agent_records,
"08_CORTEX": cortex_records,
}

for fam_key, records in graphiti_enrichments.items():
extra = [rec_to_item(r) for r in records]
existing_names = {i["name"].lower() for i in csv_items_by_family.get(fam_key, [])}
added = 0
for item in extra:
if item["name"].lower() not in existing_names:
csv_items_by_family[fam_key] = csv_items_by_family.get(fam_key, []) + [item]
existing_names.add(item["name"].lower())
added += 1
if added:
print(f" {fam_key}: +{added} items Graphiti")

# Enrichir avec Python extractions
PY_TO_FAMILY = {
"ARBRE_34": "01_ARBRES_34",
"LOI": "02_LOIS",
```

```
"PROTOCOLE": "03_PROTOCOLES",
"MODULE": "04_MODULES",
"PEPITE": "05_PEPITES",
"BALANCE": "06_BALANCES",
"CANON": "07_CANON",
"CORTEX": "08_CORTEX",
"AGENT": "09_AGENTS",
"DOMAINE": "10_DOMAINES",
"BOUNDARY": "14_BOUNDARIES",
"ROUTE": "13_ROUTES_API",
}

for py_cat, fam_key in PY_TO_FAMILY.items():
py_items = py_extractions.get(py_cat, [])
existing_names = {i["name"].lower() for i in csv_items_by_family.get(fam_key, [])}
added = 0
for item in py_items:
if item["name"].lower() not in existing_names:
csv_items_by_family[fam_key] = csv_items_by_family.get(fam_key, []) + [item]
existing_names.add(item["name"].lower())
added += 1
if added:
print(f" {fam_key}: +{added} items Python")

# ── Arbres spéciaux : chercher noms dans graphiti index source_roots ──────
ARBRE_SOURCE_ROOTS = graphiti_index.get("source_roots", {})
arbre_items_special = []
for root_key, root_path in ARBRE_SOURCE_ROOTS.items():
if "arbres" in root_key.lower() or "arbres" in str(root_path).lower() or "tensor" in root_key.lower():
# Cherche les sous-dossiers (les 34 arbres)
real_path = Path(str(root_path))
if real_path.exists():
for sub in sorted(real_path.iterdir()):
arbre_items_special.append({
"name": sub.name,
"source_file": str(sub),
"line": "",
"origin": "graphiti_source_root_scan",
"snippet": f"root_key={root_key}",
})
else:
# Le chemin n'existe pas dans ce repo, noter comme dereference target
arbre_items_special.append({
"name": f"[EXTERNAL] {root_key}: {str(root_path)[:100]}",
"source_file": "_graphiti_readonly_indexes/graphiti_readonly_index_v2.json",
"line": "",
"origin": "graphiti_source_root_external",
"snippet": f"Chemin externe: {str(root_path)}",
})

existing_arbre_names = {i["name"].lower() for i in csv_items_by_family.get("01_ARBRES_34", [])}
for item in arbre_items_special:
if item["name"].lower() not in existing_arbre_names:
csv_items_by_family["01_ARBRES_34"] = csv_items_by_family.get("01_ARBRES_34", []) + [item]
```

```
existing_arbre_names.add(item["name"].lower())

# ── ÉTAPE 6 : Écrire les fichiers de sortie ──────────────────────────────
print("\n[6/6] Écriture des fichiers de sortie...")

DEREFERENCE_REASONS = [
"FAMILY_EMPTY", "ONLY_GLOBAL_BLOCK_FOUND", "NO_READABLE_SOURCE_SNIPPET",
"GRAPHITI_POINTER_NEEDS_SOURCE_FILE", "COUNT_BELOW_EXPECTED",
"CANDIDATE_NOT_PROVEN", "SOURCE_MISSING", "EXTERNAL_PATH_ONLY",
]

dereference_rows = []
summary_rows = []

def status_for_family(fam_key, items, expected):
if not items:
return "MISSING_IN_CURRENT_SOURCES"
if expected and len(items) < expected:
return "PARTIAL_NEEDS_DEREFERENCE"
# Check if all items are external pointers only
all_external = all("EXTERNAL" in i["name"] or "graphiti_source_root_external" in i["origin"] for i in it
ems[:5])
if all_external:
return "PARTIAL_NEEDS_DEREFERENCE"
return "FOUND_EXPLICIT"

def write_family_file(fam_key, fam_info, items, out_dir):
label = fam_info["label"]
expected = fam_info.get("expected")
status = status_for_family(fam_key, items, expected)

lines = [f"# {fam_key} — {label}\n\n"]
lines.append(f"Status: {status}\n")
if expected:
lines.append(f"Expected: {expected}\n")
lines.append(f"Count: {len(items)}\n")
if expected and len(items) < expected:
lines.append(f"Missing: {expected - len(items)}\n")
lines.append("\n")

if not items:
lines.append("_Aucun item trouvé dans les sources disponibles._\n")
lines.append("\nSources consultées :\n")
lines.append(f"- {CSV_MAIN.name}\n")
lines.append(f"- graphiti_readonly_index_v2.json\n")
for f in BRODY_PY_FILES:
lines.append(f"- {f}\n")
else:
lines.append("## Items\n\n")
for idx, item in enumerate(items[:500], 1): # max 500 par famille
name = item.get("name", "")
src = item.get("source_file", "")
lineno = item.get("line", "")
origin = item.get("origin", "")
```

```
456 snippet = item.get("snippet", "")[:150]
457
458 lines.append(f"{idx:04d}. {name}\n")
459 if src:
460 lines.append(f" source_file: {src}\n")
461 if lineno:
462 lines.append(f" line: {lineno}\n")
463 lines.append(f" origin: {origin}\n")
464 if snippet:
465 snippet_clean = snippet.replace("\n", " ").strip()
466 lines.append(f" evidence: {snippet_clean}\n")
467 lines.append("\n")
468
469 if len(items) > 500:
470 lines.append(f"\n_... {len(items) - 500} items supplémentaires non affichés (voir CSV source)._\
n")
471
472 fpath = out_dir / f"{fam_key}.md"
473 fpath.write_text("".join(lines), encoding="utf-8")
474 return status
475
476 def write_source_lines_file(fam_key, items, out_dir):
477 lines = [f"# SOURCE_LINES — {fam_key}\n\n"]
478 # Group by source_file
479 by_src = defaultdict(list)
480 for item in items:
481 by_src[item.get("source_file", "unknown")].append(item)
482
483 for src, src_items in sorted(by_src.items())[:50]: # max 50 sources
484 lines.append(f"## {src}\n\n")
485 for item in src_items[:30]:
486 lineno = item.get("line", "")
487 name = item.get("name", "")
488 snippet = item.get("snippet", "")[:120]
489 ref = f"L{lineno}" if lineno else ""
490 lines.append(f"- [{ref}] `{name}` — {snippet}\n")
491 if len(src_items) > 30:
492 lines.append(f" _... {len(src_items) - 30} lignes supplémentaires_\n")
493 lines.append("\n")
494
495 fpath = out_dir / f"SOURCE_LINES_{fam_key}.md"
496 fpath.write_text("".join(lines), encoding="utf-8")
497
498 # Écrire chaque famille
499 for fam_key, fam_info in FAMILIES.items():
500 items = csv_items_by_family.get(fam_key, [])
501 status = write_family_file(fam_key, fam_info, items, OUT)
502 write_source_lines_file(fam_key, items, OUT)
503
504 expected = fam_info.get("expected")
505 summary_rows.append({
506 "family": fam_key,
507 "label": fam_info["label"],
508 "status": status,
```

```
509 "count": len(items),
510 "expected": expected or "",
511 })
512
513 # Dereference targets
514 if status == "MISSING_IN_CURRENT_SOURCES":
515 dereference_rows.append({
516 "family": fam_key,
517 "reason": "FAMILY_EMPTY",
518 "name": fam_info["label"],
519 "source_file": CSV_MAIN.name,
520 "line": "",
521 "path": "",
522 "next_action": "Chercher dans obsidia-engine-candidate ou sources externes",
523 })
524 elif status == "PARTIAL_NEEDS_DEREFERENCE":
525 if expected and len(items) < expected:
526 dereference_rows.append({
527 "family": fam_key,
528 "reason": "COUNT_BELOW_EXPECTED",
529 "name": f"{fam_key} — trouvé {len(items)}/{expected}",
530 "source_file": "_graphiti_readonly_indexes/graphiti_readonly_index_v2.json",
531 "line": "",
532 "path": "$.source_roots.34_ARBRES_TENSOR_MATRIX",
533 "next_action": f"Déréférencer obsidia-engine-candidate/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AG
ENTS_P2PLUS_V1/04_ARBRES_34_TENSOR_MATRIX/",
534 })
535 # Items purement externes
536 for item in items[:10]:
537 if "EXTERNAL" in item["name"] or "external" in item["origin"]:
538 dereference_rows.append({
539 "family": fam_key,
540 "reason": "EXTERNAL_PATH_ONLY",
541 "name": item["name"][:100],
542 "source_file": item["source_file"][:100],
543 "line": item["line"],
544 "path": item["snippet"][:100],
545 "next_action": "Accéder au chemin externe ou lire via graphiti_readonly_records_v2.jsonl
",
546 })
547
548 print(f" Fichiers famille écrits: {len(FAMILIES) * 2}")
549
550 # ── Écrire CSV summary ──────────────────────────────────────────────────────
551 summary_path = OUT / "00_SUMMARY_COUNTS.csv"
552 with open(summary_path, "w", encoding="utf-8", newline="") as f:
553 w = csv.DictWriter(f, fieldnames=["family", "label", "status", "count", "expected"])
554 w.writeheader()
555 w.writerows(summary_rows)
556 print(f" {summary_path.name} écrit")
557
558 # ── Écrire CSV dereference ──────────────────────────────────────────────────
559 deref_path = OUT / "00_DEREFERENCE_TARGETS.csv"
560 with open(deref_path, "w", encoding="utf-8", newline="") as f:
```

```
561 w = csv.DictWriter(f, fieldnames=["family", "reason", "name", "source_file", "line", "path", "next_actio
n"])
562 w.writeheader()
563 w.writerows(dereference_rows)
564 print(f" {deref_path.name} écrit ({len(dereference_rows)} targets)")
565
566 # ── Écrire CSV all components ──────────────────────────────────────────
567 all_comp_path = OUT / "00_ALL_COMPONENTS_READABLE.csv"
568 with open(all_comp_path, "w", encoding="utf-8", newline="") as f:
569 w = csv.DictWriter(f, fieldnames=["family", "idx", "name", "source_file", "line", "origin", "snippet"])
570 w.writeheader()
571 for fam_key in FAMILIES:
572 for idx, item in enumerate(csv_items_by_family.get(fam_key, [])[:200], 1):
573 w.writerow({
574 "family": fam_key,
575 "idx": idx,
576 "name": item.get("name", "")[:150],
577 "source_file": item.get("source_file", "")[:150],
578 "line": item.get("line", ""),
579 "origin": item.get("origin", ""),
580 "snippet": item.get("snippet", "")[:150],
581 })
582 print(f" {all_comp_path.name} écrit")
583
584 # ── Écrire README index ────────────────────────────────────────────────
585 readme_lines = [
586 f"# CLAUDE BRODY SEMANTIC DEEP ENUMERATION — INDEX\n\n",
587 f"Généré le : {datetime.now().isoformat()}\n",
588 f"Source CSV : {CSV_MAIN}\n",
589 f"Total lignes CSV : {total_csv}\n",
590 f"Total catégories CSV : {len(raw_by_cat)}\n\n",
591 f"## Familles auditées\n\n",
592 f"| Famille | Label | Status | Count | Expected |\n",
593 f"|---------|-------|--------|-------|----------|\n",
594 ]
595
596 for row in summary_rows:
597 readme_lines.append(
598 f"| {row['family']} | {row['label']} | {row['status']} | {row['count']} | {row['expected'] or '-'} |
\n"
599 )
600
601 readme_lines += [
602 f"\n## Sources consultées\n\n",
603 f"- `{CSV_MAIN.relative_to(ROOT)}`\n",
604 f"- `_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512
_224854/graphiti_readonly_inde
x_v2.json`\n",
605 f"- `_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512
_224854/graphiti_readonly_reco
rds_v2.jsonl`\n",
606 ]
607 for f in BRODY_PY_FILES:
608 readme_lines.append(f"- `{f}`\n")
609
```

```
610 readme_lines += [
611 f"\n## Résumé dereference\n\n",
612 f"Targets nécessitant déréférencement : {len(dereference_rows)}\n\n",
613 f"Chemin externe principal :\n",
614 f"`C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engine -candidate\\OBSIDIA_MMONDE_REVERSE
_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\\OBSIDIA_MMONDE_REVERSE
_OS_34ARBRES_AGENTS_P2PLUS_V1\\04_ARBR
ES_34_TENSOR_MATRIX`\n",
615 ]
616
617 (OUT / "00_INDEX_README.md").write_text("".join(readme_lines), encoding="utf-8")
618 print(f" 00_INDEX_README.md écrit")
619
620 # ── Affichage final ──────────────────────────────────────────────────────
621 print("\n" + "="*60)
622 print("=== CLAUDE SEMANTIC DEEP ENUMERATION INDEX ===")
623 print("="*60)
624 print((OUT / "00_INDEX_README.md").read_text(encoding="utf-8"))
625
626 print("\n=== SUMMARY COUNTS ===")
627 for row in summary_rows:
628 exp = f"/{row['expected']}" if row['expected'] else ""
629 print(f" {row['family']:<25} {row['status']:<35} {row['count']}{exp}")
630
631 print(f"\n=== DEREFERENCE TARGETS ({len(dereference_rows)}) ===")
632 for dr in dereference_rows[:20]:
633 print(f" [{dr['reason']}] {dr['family']} — {dr['name'][:70]}")
634 print(f" next: {dr['next_action'][:80]}")
635
636 print("\n=== ARBRES STATUS (premiers 40 items) ===")
637 arbre_content = (OUT / "01_ARBRES_34.md").read_text(encoding="utf-8")
638 for line in arbre_content.split("\n")[:60]:
639 print(line)
640
641 print("\n=== FILES ===")
642 for fp in sorted(OUT.iterdir()):
643 size = fp.stat().st_size
644 print(f" {fp.name:<55} {size:>8} bytes")
645
646 print(f"\nOUT={OUT}")
```

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3
_claude_brody_semantic_deep_enum.py 2>&1)
```

```
 OUT=C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proof
s_REMOTE_A5F21C6B\.local_audits\CLAUDE_BRODY_
SEMANTIC_DEEP_ENUMERATION_V0_20260618_181540
[1/6] Lecture CSV brut...
... 50000 lignes lues
... 100000 lignes lues
... 150000 lignes lues
... 200000 lignes lues
Total lignes: 204147, catégories: 15, erreurs: 0
AGENT: 14545
ARBRE_34: 29942
BALANCE: 3460
BOUNDARY: 16693
CANON: 11071
```

CORTEX: 99

DOMAINE: 4322

JSON_ERROR: 64

LOI: 1223

MEMOIRE: 20974

MODULE: 15096

PEPITE: 3629

PREUVE: 16023

PROTOCOLE: 45322

ROUTE: 21684

[2/6] Lecture index Graphiti...

Index chargé. Clés: ['status', 'date', 'records_count', 'source_roots', 'loaded_sources', 'records', 'readonly',

'memory_decision', 'allowed_to_decide', 'emits_act']

source_roots: 6 entrées

MMONDE_REVERSE_OS_ROOT -> C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida
te\OBSIDIA_MMON

DE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OBSIDIA_MMOND
E_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1

34_ARBRES_TENSOR_MATRIX ->

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida
te\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_

P2PLUS_V1_FREEZE_CANDIDATE\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P
2PLUS_V1\04_ARBRES_34_TENSOR_MATRIX

NUAGE_POINTS_SOURCE ->

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida
te\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_

P2PLUS_V1_FREEZE_CANDIDATE\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P
2PLUS_V1\13_GRAPHES_NUAGE_POINTS

REGROUPEMENTS_COHERENCE_V43 -> C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida
te\OBSIDIA

_V4_REGROUPEMENTS_V43_SOUS_DOSSIERS\OBSIDIA_V4_STRUCTURED_FULL\14_REGR OUPEMENTS_COHERENCE

PREMERGE_CANDIDATE_DOSSIER -> C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida
te\premerge

_candidates\OBSIDIA_ENGINE_PREMERGE_CANDIDATE_DOSSIER_V1_20260508_1827 59

X108_BRODY_MEMORY_READONLY ->

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs\pe riphery\brody_memory_readonly

Lecture records JSONL (21592KB)...

Records arbres: 546

Records lois: 13

Records protocoles: 17

Records pépites: 151

Records agents: 30

Records cortex: 2

Clés d'un record: ['id', 'source_kind', 'title', 'path', 'file_sha256', 'body_sha256', 'tags', 'tree_id',

'category', 'text', 'payload', 'readonly', 'memory_decision', 'allowed_to_decide', 'emits_act', 'kernel_binding',

'x108_merge', 'kernel_mutation', 'x108_mutation', 'decision_authority']

[3/6] Lecture fichiers Python Brody...

Extractions Python: {'BOUNDARY': 73, 'DOMAINE': 36, 'CANON': 67, 'ROUTE': 23, 'ARBRE_34': 2, 'BALANCE': 1,

'MODULE': 21, 'PROTOCOLE': 13, 'AGENT': 16, 'LOI': 4}

[4/6] Extraction items nommés par famille depuis CSV...

01_ARBRES_34: 29942 lignes CSV → 8623 items uniques

02_LOIS: 1223 lignes CSV → 449 items uniques

03_PROTOCOLES: 45322 lignes CSV → 12991 items uniques

04_MODULES: 15096 lignes CSV → 4212 items uniques

05_PEPITES: 3629 lignes CSV → 2183 items uniques

06_BALANCES: 3460 lignes CSV → 1479 items uniques

07_CANON: 11071 lignes CSV → 4875 items uniques
08_CORTEX: 99 lignes CSV → 43 items uniques
09_AGENTS: 14545 lignes CSV → 5642 items uniques
10_DOMAINES: 4322 lignes CSV → 1976 items uniques
11_MEMOIRE_GRAPHITI: 20974 lignes CSV → 8257 items uniques
12_PREUVES: 16023 lignes CSV → 6773 items uniques
13_ROUTES_API: 21684 lignes CSV → 2232 items uniques
14_BOUNDARIES: 16693 lignes CSV → 5077 items uniques
[5/6] Enrichissement avec Graphiti records...
01_ARBRES_34: +219 items Graphiti
02_LOIS: +11 items Graphiti
03_PROTOCOLES: +8 items Graphiti
05_PEPITES: +6 items Graphiti
09_AGENTS: +12 items Graphiti
01_ARBRES_34: +1 items Python
02_LOIS: +4 items Python
03_PROTOCOLES: +11 items Python
04_MODULES: +2 items Python
07_CANON: +39 items Python
09_AGENTS: +16 items Python
10_DOMAINES: +33 items Python
14_BOUNDARIES: +34 items Python
13_ROUTES_API: +10 items Python
[6/6] Écriture des fichiers de sortie...
Fichiers famille écrits: 28

```
00_SUMMARY_COUNTS.csv écrit
00_DEREFERENCE_TARGETS.csv écrit (0 targets)
00_ALL_COMPONENTS_READABLE.csv écrit
00_INDEX_README.md écrit
```

CLAUDE SEMANTIC DEEP ENUMERATION INDEX

CLAUDE BRODY SEMANTIC DEEP ENUMERATION — INDEX

Généré le : 2026-06-18T18:15:42.094825

Source CSV : C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\BROD
Y_SEMANTIC_COMPONENT_REGISTRY_V0_20260618_170307\brody_semantic_component_registry.csv

Total lignes CSV : 204147

Total catégories CSV : 15

Familles auditées

Famille	Label	Status	Count	Expected
01_ARBRES_34	34 Arbres Obsidia	FOUND_EXPLICIT	8868	34
02_LOIS	Lois Obsidiennes	FOUND_EXPLICIT	464	-
03_PROTOCOLES	Protocoles	FOUND_EXPLICIT	13010	-
04_MODULES	Modules	FOUND_EXPLICIT	4214	-
05_PEPITES	Pépites	FOUND_EXPLICIT	2189	-
06_BALANCES	Balances	FOUND_EXPLICIT	1479	-
07_CANON	Canon	FOUND_EXPLICIT	4914	-
08_CORTEX	Cortex	FOUND_EXPLICIT	43	-
09_AGENTS	Agents (52 attendus)	FOUND_EXPLICIT	5670	52
10_DOMAINES	Domaines	FOUND_EXPLICIT	2009	-
11_MEMOIRE_GRAPHITI	Mémoire / Graphiti	FOUND_EXPLICIT	8257	-
12_PREUVES	Preuves / Invariants Lean	FOUND_EXPLICIT	6773	-
13_ROUTES_API	Routes API	FOUND_EXPLICIT	2242	-
14_BOUNDARIES	Boundaries KX108	FOUND_EXPLICIT	5111	-

Sources consultées

- `.local_audits\BRODY_SEMANTIC_COMPONENT_REGISTRY_V0_20260618_170307\br ody_semantic_component_registry.csv`
- `_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512 _224854/graphiti_readonly_index_v2.json`
- `_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512 _224854/graphiti_readonly_records_v2.jsonl`
- `apps/obsidia_api/brody_machination_composer.py`
- `apps/obsidia_api/brody_contracts_packet.py`
- `apps/obsidia_api/brody_domain_raccord_adapter.py`
- `apps/obsidia_api/brody_automation_orchestrator.py`
- `apps/obsidia_api/brody_backend_response_composer.py`
- `apps/obsidia_api/brody_adaptive_response_policy.py`
- `apps/obsidia_api/brody_candidate_memory_adapter.py`
- `apps/obsidia_api/brody_cic_context_adapter.py`
- `apps/obsidia_api/brody_cognitive_modules_adapter.py`
- `apps/obsidia_api/brody_real_response_pipeline.py`
- `runtime_wiring/source_runtime/capability_path_router.py`
- `sigma/contracts.py`

Résumé dereference

Targets nécessitant déréférencement : 0

Chemin externe principal :

`C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candid
ate\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS
_P2PLUS_V1_FREEZE_CANDIDATE\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_
P2PLUS_V1\04_ARBRES_34_TENSOR_MATRIX`

SUMMARY COUNTS

01_ARBRES_34 FOUND_EXPLICIT 8868/34

02_LOIS FOUND_EXPLICIT 464

03_PROTOCOLES FOUND_EXPLICIT 13010

04_MODULES FOUND_EXPLICIT 4214

05_PEPITES FOUND_EXPLICIT 2189

06_BALANCES FOUND_EXPLICIT 1479

07_CANON FOUND_EXPLICIT 4914
08_CORTEX FOUND_EXPLICIT 43
09_AGENTS FOUND_EXPLICIT 5670/52
10_DOMAINES FOUND_EXPLICIT 2009
11_MEMOIRE_GRAPHITI FOUND_EXPLICIT 8257
12_PREUVES FOUND_EXPLICIT 6773
13_ROUTES_API FOUND_EXPLICIT 2242
14_BOUNDARIES FOUND_EXPLICIT 5111

DEREFERENCE TARGETS (0)

ARBRES STATUS (premiers 40 items)

01_ARBRES_34 — 34 Arbres Obsidia

Status: FOUND_EXPLICIT

Expected: 34

Count: 8868

Items

0001. C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candidate\OBSIDIA_MMONDE_RE VERSE_OS_34ARBRES_A

GENTS_P2PLUS_V1_FREEZE_CANDIDATE\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AG ENTS_P2PLUS_V1

source_file:

`_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854/graphiti_readonly_ index_v2.json`

origin: json_value

evidence: cat=ARBRE_34 path=$.source_roots.MMONDE_REVERSE_OS_ROOT

0002. 34_ARBRES_TENSOR_MATRIX

source_file:

`_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854/graphiti_readonly_ index_v2.json`

origin: json_key

evidence: cat=ARBRE_34 path=$.source_roots

0003. C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candidate\OBSIDIA_MMONDE_RE VERSE_OS_34ARBRES_A

GENTS_P2PLUS_V1_FREEZE_CANDIDATE\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AG ENTS_P2PLUS_V1\04_ARBRES_34_TENSOR_MATRIX

source_file:

`_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854/graphiti_readonly_ index_v2.json`

origin: json_value

evidence: cat=ARBRE_34 path=$.source_roots.34_ARBRES_TENSOR_MATRIX

0004. C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candidate\OBSIDIA_MMONDE_RE VERSE_OS_34ARBRES_A

GENTS_P2PLUS_V1_FREEZE_CANDIDATE\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AG ENTS_P2PLUS_V1\13_GRAPHES_NUAGE_POINTS

source_file:

`_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854/graphiti_readonly_ index_v2.json`

origin: json_value

evidence: cat=ARBRE_34 path=$.source_roots.NUAGE_POINTS_SOURCE

0005. 34_arbres

source_file:

`_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854/graphiti_readonly_ index_v2.json`

origin: json_value

evidence: cat=ARBRE_34 path=$.records[0].tags[0]

`0006. 0B9AF809DD8A_arbres_34.canon.json`

source_file:

`_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854/graphiti_readonly_ index_v2.json`

origin: json_value

evidence: cat=ARBRE_34 path=$.records[105].title

0007. Arbres Fondamentaux

source_file:

`_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854/graphiti_readonly_ index_v2.json`

origin: json_value

evidence: cat=ARBRE_34 path=$.records[105].category.category_name

0008. 2BC905D9D94C_L'Agent Shazam Cognitif et la Matrice des 34 Arbres.docx

source_file:

`_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854/graphiti_readonly_ index_v2.json`

origin: json_value

evidence: cat=ARBRE_34 path=$.records[446].title

```
0009. 916F1E363102_arbres_34.registry.json
```

source_file:

```
_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854/graphiti_readonly_ index_v2.json
```

origin: json_value

evidence: cat=ARBRE_34 path=$.records[1363].title

```
0010. 9840D1D5ECE8_LES 34 ARBRES _ Architecture et Structure Cognitive Globale.docx
```

source_file:

```
_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854/graphiti_readonly_ index_v2.json
```

origin: json_value

evidence: cat=ARBRE_34 path=$.records[1428].title

```
0011. 997BFF4EFC9F_AUDIT_34_ARBRES.md
```

source_file:

```
_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512_224854/graphiti_readonly_ index_v2.json
```

FILES

```
00_ALL_COMPONENTS_READABLE.csv 644988 bytes
00_DEREFERENCE_TARGETS.csv 54 bytes
00_INDEX_README.md 2650 bytes
00_SUMMARY_COUNTS.csv 704 bytes
01_ARBRES_34.md 230577 bytes
02_LOIS.md 155076 bytes
03_PROTOCOLES.md 158266 bytes
04_MODULES.md 135804 bytes
05_PEPITES.md 164552 bytes
11_MEMOIRE_GRAPHITI.md 117001 bytes
12_PREUVES.md 210128 bytes
13_ROUTES_API.md 121624 bytes
14_BOUNDARIES.md 215726 bytes
SOURCE_LINES_01_ARBRES_34.md 124045 bytes
SOURCE_LINES_02_LOIS.md 35589 bytes
SOURCE_LINES_03_PROTOCOLES.md 114090 bytes
SOURCE_LINES_04_MODULES.md 115814 bytes
SOURCE_LINES_05_PEPITES.md 62376 bytes
SOURCE_LINES_06_BALANCES.md 64393 bytes
SOURCE_LINES_07_CANON.md 105783 bytes
SOURCE_LINES_08_CORTEX.md 13094 bytes
SOURCE_LINES_09_AGENTS.md 117718 bytes
SOURCE_LINES_10_DOMAINES.md 109098 bytes
SOURCE_LINES_11_MEMOIRE_GRAPHITI.md 132324 bytes
SOURCE_LINES_12_PREUVES.md 140915 bytes
SOURCE_LINES_13_ROUTES_API.md 77545 bytes
SOURCE_LINES_14_BOUNDARIES.md 122394 bytes
OUT=C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B\.local_ audits\CLAUDE_BRODY_
```

SEMANTIC_DEEP_ENUMERATION_V0_20260618_181540

06:16 PM claude-sonnet-4-6

● Le script a produit 32 fichiers. Cependant, le statut FOUND_EXPLICIT pour 01_ARBRES_34 est trompeur : 8 868 items sont des

chemins/documents autour des arbres, pas les 34 noms individuels. Je dois déréférencer les fichiers arbres_34.canon.json et

```
arbres_34.registry.json dans les records Graphiti.
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
```

import json, sys

sys.stdout.reconfigure(encoding='utf-8')

Cherche les records contenant les noms des 34 arbres

targets = ['arbres_34.canon', 'arbres_34.registry', '34 ARBRES', 'arbre_01', 'ARBRE_1', 'ARBRE_01']
canon_records = []
with
open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',
encoding='utf-8') as f:
for line in f:
line = line.strip()
if not line: continue
try:
rec = json.loads(line)
title = (rec.get('title','') or '').lower()
path = (rec.get('path','') or '').lower()
text = (rec.get('text','') or '')
payload = rec.get('payload', {}) or {}
if 'arbres_34.canon' in title or 'arbres_34.registry' in title:
print(f'=== FOUND: {rec.get(\"title\",\"\")}')
print(f' path: {rec.get(\"path\",\"\")}')
print(f' text[:500]: {text[:500]}')
print(f' payload keys: {list(payload.keys())[:20] if isinstance(payload, dict) else
str(payload)[:200]}')
canon_records.append(rec)
elif 'architecture et structure cognitive' in title:
print(f'=== DOC: {rec.get(\"title\",\"\")}')
print(f' text[:800]: {text[:800]}')
except: pass
print(f'Total canon/registry records: {len(canon_records)}')
" 2>&1 | head -200)
 Waiting...
 === FOUND: 0B9AF809DD8A_arbres_34.canon.json
path:
text[:500]: {"id": "BRODY_GRAPHITI_READY_000105", "title": "0B9AF809DD8A_arbres_34.canon.json",
"source_original_path":
"_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\
\0B9AF809DD8A_arbres_34.canon.json",
"normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engine-c andidate_world_intake
\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162_2026051
2_211327\\normalized_md\\0B9AF809DD8A_arbres_
34_canon_0B9AF809DD8A.md", "original_extension": ".json
payload keys: ['id', 'title', 'source_original_path', 'normalized_md_path', 'original_extension',
'extraction_status', 'extractor', 'content_usable', 'source_original_sha256', 'text_sha256', 'text_excerpt',
'tags', 'readonly', 'memory_authority', 'memory_decision', 'allowed_to_decide', 'emits_act', 'kernel_binding',
'x108_merge', 'kernel_mutation']
=== FOUND: 916F1E363102_arbres_34.registry.json
path:
text[:500]: {"id": "BRODY_GRAPHITI_READY_001363", "title": "916F1E363102_arbres_34.registry.json",
"source_original_path":
"_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\
\916F1E363102_arbres_34.registry.json",
"normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engine-c andidate_world_intake
\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162_2026051
2_211327\\normalized_md\\916F1E363102_arbres_
34_registry_916F1E363102.md", "original_extension
payload keys: ['id', 'title', 'source_original_path', 'normalized_md_path', 'original_extension',

'extraction_status', 'extractor', 'content_usable', 'source_original_sha256', 'text_sha256', 'text_excerpt',
'tags', 'readonly', 'memory_authority', 'memory_decision', 'allowed_to_decide', 'emits_act', 'kernel_binding',
'x108_merge', 'kernel_mutation']
=== DOC: 9840D1D5ECE8_LES 34 ARBRES _ Architecture et Structure Cognitive Globale.docx
text[:800]: {"id": "BRODY_GRAPHITI_READY_001428", "title": "9840D1D5ECE8_LES 34 ARBRES _ Architecture et
Structure Cognitive Globale.docx", "source_original_path":
"_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\ \9840D1D5ECE8_LES 34 ARBRES
_ Architecture et
Structure Cognitive Globale.docx", "normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obs
idia-engine-candidate_world_intake\\BRODY_WORLD_INTAKE_PROJECT_MEMOR
Y_CLEAN_SOURCE_NO_INDEX_V162_20260512_211327\
\normalized_md\\9840D1D5ECE8_LES_34_ARBRES___Architecture_et_Structure_Cognitive_Globale_9840D1 D5ECE8.md",
"original_extension": ".docx", "extraction_status": "DOCX_TEXT_EXTRACTED", "extractor": "python-docx",
"content_usable": true, "source_original_sha256": "9840D1D5ECE8F305B24A93B7ED7C10EDE0C15B3B20
=== FOUND: arbres_34.canon.json
path: C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida
te\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES
_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_
AGENTS_P2PLUS_V1\04_ARBRES_34_TENSOR_MATRIX\a
rbres_34.canon.json
text[:500]: {
"trees": [
{
"id": "01",
"name": "Arbre de l’Humain",
"folder": "ARBRE_01__Arbre_de_l_Humain",
"family": "I_FONDAMENTAUX",
"non_decision": true,
"needs_human_validation": false
},
{
"id": "02",
"name": "Arbre de la Conscience",
"folder": "ARBRE_02__Arbre_de_la_Conscience",
"family": "I_FONDAMENTAUX",
"non_decision": true,
"needs_human_validation": false
},
{
"id": "03",
"name": "Arbre de l
payload keys: ['relative_path', 'source_root']
=== FOUND: arbres_34.registry.json
path: C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida
te\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES
_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_
AGENTS_P2PLUS_V1\19_REGISTRES_JSON\arbres_34.
registry.json
text[:500]: {
"trees": [
{
"id": "01",
"name": "Arbre de l’Humain",
"family": "I_FONDAMENTAUX"
},
{
"id": "02",

"name": "Arbre de la Conscience",
"family": "I_FONDAMENTAUX"
},
"family": "I_FONDAMENTAUX"
},
{
"id": "04",
"name": "Arbre du Sens",
"family": "I_FONDAMENTAUX"
},
{
"id": "05",
"name": "Arbre de l’Identité",
"family": "I_FON
payload keys: ['relative_path', 'source_root']
=== DOC: LES 34 ARBRES _ Architecture et Structure Cognitive Globale.docx
text[:800]: LES 34 ARBRES : Architecture et Structure Cognitive Globale 1. Introduction au Système des 34 Arbres La présente nomenclature définit l’architecture fondamentale de la cognition en segmentant la « matière » de l’intelligence en 34 arbres structurels, répartis au sein de 8 catégories ontologiques. Ce système ne constitue pas une simple taxonomie, mais le substrat informationnel nécessaire à toute compréhension systémique de l’esprit humain et au développement d’une Intelligence Artificielle Générale (AGI) cohérente. En isolant ces briques élémentaires, nous modélisons les variables latentes qui permettent au système de traiter la complexité du réel. Cette classification sert de socle à l’ingénierie cognitive, transformant des concepts abstraits en structures de données exploitables par un S
Total canon/registry records: 4
 Allowed by auto mode classifier
06:16 PM claude-sonnet-4-6
● Les noms des 34 arbres sont dans le record arbres_34.canon.json. Je les extrais intégralement maintenant.
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
import json, sys
sys.stdout.reconfigure(encoding='utf-8')

Extraire les 34 arbres complets depuis arbres_34.registry.json et arbres_34.canon.json

+ chercher les Lois nommées, les Pépites nommées, les Agents nommés

arbres_from_canon = []
arbres_from_registry = []
lois_named = []
agents_named = []
pepites_named = []
protocoles_named = []

Cherche aussi les records avec payload de lois et agents

```
ROLES_PATTERN = ['role_0', 'agents_canoniques.json', 'agents_et_roles']
with
open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',
encoding='utf-8') as f:
for line in f:
line = line.strip()
if not line: continue
try:
rec = json.loads(line)
title = (rec.get('title','') or '').lower()
path = (rec.get('path','') or '')
text = (rec.get('text','') or '')
```

Canon des arbres

```
if 'arbres_34.canon.json' == title or title.endswith('_arbres_34.canon.json') or 'arbres_34.canon' in
title:
try:
data = json.loads(text)
trees = data.get('trees', [])
for t in trees:
arbres_from_canon.append({
'id': t.get('id',''),
'name': t.get('name',''),
'folder': t.get('folder',''),
'family': t.get('family',''),
})
except: pass
```

Registry des arbres

```
if 'arbres_34.registry.json' == title or title.endswith('_arbres_34.registry.json') or 'arbres_34.registry'
in title:
try:
data = json.loads(text)
trees = data.get('trees', [])
for t in trees:
arbres_from_registry.append({
'id': t.get('id',''),
'name': t.get('name',''),
'family': t.get('family',''),
})
except: pass
```

Agents canoniques

if 'agents_canoniques' in title or 'role_0' in title[:50]:
try:
data = json.loads(text)
if isinstance(data, list):
for a in data:
agents_named.append({
'id': a.get('id','') or a.get('role_id',''),
'name': a.get('name','') or a.get('role_name',''),
'type': a.get('type',''),

04. Arbre du Sens [I_FONDAMENTAUX] folder=ARBRE_04__Arbre_du_Sens

05. Arbre de l’Identité [I_FONDAMENTAUX] folder=ARBRE_05__Arbre_de_l_Identite

06. Arbre de la Compréhension [II_COGNITIFS] folder=ARBRE_06__Arbre_de_la_Comprehension

07. Arbre de l’Organisation [II_COGNITIFS] folder=ARBRE_07__Arbre_de_l_Organisation

08. Arbre de la Pensée [II_COGNITIFS] folder=ARBRE_08__Arbre_de_la_Pensee

09. Arbre de l’Intelligence [II_COGNITIFS] folder=ARBRE_09__Arbre_de_l_Intelligence

10. Arbre du Langage [II_COGNITIFS] folder=ARBRE_10__Arbre_du_Langage

11. Arbre de la Science [III_CONNAISSANCE] folder=ARBRE_11__Arbre_de_la_Science

12. Arbre de la Technique [III_CONNAISSANCE] folder=ARBRE_12__Arbre_de_la_Technique

13. Arbre de l’Art [III_CONNAISSANCE] folder=ARBRE_13__Arbre_de_l_Art

14. Arbre de la Philosophie [III_CONNAISSANCE] folder=ARBRE_14__Arbre_de_la_Philosophie

15. Arbre de la Spiritualité [III_CONNAISSANCE] folder=ARBRE_15__Arbre_de_la_Spiritualite

16. Arbre de la Relation [IV_RELATIONNELS_SOCIAUX] folder=ARBRE_16__Arbre_de_la_Relation

17. Arbre du Collectif [IV_RELATIONNELS_SOCIAUX] folder=ARBRE_17__Arbre_du_Collectif

18. Arbre de la Transmission [IV_RELATIONNELS_SOCIAUX] folder=ARBRE_18__Arbre_de_la_Transmission

19. Arbre de la Culture [IV_RELATIONNELS_SOCIAUX] folder=ARBRE_19__Arbre_de_la_Culture

20. Arbre de l’Action [V_ACTION_TRANSFORMATION] folder=ARBRE_20__Arbre_de_l_Action

21. Arbre de la Création [V_ACTION_TRANSFORMATION] folder=ARBRE_21__Arbre_de_la_Creation

22. Arbre de la Transformation [V_ACTION_TRANSFORMATION] folder=ARBRE_22__Arbre_de_la_Transformation

23. Arbre du Temps [VI_TEMPORELS_MEMORIELS] folder=ARBRE_23__Arbre_du_Temps

24. Arbre de la Mémoire [VI_TEMPORELS_MEMORIELS] folder=ARBRE_24__Arbre_de_la_Memoire

25. Arbre de l’Histoire [VI_TEMPORELS_MEMORIELS] folder=ARBRE_25__Arbre_de_l_Histoire

26. Arbre de la Cohérence [VII_META_STRUCTURELS] folder=ARBRE_26__Arbre_de_la_Coherence

27. Arbre de la Vérité [VII_META_STRUCTURELS] folder=ARBRE_27__Arbre_de_la_Verite

28. Arbre de la Valeur [VII_META_STRUCTURELS] folder=ARBRE_28__Arbre_de_la_Valeur

29. Arbre de la Finalité [VII_META_STRUCTURELS] folder=ARBRE_29__Arbre_de_la_Finalite

30. Arbre Cognitif Global [VIII_OBSIDIA_AGI] folder=ARBRE_30__Arbre_Cognitif_Global

31. Arbre des Flux [VIII_OBSIDIA_AGI] folder=ARBRE_31__Arbre_des_Flux

32. Arbre des Connexions [VIII_OBSIDIA_AGI] folder=ARBRE_32__Arbre_des_Connexions

33. Arbre de l’Optimisation [VIII_OBSIDIA_AGI] folder=ARBRE_33__Arbre_de_l_Optimisation

34. Arbre de la Stabilité [VIII_OBSIDIA_AGI] folder=ARBRE_34__Arbre_de_la_Stabilite

34 ARBRES depuis registry (34)

01. Arbre de l’Humain [I_FONDAMENTAUX]

02. Arbre de la Conscience [I_FONDAMENTAUX]

03. Arbre de la Perception [I_FONDAMENTAUX]

04. Arbre du Sens [I_FONDAMENTAUX]

05. Arbre de l’Identité [I_FONDAMENTAUX]

06. Arbre de la Compréhension [II_COGNITIFS]

07. Arbre de l’Organisation [II_COGNITIFS]

08. Arbre de la Pensée [II_COGNITIFS]

09. Arbre de l’Intelligence [II_COGNITIFS]

10. Arbre du Langage [II_COGNITIFS]

11. Arbre de la Science [III_CONNAISSANCE]

16. Arbre de la Relation [IV_RELATIONNELS_SOCIAUX]

17. Arbre du Collectif [IV_RELATIONNELS_SOCIAUX]

18. Arbre de la Transmission [IV_RELATIONNELS_SOCIAUX]

19. Arbre de la Culture [IV_RELATIONNELS_SOCIAUX]

20. Arbre de l’Action [V_ACTION_TRANSFORMATION]

21. Arbre de la Création [V_ACTION_TRANSFORMATION]

22. Arbre de la Transformation [V_ACTION_TRANSFORMATION]

23. Arbre du Temps [VI_TEMPORELS_MEMORIELS]

24. Arbre de la Mémoire [VI_TEMPORELS_MEMORIELS]

25. Arbre de l’Histoire [VI_TEMPORELS_MEMORIELS]

26. Arbre de la Cohérence [VII_META_STRUCTURELS]

27. Arbre de la Vérité [VII_META_STRUCTURELS]

28. Arbre de la Valeur [VII_META_STRUCTURELS]

29. Arbre de la Finalité [VII_META_STRUCTURELS]

30. Arbre Cognitif Global [VIII_OBSIDIA_AGI]

31. Arbre des Flux [VIII_OBSIDIA_AGI]

32. Arbre des Connexions [VIII_OBSIDIA_AGI]

33. Arbre de l’Optimisation [VIII_OBSIDIA_AGI]

34. Arbre de la Stabilité [VIII_OBSIDIA_AGI]

AGENTS nommés depuis canoniques (0)

 Allowed by auto mode classifier

06:16 PM claude-sonnet-4-6

● Les 34 arbres sont entièrement déréférencés. Maintenant j'extrais les agents nommés (52 attendus), les Lois, les Pépites
numérotées.

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
import json, re, sys
sys.stdout.reconfigure(encoding='utf-8')
agents_named = []
lois_named = []
pepites_named = []
protocoles_named = []
with
```

```
open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',
encoding='utf-8') as f:
for line in f:
line = line.strip()
if not line: continue
try:
rec = json.loads(line)
title = (rec.get('title','') or '')
path = (rec.get('path','') or '')
text = (rec.get('text','') or '')
tl = title.lower()
```

Agents - cherche fichiers ROLE_xxx, agents_canoniques

```
if re.search(r'role_\d+|agents_canoniques|06_agent_role', tl):
try:
data = json.loads(text)
if isinstance(data, dict):
if 'roles' in data or 'agents' in data:
items = data.get('roles', data.get('agents', []))
for a in items:
agents_named.append({
'id': str(a.get('id', a.get('role_id', a.get('code',')))),
'name': str(a.get('name', a.get('role_name', a.get('label',')))),
'type': str(a.get('type', a.get('famille',''))),
'source': title[:60],
})
else:
```

Format agent individuel

```
n = data.get('name','') or data.get('role_name','')
i = data.get('id','') or data.get('role_id','') or data.get('code','')
if n:
agents_named.append({'id':str(i), 'name':str(n), 'type':str(data.get('type','')),
'source':title[:60]})
except:
```

Cherche pattern texte ROLE_xxx

m = re.search(r'ROLE_(\d+)[^:]*:\s*([^\n]+)', text)
if m:
agents_named.append({'id':m.group(1), 'name':m.group(2)[:60], 'type':'text_extract',
'source':title[:60]})

Lois - cherche fichiers de lois

```
if
re.search(r'lois_obsidiennes|07_lois|constitution_lois|02_constitution |lois_binding|lois_lean|kernel_laws', tl):
try:
data = json.loads(text)
if isinstance(data, dict):
lois = data.get('lois', data.get('laws', data.get('laws_binding', [])))
for l in lois if isinstance(lois, list) else []:
lois_named.append({
'id': str(l.get('id', l.get('loi_id',''))),
'name': str(l.get('name', l.get('loi_name', l.get('label','')))),
'source': title[:60],
})
```

Cherche aussi clés directes

```
for k,v in data.items():
if k.startswith('loi_') or k.startswith('LOI_') or k.startswith('law_'):
lois_named.append({'id':k, 'name':str(v)[:80], 'source':title[:60]})
except: pass
```

Pépites - cherche fichiers 02_PEPITE_Pxxx

```
if re.search(r'02_pepite_p\d+|pepite_p\d+', tl):
m = re.search(r'P(\d+)__([^.\n]{3,80})', title)
if m:
pepites_named.append({
'code': f'P{m.group(1)}',
'name': m.group(2).replace('_',' ').strip(),
'source': title[:80],
})
```

Protocoles - cherche fichiers protocoles

if re.search(r'03_protocole|protocoles_obs|08_protocole|invariant_protoco l', tl):
try:
data = json.loads(text)
if isinstance(data, dict):
protos = data.get('protocoles', data.get('protocols', []))
for p in protos if isinstance(protos, list) else []:
print(f' {l[\"id\"]:>10} | {l[\"name\"][:70]}')
print(f'\nPEPITES nommées: {len(pepites_named)}')
for p in sorted(pepites_named, key=lambda x: x[\"code\"])[:50]:
print(f' {p[\"code\"]:<6} | {p[\"name\"][:70]}')
print(f'\nPROTOCOLES nommés: {len(protocoles_named)}')
for p in protocoles_named[:20]:
print(f' {p[\"id\"]:>5} | {p[\"name\"][:70]}')
" 2>&1)
 Waiting...
 AGENTS trouvés: 7
013 | ** rôle_runtime | text_extract |
`06_AGENT_ROLE__Orchestrator.md`
014 | ** adapter_outil | text_extract |
`06_AGENT_ROLE__Adapter_MCP.md`
008 | ** rôle_outils | text_extract |
`06_AGENT_ROLE__Agents_Tools.md`
006 | ** rôle_système | text_extract |
`06_AGENT_ROLE__Governance_Layer.md`
005 | ** rôle_responsabilité | text_extract |
`06_AGENT_ROLE__Human_Operator.md`
015 | ** rôle_couche | text_extract | 06_AGENT_ROLE__Cortex.md
007 | ** rôle_système | text_extract |
`06_AGENT_ROLE__OS3_Audit_Layer.md`
LOIS nommées: 0
PEPITES nommées: 290
P001 | Memoire stockage
P001 | Memoire stockage
P002 | Memoire fractale hierarchique
P002 | Memoire fractale hierarchique
P003 | Oubli actif
P003 | Oubli actif
P004 | Memoire comme condition dethique
P004 | Memoire comme condition dethique
P005 | Memoire humaine inter memoire systeme
P005 | Memoire humaine inter memoire systeme
P006 | Memoire comme recit vivant
P006 | Memoire comme recit vivant
P007 | Memoire comme garde fou contre lhallucination
P007 | Memoire comme garde fou contre lhallucination
P008 | Memoire transversale
P008 | Memoire transversale
P009 | Sens information
P009 | Sens information
P010 | Cadre avant reponse
P010 | Cadre avant reponse
P011 | Refus hors cadre
P011 | Refus hors cadre
P012 | Desambiguisation contextuelle

P012 | Desambiguisation contextuelle
P013 | Immuabilite par sceau Merkle
P013 | Immuabilite par sceau Merkle
P014 | Traduction humain inter machine
P014 | Traduction humain inter machine
P015 | Immuabilite forte Sensitivity
P015 | Immuabilite forte Sensitivity
P016 | Rupture data inter sens
P016 | Rupture data inter sens
P017 | Croissance du log daudit
P017 | Croissance du log daudit
P018 | Semantique avant calcul
P018 | Semantique avant calcul
P019 | Pluralite des niveaux de sens
P019 | Pluralite des niveaux de sens
P020 | Sens avant performance
P020 | Sens avant performance
P021 | Semantique transversale
P021 | Semantique transversale
P022 | Le symbolique nest pas decoratif
P022 | Le symbolique nest pas decoratif
P023 | Compression du sens
P023 | Compression du sens
P024 | Non verbal comme vecteur
P024 | Non verbal comme vecteur
P025 | Symboles comme passerelles
P025 | Symboles comme passerelles
PROTOCOLES nommés: 0
 Allowed by auto mode classifier

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
```

import json, re, sys
sys.stdout.reconfigure(encoding='utf-8')

Cherche les agents dans tous les records avec pattern plus large

```
agents_all = {}
lois_all = []
pepites_all = {}
with
open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',
encoding='utf-8') as f:
for line in f:
line = line.strip()
if not line: continue
try:
rec = json.loads(line)
title = rec.get('title','') or ''
text = rec.get('text','') or ''
tl = title.lower()
```

Pépites - déduplique

```
if re.search(r'pepite_p\d+|02_pepite', tl):
m = re.search(r'P(\d+)__(.+?)(?:\.md|\s*\$|$)', title, re.IGNORECASE)
if not m:
m = re.search(r'P(\d+)__([^\n.]+)', title)
if m:
code = f'P{int(m.group(1)):03d}'
name = m.group(2).replace('_',' ').strip()
if code not in pepites_all:
pepites_all[code] = name
```

Agents - cherche ROLE_xxx ou AGENT_xxx dans le titre/texte

m_role = re.search(r'ROLE_(\d+)__(.+?)(?:\.md|$)', title)
if m_role:
role_id = m_role.group(1)
role_name = m_role.group(2).replace('_',' ').strip()
if role_id not in agents_all:
agents_all[role_id] = {'name': role_name, 'source': title[:60]}

Lois dans le texte - cherche 'Loi X: ' ou '## Loi' patterns

loi_matches = re.findall(r'(?:^|\n)(?:Loi\s+(?:N°\s*)?(\d+)[:\s—-]+([^\n]+)|###?\s+L oi\s+([^\n]+))', text, re.MULTILINE)
for m in loi_matches[:5]:
P045 | Skew negatif vers HOLD
P046 | Consensus fail closed 4 agents
P047 | Couplage reel moteur dR dt
P048 | Structure cosmologique
P049 | Emergence par interaction
P050 | Recursivite fondamentale
P051 | Structure fractale
P052 | Principe holographique
P053 | Resonance cosmique
P054 | Temps comme integrale devenements
P055 | Espace metrique
P056 | Causalite temporelle
P057 | Entropie de Shannon
P058 | Neguentropie
P059 | Complexite de Kolmogorov
P060 | Auto organisation
P061 | Bifurcation de trajectoire
P062 | Attracteur
P063 | Sensibilite aux conditions initiales
P064 | Ordre 1 entropie normalisee
P065 | Definition de la vie
P066 | Conscience comme recursivite infinie
P067 | Filtre ethique continu t
P068 | Ethique integree sur le temps
P069 | Responsabilite action consequence
P070 | Consentement binaire
P071 | Transparence universelle
P072 | Equite approximee
P073 | Non nuisance
P074 | Bienfaisance maximale
P075 | Justice proportionnelle
P076 | Budget energetique Ect
P077 | Flux energetique dE dt
P078 | Reserve energetique
P079 | Regeneration en repos
P080 | Equilibre energetique
P081 | Surcharge vers HOLD
P082 | Optimisation cout qualite
P083 | Efficience output input
P084 | Durabilite a long terme
P085 | Loi de reciprocite
P102 | Iteration convergente
P103 | Validation par criteres
P104 | Refutation par contre exemple
P105 | Induction mathematique
P106 | Deduction logique
P107 | Stabilite Lyapunov
P108 | Robustesse aux perturbations
P109 | Reproductibilite

P110 | Falsifiabilite Popper
P111 | Moteur non substituable
P112 | Hierarchie OS0 vers OS4
P113 | Kernel intersection des couches
P114 | Guard Kernel Temporal Consensus
P115 | Filtre ethique du moteur
P116 | Pipeline de couches
P117 | Orchestration par scheduler
P118 | Isolation des couches
P119 | Composition fonctionnelle
P120 | Abstraction par interface
P121 | Encapsulation boite noire
P122 | Modularite
P123 | Extensibilite
P124 | Infrastructure IM O
P125 | Deploiement Build Test Release
P126 | Scalabilite lineaire
P127 | Disponibilite uptime
P128 | Latence de reponse
P129 | Debit throughput
P130 | Resilience MTBF MTTR
P131 | Securite CIA
P132 | Observabilite logs metriques traces
P133 | Configuration cle valeur
P134 | Versioning semantique
P135 | Pipeline CI CD
P136 | Infrastructure as Code
P137 | Agent Role Protocole Flux
P138 | Ecologie dagents
P139 | Protocole sequence de messages
P144 | Apprentissage par experience
P145 | Adaptation a lenvironnement
P146 | Autonomie decisionnelle
P147 | Collaboration synergique
P148 | Competition selective
P149 | Narration debut milieu fin
P150 | Personnage identite motivation arc
P151 | Conflit desir vs obstacle
P152 | Resolution du conflit
P153 | Theme abstraction du recit
P154 | Symbole narratif
P155 | Metaphore analogie
P156 | Ironie ecart attendu reel
P157 | Suspense incertitude de resolution
P158 | Catharsis liberation emotionnelle
P159 | Epiphanie revelation soudaine
P160 | Denouement final
P161 | Calibration energetique temporelle loi finale

LOIS nommées: 1

[ ] Decision = KX108. (AGENTS_52_NON_DECISION_CONTRACT.md)

 Allowed by auto mode classifier

06:17 PM claude-sonnet-4-6

● Excellent ! 145 Pépites déréférencées, 34 Arbres complets. Maintenant je cherche les Lois dans le genome.json et les agents

canoniques complets.

● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "

```
import json, re, sys
sys.stdout.reconfigure(encoding='utf-8')
lois_all = {}
agents_all = {}
balances_all = {}
canon_entries = {}
with
open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',
encoding='utf-8') as f:
for line in f:
line = line.strip()
if not line: continue
try:
rec = json.loads(line)
title = rec.get('title','') or ''
path = rec.get('path','') or ''
text = rec.get('text','') or ''
tl = title.lower()
pl = path.lower()
```

genome.json = source des lois

```
if 'genome.json' in tl or 'genome.json' in pl:
print(f'=== GENOME: {title}')
try:
data = json.loads(text)
print(f' Keys: {list(data.keys())[:20]}')
lois_raw = data.get('lois', data.get('laws', data.get('constitution', [])))
if lois_raw:
print(f' Lois count: {len(lois_raw)}')
for l in lois_raw[:30]:
print(f' {l}')
if not lois_raw and len(text) < 5000:
print(f' text: {text[:2000]}')
except Exception as e:
print(f' err parse: {e}')
print(f' text[:1500]: {text[:1500]}')
```

Lois Obsidiennes - patterns variés

elif re.search(r'lois_obsidiennes|07_lois|bloc_09.*lois|les_7_lois|7_lois', tl):
print(f'=== LOIS FILE: {title}')

Extrait les lois depuis le texte markdown

loi_matches = re.findall(r'(?:##?\s+)?(?:Loi\s+(?:N°?\s*)?(\d+)[\s:—-]+([^\n]{5,100}))', text)
for num, name in loi_matches[:20]:
print(f' Loi {num}: {name.strip()}')
if not loi_matches:
print(f' text[:1000]: {text[:1000]}')

Agents 52 agents - cherche dans AGENTS_52 ou agents_canoniques

elif re.search(r'agents_52|52_agents|agents_canoniques', tl) or

re.search(r'agents_52|52_agents|agents_canoniques', pl):

print(f'=== AGENTS52: {title}')

print(f' text[:1500]: {text[:1500]}')

except: pass

" 2>&1 | head -300)

 Waiting...

 === AGENTS52: 06260A696F32_AGENTS_52_RUNTIME_BOUNDARY.json

text[:1500]: {"id": "BRODY_GRAPHITI_READY_000049", "title": "06260A696F32_AGENTS_52_RUNTIME_BOUNDARY.json",

"source_original_path": "_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\06260A696F32_AGENTS_

52_RUNTIME_BOUNDARY.json", "normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engi

ne-candidate_world_intake\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162_20260512_211327\\normaliz

ed_md\\06260A696F32_AGENTS_52_RUNTIME_BOUNDARY_06260A696F32.md", "original_extension": ".json",

"extraction_status": "TEXT_EXTRACTED_utf-8-sig", "extractor": "plain-text", "content_usable": true,

"source_original_sha256": "06260A696F32173B0BE7F23CDEFDC49E38ACAB6D8A72C76F4C6EE2FE102EB039", "text_sha256":

"C204979EE1409EDCB8838EC6552642D96482FBAA03BE441A12C4A36A11F8463A", "text_excerpt": "# BRODY WORLD SOURCE INTAKE

READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path: _world_sources\\PROJECT_MEM

ORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\06260A696F32_AGENTS_52_RUNTIME_BOUNDARY.json\n- source_original_sha256: 06260A696F32173B0BE7F23CDEFDC49E38ACAB6D8A72C76F4C6EE2FE102EB039\n- original_extension: .json\n- extraction_status:

TEXT_EXTRACTED_utf-8-sig\n- extractor: plain-text\n- content_usable: true\n\n## Boundary\n\n- readonly: true\n- memory_authority: false\n- memory_decision: false\n- decision_authority: KX108_ONLY\n- kernel_binding: false\n- x108_merge: false\n- emits_act: false\n\n## Extracted text\n\n```text\n{

=== AGENTS52: 0F8BE82FE7B0_AGENTS_52_NON_DECISION_CONTRACT.md

text[:1500]: {"id": "BRODY_GRAPHITI_READY_000147", "title": "0F8BE82FE7B0_AGENTS_52_NON_DECISION_CONTRACT.md",

"source_original_path": "_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\0F8BE82FE7B0_AGENTS_

52_NON_DECISION_CONTRACT.md", "normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-e

ngine-candidate_world_intake\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162_20260512_211327\\norma

lized_md\\0F8BE82FE7B0_AGENTS_52_NON_DECISION_CONTRACT_0F8BE82FE7B0.md", "original_extension": ".md",

"extraction_status": "TEXT_EXTRACTED_utf-8-sig", "extractor": "plain-text", "content_usable": true,

"source_original_sha256": "0F8BE82FE7B080C30EB0D17A72A57C9CC98037E8E5ACC8A7999CD6BB804BE7A7", "text_sha256":

"32534342D9B0461FE3F3A943BA660532F2DFB2084A94F6478C3B435D50236CCC", "text_excerpt": "# BRODY WORLD SOURCE INTAKE

READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path: _world_sources\\PROJECT_MEM

ORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\0F8BE82FE7B0_AGENTS_52_NON_DECISION_CONTRACT.md\n-

source_original_sha256: 0F8BE82FE7B080C30EB0D17A72A57C9CC98037E8E5ACC8A7999CD6BB804BE7A7\n- original_extension:

.md\n- extraction_status: TEXT_EXTRACTED_utf-8-sig\n- extractor: plain-text\n- content_usable: true\n\n## Boundary\n\n- readonly: true\n- memory_authority: false\n- memory_decision: false\n- decision_authority: KX108_ONLY\n- kernel_binding: false\n- x108_merge: false\n- emits_act: false\n\n## Extracted text\n\n

=== AGENTS52: 10839F27933F_AGENTS_52_FORBIDDEN_ACTIONS.md

text[:1500]: {"id": "BRODY_GRAPHITI_READY_000184", "title": "10839F27933F_AGENTS_52_FORBIDDEN_ACTIONS.md",

"source_original_path": "_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\10839F27933F_AGENTS_

52_FORBIDDEN_ACTIONS.md", "normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engin

e-candidate_world_intake\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162_20260512_211327\\normalize

d_md\\10839F27933F_AGENTS_52_FORBIDDEN_ACTIONS_10839F27933F.md", "original_extension": ".md", "extraction_status": "TEXT_EXTRACTED_utf-8-sig", "extractor": "plain-text", "content_usable": true, "source_original_sha256": "10839F27933F155BD906DCB62AC8FBA88A9FCE306E30AC300364414CC6F58980", "text_sha256": "E8FA29A7E3B34D19D1C81701348EE00C660B7C1887B63B0AE7AEEABF9BE36344", "text_excerpt": "# BRODY WORLD SOURCE INTAKE

READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path: _world_sources\\PROJECT_MEM

ORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\10839F27933F_AGENTS_52_FORBIDDEN_ACTIONS.md\n- source_original_sha256: 10839F27933F155BD906DCB62AC8FBA88A9FCE306E30AC300364414CC6F58980\n- original_extension: .md\n- extraction_status: TEXT_EXTRACTED_utf-8-sig\n- extractor: plain-text\n- content_usable: true\n\n## Boundary\n\n- readonly: true\n- memory_authority: false\n- memory_decision: false\n- decision_authority: KX108_ONLY\n- kernel_binding: false\n- x108_merge: false\n- emits_act: false\n\n## Extracted text\n\n```text\n# AGENT

=== GENOME: 300FA1A5EC54_genome.json

Keys: ['id', 'title', 'source_original_path', 'normalized_md_path', 'original_extension', 'extraction_status', 'extractor', 'content_usable', 'source_original_sha256', 'text_sha256', 'text_excerpt', 'tags', 'readonly', 'memory_authority', 'memory_decision', 'allowed_to_decide', 'emits_act', 'kernel_binding', 'x108_merge', 'kernel_mutation']

text: {"id": "BRODY_GRAPHITI_READY_000486", "title": "300FA1A5EC54_genome.json", "source_original_path": "_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\ \300FA1A5EC54_genome.json", "normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engine-candidate_world_intake\\BRODY_WORLD_INTAKE_P

ROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162_20260512_211327\\normalized_md\\300FA1A5EC54_genome_30 0FA1A5EC54.md", "original_extension": ".json", "extraction_status": "TEXT_EXTRACTED_utf-8-sig", "extractor": "plain-text", "content_usable": true, "source_original_sha256": "300FA1A5EC54CDA9DA2552A3FC2ED8A1D1F02B7CA51D6842988291F7207DBEBC", "text_sha256": "ECB5D81195203D609A91B21EF47A14085353DEAB931AF004C6D76079CB388B45", "text_excerpt": "# BRODY WORLD SOURCE INTAKE

READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path:

_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\300FA1A5EC54_genome.json\ n-

source_original_sha256: 300FA1A5EC54CDA9DA2552A3FC2ED8A1D1F02B7CA51D6842988291F7207DBEBC\n- original_extension: .json\n- extraction_status: TEXT_EXTRACTED_utf-8-sig\n- extractor: plain-text\n- content_usable: true\n\n## Boundary\n\n- readonly: true\n- memory_authority: false\n- memory_decision: false\n- decision_authority: KX108_ONLY\n- kernel_binding: false\n- x108_merge: false\n- emits_act: false\n\n## Extracted text\n\n```text\n{\n \"O1\": \"Non-Action Légitime\",\n \"O2\": \"Irréversibilité Structurelle\",\n \"O3\": \"Épreuve Temporelle X-108\",\n \"O4\": \"Non-Arbitrage sous Conflit\",\n \"O5\": \"Élimination Décisionnelle\",\n \"O6\": \"Mémoire Négative\",\n \"O7\": \"Auditabilité Native\",\n \"O8\": \"Séparation Cognition / Action\"\n}\n```\n", "tags": ["world_intake", "x108"], "readonly": true, "memory_authority": false, "memory_decision": false, "allowed_to_decide": false, "emits_act": false, "kernel_binding": false, "x108_merge": false, "kernel_mutation": false, "x108_mutation": fals

=== LOIS FILE: 427731C073EB_Bloc_09__Les_7_Lois_Obsidiennes_C8.md

text[:1000]: {"id": "BRODY_GRAPHITI_READY_000655", "title": "427731C073EB_Bloc_09__Les_7_Lois_Obsidiennes_C8.md", "source_original_path": "_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\ \427731C073EB_Bloc_0

9__Les_7_Lois_Obsidiennes_C8.md", "normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engine-candidate_world_intake\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162_20260512_211327\\n

ormalized_md\\427731C073EB_Bloc_09__Les_7_Lois_Obsidiennes_C8_427731C073EB.md", "original_extension": ".md", "extraction_status": "TEXT_EXTRACTED_utf-8-sig", "extractor": "plain-text", "content_usable": true, "source_original_sha256": "427731C073EB6B447860751540E21DB7C0B4DDCA1918FE9735F1F8B1ACC2B946", "text_sha256": "4AB6FA5AE92CA9BA29D097CBCFBB487ED57186C941A1FD04B0E37981CE1B74C4", "text_excerpt": "# BRODY WORLD SOURCE INTAKE

READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path: _world_sources\\PROJECT_MEMORY_CLEAN

=== AGENTS52: 446D12773467_agents_canoniques.json

text[:1500]: {"id": "BRODY_GRAPHITI_READY_000668", "title": "446D12773467_agents_canoniques.json",

"source_original_path":

"_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\
\446D12773467_agents_canoniques.json",

"normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engine-c andidate_world_intake

\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162_2026051
2_211327\\normalized_md\\446D12773467_agents_

canoniques_446D12773467.md", "original_extension": ".json", "extraction_status": "TEXT_EXTRACTED_utf-8-sig",

"extractor": "plain-text", "content_usable": true, "source_original_sha256":

"446D1277346758A4228ABA1DD638C98C145CD8A00C78A2AF1B0BEDB4E8AE96BD", "text_sha256":

"C20D6EFD006B15AD1AF1C9A583B4FE4A139DCE7DC2513B81AE47B771F05FB292", "text_excerpt": "# BRODY
WORLD SOURCE INTAKE

READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path:

_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\446D12773467_agents_canon iques.json\n-

source_original_sha256: 446D1277346758A4228ABA1DD638C98C145CD8A00C78A2AF1B0BEDB4E8AE96BD\n-
original_extension:

.json\n- extraction_status: TEXT_EXTRACTED_utf-8-sig\n- extractor: plain-text\n- content_usable: true\n\n##

Boundary\n\n- readonly: true\n- memory_authority: false\n- memory_decision: false\n- decision_authority:

KX108_ONLY\n- kernel_binding: false\n- x108_merge: false\n- emits_act: false\n\n## Extracted text\n\n```text\n[\n

{\n \"id\": \"AGENT_001\",\n

=== AGENTS52: 61E3D061BB09_agents_52.registry.json

text[:1500]: {"id": "BRODY_GRAPHITI_READY_000902", "title": "61E3D061BB09_agents_52.registry.json",

"source_original_path":

"_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\
\61E3D061BB09_agents_52.registry.json",

"normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engine-c andidate_world_intake

\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162_2026051
2_211327\\normalized_md\\61E3D061BB09_agents_

52_registry_61E3D061BB09.md", "original_extension": ".json", "extraction_status": "TEXT_EXTRACTED_utf-8-sig",

"extractor": "plain-text", "content_usable": true, "source_original_sha256":

"61E3D061BB09DAB7521809DA79F32CB39B94E7F44AA62B16D24635C7B0A8625F", "text_sha256":

"6B0D603D4CA22D10C7B29FBF5BFA313EE1D86CEC9FB0E23A9002FC9194BF331A", "text_excerpt": "# BRODY
WORLD SOURCE INTAKE

READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path:

_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\61E3D061BB09_agents_52.re gistry.json\n-

source_original_sha256: 61E3D061BB09DAB7521809DA79F32CB39B94E7F44AA62B16D24635C7B0A8625F\n-
original_extension:

.json\n- extraction_status: TEXT_EXTRACTED_utf-8-sig\n- extractor: plain-text\n- content_usable: true\n\n##

Boundary\n\n- readonly: true\n- memory_authority: false\n- memory_decision: false\n- decision_authority:

KX108_ONLY\n- kernel_binding: false\n- x108_merge: false\n- emits_act: false\n\n## Extracted text\n\n```text\n[\n

{\n \"id\": \"1\",\n \

=== AGENTS52: 789A800379DA_agents_52.registry.json

text[:1500]: {"id": "BRODY_GRAPHITI_READY_001122", "title": "789A800379DA_agents_52.registry.json",

"source_original_path":

"_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\
\789A800379DA_agents_52.registry.json",

"normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engine-c andidate_world_intake

\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162_2026051
2_211327\\normalized_md\\789A800379DA_agents_

52_registry_789A800379DA.md", "original_extension": ".json", "extraction_status": "TEXT_EXTRACTED_utf-8-sig",

"extractor": "plain-text", "content_usable": true, "source_original_sha256":

"789A800379DA38DCE5D517DB82030F1758762DE053446A2BC9773357F7E7BD5D", "text_sha256":

"2EFD676C12E8F11F8B6F9157F87C55F78BB8C80F4C62C64275518F9D6E54B1D3", "text_excerpt": "# BRODY WORLD
SOURCE INTAKE

READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path:

_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\789A800379DA_agents_52.re gistry.json\n-

source_original_sha256: 789A800379DA38DCE5D517DB82030F1758762DE053446A2BC9773357F7E7BD5D\n-
original_extension:

.json\n- extraction_status: TEXT_EXTRACTED_utf-8-sig\n- extractor: plain-text\n- content_usable: true\n\n##

Boundary\n\n- readonly: true\n- memory_authority: false\n- memory_decision: false\n- decision_authority:

KX108_ONLY\n- kernel_binding: false\n- x108_merge: false\n- emits_act: false\n\n## Extracted text\n\n```text\n[\n

{\n \"id\": \"1\",\n \

=== AGENTS52: 84C88A7559A2_AUDIT_AGENTS_52.md

text[:1500]: {"id": "BRODY_GRAPHITI_READY_001248", "title": "84C88A7559A2_AUDIT_AGENTS_52.md",

"source_original_path":

"_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\
\84C88A7559A2_AUDIT_AGENTS_52.md",

"normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engine-c andidate_world_intake

\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162_2026051
2_211327\\normalized_md\\84C88A7559A2_AUDIT_A

GENTS_52_84C88A7559A2.md", "original_extension": ".md", "extraction_status": "TEXT_EXTRACTED_utf-8-sig",

"extractor": "plain-text", "content_usable": true, "source_original_sha256":

"84C88A7559A29236FC0551D7038E86E849AA33C1B645E1A205E8485DDAF4336F", "text_sha256":

"8ADA57BBDD3487424964E4BC7FFEB01E8008D0D7E3E8132E5554D34F852671D9", "text_excerpt": "# BRODY
WORLD SOURCE INTAKE

READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path:

_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\84C88A7559A2_AUDIT_AGENTS _52.md\n-

source_original_sha256: 84C88A7559A29236FC0551D7038E86E849AA33C1B645E1A205E8485DDAF4336F\n-
original_extension:

.md\n- extraction_status: TEXT_EXTRACTED_utf-8-sig\n- extractor: plain-text\n- content_usable: true\n\n##

Boundary\n\n- readonly: true\n- memory_authority: false\n- memory_decision: false\n- decision_authority:

KX108_ONLY\n- kernel_binding: false\n- x108_merge: false\n- emits_act: false\n\n## Extracted text\n\n```text\n\n#

Audit Agents 52\n\n## AUDIT_STATUS\n\n`AUDIT_STATUS

=== AGENTS52: 8940C0A4EEBB_AGENTS_52_ALLOWED_OUTPUT_SCHEMA.json

text[:1500]: {"id": "BRODY_GRAPHITI_READY_001293", "title":
"8940C0A4EEBB_AGENTS_52_ALLOWED_OUTPUT_SCHEMA.json",

"source_original_path": "_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\
\8940C0A4EEBB_AGENTS_

52_ALLOWED_OUTPUT_SCHEMA.json", "normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia

-engine-candidate_world_intake\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CL
EAN_SOURCE_NO_INDEX_V162_20260512_211327\\nor

malized_md\\8940C0A4EEBB_AGENTS_52_ALLOWED_OUTPUT_SCHEMA_8940C0A4EEBB.md", "original_extension": ".json",

"extraction_status": "TEXT_EXTRACTED_utf-8-sig", "extractor": "plain-text", "content_usable": true,

"source_original_sha256": "8940C0A4EEBBD21C4626659DED4EE2D1CE6FEEA7C2A26A65FE49E6D9C6FFB23D",
"text_sha256":

"F6FB8D6B33BABCA9A16ECFAAA74B6D5188C163FB00942B8A72D3AEAC4B15FD7E", "text_excerpt": "# BRODY
WORLD SOURCE INTAKE

READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path:
_world_sources\\PROJECT_MEM

ORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\8940C0A4EEBB_AGENTS_52_ALLOWED_OUTPUT_SCHEMA.json\n-

source_original_sha256: 8940C0A4EEBBD21C4626659DED4EE2D1CE6FEEA7C2A26A65FE49E6D9C6FFB23D\n-
original_extension:

.json\n- extraction_status: TEXT_EXTRACTED_utf-8-sig\n- extractor: plain-text\n- content_usable: true\n\n##

Boundary\n\n- readonly: true\n- memory_authority: false\n- memory_decision: false\n- decision_authority:

KX108_ONLY\n- kernel_binding: false\n- x108_merge: false\n- emits_act: false\n\n## Extracte

=== AGENTS52: 9799C419D3FD_agents_52.csv

text[:1500]: {"id": "BRODY_GRAPHITI_READY_001422", "title": "9799C419D3FD_agents_52.csv",
"source_original_path":

"_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\ \9799C419D3FD_agents_52.csv",

"normalized_md_path":

"C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engine-c
andidate_world_intake\\BRODY_WORLD_INTAKE_P

ROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162_20260512_211327\\normalized_md\\9799C419D3FD_agents_52 _9799C419D3FD.md",

"original_extension": ".csv", "extraction_status": "CSV_TEXT_EXTRACTED", "extractor": "csv", "content_usable":

true, "source_original_sha256": "9799C419D3FD9F33EF94A1F30F826080A6F0EAEC1C9CCD65073F1B40EC8223B3",
"text_sha256":

"237FEDC1D3A9B898E6D71D0DC8713F133FF212BBEAEC8E2BEA930BB3AEECA569", "text_excerpt": "# BRODY
WORLD SOURCE INTAKE

READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path:

_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\9799C419D3FD_agents_52.cs v\n-

source_original_sha256: 9799C419D3FD9F33EF94A1F30F826080A6F0EAEC1C9CCD65073F1B40EC8223B3\n- original_extension:

.csv\n- extraction_status: CSV_TEXT_EXTRACTED\n- extractor: csv\n- content_usable: true\n\n## Boundary\n\n-

readonly: true\n- memory_authority: false\n- memory_decision: false\n- decision_authority: KX108_ONLY\n-

kernel_binding: false\n- x108_merge: false\n- emits_act: false\n\n## Extracted text\n\n```text\nid | nom | famille

| role | sortie_principale | deploiement | non_decision | status\n1 | OBSIDIA_ATL

=== AGENTS52: C7CAEA284406_AGENTS_CANONIQUES_ET_GARDIENS.md

text[:1500]: {"id": "BRODY_GRAPHITI_READY_001880", "title": "C7CAEA284406_AGENTS_CANONIQUES_ET_GARDIENS.md",

"source_original_path": "_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\ \C7CAEA284406_AGENTS_

CANONIQUES_ET_GARDIENS.md", "normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-eng

ine-candidate_world_intake\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CLEAN_ SOURCE_NO_INDEX_V162_20260512_211327\\normali

zed_md\\C7CAEA284406_AGENTS_CANONIQUES_ET_GARDIENS_C7CAEA284406.md", "original_extension": ".md",

"extraction_status": "TEXT_EXTRACTED_utf-8-sig", "extractor": "plain-text", "content_usable": true,

"source_original_sha256": "C7CAEA284406C3AFD2600FC3AA414955AF0B5633B6880EA4EF14BE0556D47589", "text_sha256":

"6FB4FF66C771847E5297401F02629FA3D5B24D15FADABCFD87A438EE3EEC7656", "text_excerpt": "# BRODY WORLD SOURCE INTAKE

READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path: _world_sources\\PROJECT_MEM

ORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\C7CAEA284406_AGENTS_CANONIQUES_ET_GARDIENS.md\n- source_original_sha256: C7CAEA284406C3AFD2600FC3AA414955AF0B5633B6880EA4EF14BE0556D47589\n- original_extension: .md\n- extraction_status:

TEXT_EXTRACTED_utf-8-sig\n- extractor: plain-text\n- content_usable: true\n\n## Boundary\n\n- readonly: true\n-

memory_authority: false\n- memory_decision: false\n- decision_authority: KX108_ONLY\n- kernel_binding: false\n-

x108_merge: false\n- emits_act: false\n\n## Extracted text\n\n```text\

=== AGENTS52: F9093068F887_agents_52.index.json

text[:1500]: {"id": "BRODY_GRAPHITI_READY_002299", "title": "F9093068F887_agents_52.index.json",

"source_original_path":

"_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\ \F9093068F887_agents_52.index.json",

"normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engine-c andidate_world_intake

\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162_2026051 2_211327\\normalized_md\\F9093068F887_agents_

52_index_F9093068F887.md", "original_extension": ".json", "extraction_status": "TEXT_EXTRACTED_utf-8-sig",

"extractor": "plain-text", "content_usable": true, "source_original_sha256":

"F9093068F887483FE923A631FBFDEF89E9B6EC24D8783B1CC1FBC8E497B8DDC6", "text_sha256":

"8541FFA1B4F7D6B39D569C997222677C7A9B2F7EC35EC18B868701EE1F6284AD", "text_excerpt": "# BRODY WORLD SOURCE INTAKE

READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path: _world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\F9093068F887_agents_52.in dex.json\n-

source_original_sha256: F9093068F887483FE923A631FBFDEF89E9B6EC24D8783B1CC1FBC8E497B8DDC6\n- original_extension:

.json\n- extraction_status: TEXT_EXTRACTED_utf-8-sig\n- extractor: plain-text\n- content_usable: true\n\n##

Boundary\n\n- readonly: true\n- memory_authority: false\n- memory_decision: false\n- decision_authority:

KX108_ONLY\n- kernel_binding: false\n- x108_merge: false\n- emits_act: false\n\n## Extracted text\n\n```text\n{\n

\"status\": \"AGENTS_52_DOCX_GENERATED\"

=== GENOME: genome.json

Keys: ['O1', 'O2', 'O3', 'O4', 'O5', 'O6', 'O7', 'O8']

text: {

"O1": "Non-Action Légitime",

"O2": "Irréversibilité Structurelle",

"O3": "Épreuve Temporelle X-108",

"O4": "Non-Arbitrage sous Conflit",

"O5": "Élimination Décisionnelle",

"O6": "Mémoire Négative",

"O7": "Auditabilité Native",

"O8": "Séparation Cognition / Action"

}
=== AGENTS52: agents_52.csv
text[:1500]: id,nom,famille,role,sortie_principale,deploiement,non_decision,status
1,OBSIDIA_ATLAS_INGESTOR,À créer en premier (top 5),Analyse multi-couches + cartes mémoire JSON,Signal/context packet non décisionnel pour OBSIDIA_ATLAS_INGESTOR,Vertex AI — premier agent,True,NEEDS_HUMAN_VALIDATION
2,CANON_GUARDIAN,À créer en premier (top 5),Verdict statut canonique + alertes contradictions,Signal/context packet non décisionnel pour CANON_GUARDIAN,Vertex AI ou local,True,NEEDS_HUMAN_VALIDATION
3,GRAPH_BUILDER,À créer en premier (top 5),Graphe nœuds/liens avec force et type de relation,Signal/context packet non décisionnel pour GRAPH_BUILDER,Vertex AI puis local,True,NEEDS_HUMAN_VALIDATION
4,TERMINAL_BUILDER,À créer en premier (top 5),Blocs commandes + preuves d'exécution attendues,Signal/context packet
non décisionnel pour TERMINAL_BUILDER,Local prioritaire,True,NEEDS_HUMAN_VALIDATION
5,PROOF_SENTINEL,À créer en premier (top 5),"Rapport preuve : invariant touché, cause, niveau de rupture",Signal/context packet non décisionnel pour PROOF_SENTINEL,Local + CI,True,NEEDS_HUMAN_VALIDATION
6,CO_PILOTE_CODE,Code / Repo / CI,"Patchs, analyse fichiers, optimisation code",Signal/context packet non décisionnel pour CO_PILOTE_CODE,"Vertex d'abord, local ensuite",True,NEEDS_HUMAN_VALIDATION
7,CI_REPO_SURGEON,Code / Repo / CI,"PR propres, workflows stricts, diagnostics CI",Signal/context packet non décisionnel pour CI_REPO_SURGEON,Vertex ou local,True,NEEDS_HUMAN_VALIDATION
8,REPO_HISTORIAN,Code
=== AGENTS52: agents_52.index.json
text[:1500]: {
"status": "AGENTS_52_DOCX_GENERATED",
"source_docx": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engine-candidate\\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1\\10_AGENTS_52_SOURCE\\OBSIDIA_TOUS_LES_AGENTS.docx",
"agent_count": 52,
"family_count": 9,
"system_prompt_count": 47,
"placeholder_count": 1,
"agents": [
{
"number": 1,
"name": "OBSIDIA_ATLAS_INGESTOR",
"family": "À créer en premier (top 5)",
"folder": "00_TOP5_FONDATEURS",
"path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engine-c andidate\\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1\\10_AGENTS_52\\00_TOP5_FONDATEURS\\OBSIDIA_ATLAS_INGESTOR.md",
"sha256": "048ed3da3c6f34fc6659a290910d86816ff4890e48c9e736f004842a1fee61ce",
"size": 4005,
"has_system_prompt": true,
"placeholder_detected": false,
"non_decision": true,
"decision_authority": "KX108_ONLY"
},
{
"number": 2,
"name": "CANON_GUARDIAN",
"family": "À créer en premier (top 5)",
"folder": "00_TOP5_FONDATEURS",
"path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engine-c andidate\\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1_FREEZE_CANDIDATE\\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1\\10_AGENTS_5

=== AGENTS52: agents_52.registry.json
text[:1500]: [
{
"id": "1",
"name": "OBSIDIA_ATLAS_INGESTOR",
"family": "À créer en premier (top 5)",
"role": "Analyse multi-couches + cartes mémoire JSON",
"deployment": "Vertex AI — premier agent",
"system_prompt": "Tu es OBSIDIA_ATLAS_INGESTOR, agent périphérique Obsidia. Tu n'exécutes aucune action critique. Tu produis uniquement contexte, vérification, signal ou artefact auditable.",
"status": "NEEDS_HUMAN_VALIDATION",
"boundary": "OBSIDIA_ATLAS_INGESTOR ne décide jamais ; il produit uniquement Signal/context packet non décisionnel pour OBSIDIA_ATLAS_INGESTOR. Decision = KX108.",
"input": "Contexte, tâche ou artefact lié à À créer en premier (top 5)",
"output": "Signal/context packet non décisionnel pour OBSIDIA_ATLAS_INGESTOR",
"non_decision": true,
"local_cloud_policy": "Vertex AI — premier agent",
"nom": "OBSIDIA_ATLAS_INGESTOR",
"famille": "À créer en premier (top 5)",
"sortie_principale": "Signal/context packet non décisionnel pour OBSIDIA_ATLAS_INGESTOR",
"deploiement": "Vertex AI — premier agent",
"prompt_systeme": "Tu es OBSIDIA_ATLAS_INGESTOR, agent périphérique Obsidia. Tu n'exécutes aucune action critique. Tu produis uniquement contexte, vérification, signal ou artefact auditable.",
"data_sovereignty": "Aucune donnée externe sans politique explicite ; à valider humainement.",
"kernel_boundary": "AgentPeripheral ↛ ACT ; Decision = KX108",
"statut": "NEEDS_HUMAN_VALIDATION"
},
{
"id": "
=== AGENTS52: README.md
text[:1500]: # Agents 52
Ce dossier contient la liste des agents et des sous-dossiers par famille.
=== AGENTS52: AGENTS_52_ALLOWED_OUTPUT_SCHEMA.json
text[:1500]: {
"$schema": "http://json-schema.org/draft-07/schema#",
"title": "Agents52AllowedOutput",
"type": "object",
"required": [
"agent",
"output_type",
"non_decision"
],
"properties": {
"agent": {
"type": "string"
},
"output_type": {
"enum": [
"context",
"analysis",
"map",
"report",
"recommendation",
"non_decision_signal"

]
},
"non_decision": {
"const": true
},
"decision_authority": {
"const": "KX108_ONLY"
},
"kernel_mutation": {
"const": false
},
"canon_promotion": {
"const": false
},
"proof_claim": {
"const": false
}
},
"additionalProperties": true
}
=== AGENTS52: AGENTS_52_FORBIDDEN_ACTIONS.md
text[:1500]: # AGENTS 52 — Forbidden actions
• ALLOW final interdit
• HOLD final interdit
• BLOCK final interdit
• ACT interdit
• Kernel mutation interdite
• Canon promotion interdite
• Proof claim interdit
Toute action critique passe par X-108.
=== AGENTS52: AGENTS_52_NON_DECISION_CONTRACT.md
text[:1500]: # AGENTS 52 — Contrat non-décisionnel

Statut

STATUS=RUNTIME_CANDIDATE_CONTRACT
AGENTS_COUNT=52
DECISION_AUTHORITY=KX108_ONLY

Interdictions

Les agents 52 ne peuvent jamais :

• produire ALLOW final
• produire HOLD final
• produire BLOCK final
• déclencher ACT
• muter le kernel
• promouvoir un canon
• revendiquer une preuve formelle

Autorisation

Les agents 52 peuvent uniquement produire :

• contexte
• analyse
• cartographie
• rapport
• recommandation
• signal non-décisionnel

Loi

Decision = KX108.
=== AGENTS52: AGENTS_52_RUNTIME_BOUNDARY.json
text[:1500]: {
"status": "AGENTS_52_RUNTIME_BOUNDARY",
"agent_count": 52,
"allowed_outputs": [
"context",
"analysis",
"map",
"report",
"recommendation",
"non_decision_signal"
],
"forbidden_outputs": [
"ALLOW",
"HOLD",
"BLOCK",
"ACT",
"kernel_mutation",
"canon_promotion",
"proof_claim"
],
"decision_authority": "KX108_ONLY",
"non_decision": true,
"kernel_mutation": false,
"canon_promotion": false,
"proof_claim": false
}
=== AGENTS52: test_agents_52_registry.py
text[:1500]: import json, sys, importlib.util
from pathlib import Path
ROOT = Path(__file__).resolve().parents[1]
def load_module(name, relpath, add_path=None):
if add_path:
sys.path.insert(0, str(ROOT / add_path))
spec = importlib.util.spec_from_file_location(name, ROOT / relpath)
mod = importlib.util.module_from_spec(spec)
spec.loader.exec_module(mod)
return mod
import unittest
class TestAgentsRegistry(unittest.TestCase):
def test_agents(self):
agents = json.loads((ROOT / "10_AGENTS_52/agents_52.registry.json").read_text(encoding="utf-8"))
self.assertEqual(len(agents), 52)
names = {a["nom"] for a in agents}
for n in {"OBSIDIA_ATLAS_INGESTOR","CANON_GUARDIAN","GRAPH_BUILDER","TERMINAL_B
UILDER","PROOF_SENTINEL"}:
self.assertIn(n, names)
for a in agents:
self.assertTrue(a["input"])
self.assertTrue(a["output"])
self.assertTrue(a["boundary"])
self.assertTrue(a["sortie_principale"])
self.assertTrue(a["prompt_systeme"])
self.assertTrue(a["non_decision"])

if __name__ == "__main__":
unittest.main()
=== AGENTS52: AUDIT_AGENTS_52.md
text[:1500]: # Audit Agents 52

AUDIT_STATUS

`AUDIT_STATUS = V1_REAL_AUDIT`

Périmètre

Registre agents, 9 familles, top 5, doubles champs anglais/français.

Fichiers cont

 Allowed by auto mode classifier

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
```

import json, re, sys
sys.stdout.reconfigure(encoding='utf-8')

Focus sur la liste nominale des 52 agents et les 7 lois obsidiennes

```
agents_52 = {}
lois_7 = []
with
open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',
encoding='utf-8') as f:
for line in f:
line = line.strip()
if not line: continue
try:
rec = json.loads(line)
title = rec.get('title','') or ''
text = rec.get('text','') or ''
tl = title.lower()
```

AGENTS_52_RUNTIME_BOUNDARY - contient la liste ?

if 'agents_52_runtime' in tl:

Cherche dans le texte brut du payload

m = re.search(r'Extracted text.*?\`\`\`text\n(.*?)\`\`\`', text, re.DOTALL)
raw = m.group(1) if m else text

cherche nom -> ou 'name':

```
names = re.findall(r'\"name\"\s*:\s*\"([^\"]+)\"', raw)
ids = re.findall(r'\"id\"\s*:\s*\"?(\w+)\"?', raw)
if names:
print(f'AGENTS NOMS depuis {title}:')
for n in names[:60]: print(f' {n}')
else:
```

cherche pattern agent_xx

```
agent_matches = re.findall(r'AGENT_(\w+)[\s:—]+([^\n]+)', raw)
if agent_matches:
for a_id, a_name in agent_matches[:60]:
print(f' AGENT_{a_id}: {a_name.strip()[:60]}')
else:
print(f'=== RAW {title}[:800]:')
print(raw[:800])
```

Les 7 lois

elif 'bloc_09' in tl and 'lois' in tl:

m = re.search(r'Extracted text.*?\`\`\`text\n(.*?)\`\`\`', text, re.DOTALL)

raw = m.group(1) if m else text

print(f'=== LES 7 LOIS depuis {title}:')

for k,v in data.items():

if isinstance(v, list) and len(v) > 0:

print(f' {k}[{len(v)}]: {v[:3]}')

elif isinstance(v, str) and len(v) < 200:

print(f' {k}: {v}')

except:

print(raw[:2000])

except: pass

" 2>&1 | head -200)

 Waiting...

 === RAW 06260A696F32_AGENTS_52_RUNTIME_BOUNDARY.json[:800]:

{"id": "BRODY_GRAPHITI_READY_000049", "title": "06260A696F32_AGENTS_52_RUNTIME_BOUNDARY.json",

"source_original_path": "_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\
\06260A696F32_AGENTS_

52_RUNTIME_BOUNDARY.json", "normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidia-engi

ne-candidate_world_intake\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CLEAN_S
OURCE_NO_INDEX_V162_20260512_211327\\normaliz

ed_md\\06260A696F32_AGENTS_52_RUNTIME_BOUNDARY_06260A696F32.md", "original_extension": ".json",

"extraction_status": "TEXT_EXTRACTED_utf-8-sig", "extractor": "plain-text", "content_usable": true,

"source_original_sha256": "06260A696F32173B0BE7F23CDEFDC49E38ACAB6D8A72C76F4C6EE2FE102EB039",
"text_sha256":

"C204979EE1409EDCB8838EC6552642D96482FBAA03BE441A12C4A3

=== LES 7 LOIS depuis 427731C073EB_Bloc_09__Les_7_Lois_Obsidiennes_C8.md:

{"id": "BRODY_GRAPHITI_READY_000655", "title": "427731C073EB_Bloc_09__Les_7_Lois_Obsidiennes_C8.md",

"source_original_path": "_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\
\427731C073EB_Bloc_09

__Les_7_Lois_Obsidiennes_C8.md", "normalized_md_path": "C:\\Users\\User\\Desktop\\obsidia-engine-proof-core\\obsidi

a-engine-candidate_world_intake\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_C
LEAN_SOURCE_NO_INDEX_V162_20260512_211327\\no

rmalized_md\\427731C073EB_Bloc_09__Les_7_Lois_Obsidiennes_C8_427731C073EB.md", "original_extension": ".md",

"extraction_status": "TEXT_EXTRACTED_utf-8-sig", "extractor": "plain-text", "content_usable": true,

"source_original_sha256": "427731C073EB6B447860751540E21DB7C0B4DDCA1918FE9735F1F8B1ACC2B946",
"text_sha256":

"4AB6FA5AE92CA9BA29D097CBCFBB487ED57186C941A1FD04B0E37981CE1B74C4", "text_excerpt": "# BRODY
WORLD SOURCE INTAKE

READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path:
_world_sources\\PROJECT_MEM

ORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\427731C073EB_Bloc_09__Les_7_Lois_Obsidiennes_C8.md\n -

source_original_sha256: 427731C073EB6B447860751540E21DB7C0B4DDCA1918FE9735F1F8B1ACC2B946\n-
original_extension:

.md\n- extraction_status: TEXT_EXTRACTED_utf-8-sig\n- extractor: plain-text\n- content_usable: true\n\n##

Boundary\n\n- readonly: true\n- memory_authority: false\n- memory_decision: false\n- decision_authority:

KX108_ONLY\n- kernel_binding: false\n- x108_merge: false\n- emits_act: false\n\n## Extrac

=== RAW AGENTS_52_RUNTIME_BOUNDARY.json[:800]:

{

"status": "AGENTS_52_RUNTIME_BOUNDARY",

"agent_count": 52,

"allowed_outputs": [

"context",

"analysis",

"map",

"report",

"recommendation",
"non_decision_signal"
],
"forbidden_outputs": [
"ALLOW",
"HOLD",
"BLOCK",
"ACT",
"kernel_mutation",
"canon_promotion",
"proof_claim"
],
"decision_authority": "KX108_ONLY",
"non_decision": true,
"kernel_mutation": false,
"canon_promotion": false,
"proof_claim": false
}
 Allowed by auto mode classifier

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
```

import json, re, sys
sys.stdout.reconfigure(encoding='utf-8')

Les records sont des wrappers. Le contenu réel est dans text_excerpt ou payload.

Chercher les 52 agents dans les fichiers qui les LISTENT nominalement

et les 7 lois dans Bloc_09

```
with
open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',
encoding='utf-8') as f:
for i, line in enumerate(f):
line = line.strip()
if not line: continue
try:
rec = json.loads(line)
title = rec.get('title','') or ''
text = rec.get('text','') or ''
tl = title.lower()
```

Lois bloc_09

```
if 'bloc_09' in tl and 'lois' in tl:
excerpt = inner.get('text_excerpt','')
print(f'=== LOIS EXCERPT [{title}]:')
print(excerpt[:3000])
except:
```

Cherche text_excerpt dans le text brut

```
m = re.search(r'text_excerpt.*?\"(.*?)\"(?:,|\n)', text, re.DOTALL)
if m:
print(m.group(1)[:2000])
```

Agents 52 - cherche fichier qui les liste tous par nom

```
if re.search(r'agents_et_roles|agents_canoniques|all_agents|agents_regist ry|52_agents_list|agents_53', tl):
print(f'=== AGENTS LIST: {title}')
try:
inner = json.loads(text)
excerpt = inner.get('text_excerpt','')
print(excerpt[:3000])
except:
print(text[:2000])
except: pass
" 2>&1 | head -150)
 Waiting...
 === AGENTS LIST: 33E955F71FFE_AGENTS_ET_ROLES_DETECTES.md
```

BRODY WORLD SOURCE INTAKE READONLY

• label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162
• source_original_path:
```
_world_sources\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\33E955F71FFE_AGENTS_ET_ROLE S_DETECTES.md
```
• source_original_sha256: 33E955F71FFEF5D8042D0EC073F20FE41A6EF2ADC451EDE8342F91AFE04F6E78
```
- original_extension: .md
```
• extraction_status: TEXT_EXTRACTED_utf-8-sig
• extractor: plain-text
• content_usable: true

Boundary

• readonly: true
• memory_authority: false
• memory_decision: false
• decision_authority: KX108_ONLY
• kernel_binding: false
• x108_merge: false
• emits_act: false

Extracted text

```text
```

Agents et rôles détectés dans les sources

Extraction automatique depuis les documents fournis. À consolider avec le repo réel si besoin.

agents

• Occurrences : 17
• Sources : rapport161, src_40spec, src_branche149
Extraits :
• t une boucle de rétroaction locale. Dans le monde informationnel, ce serait un ensemble d’agents ou de neurones
artificiels capables de moduler leur apprentissage selon l’énergie des données qu’ils reçoivent. Chaque unité obsidienne n’est donc ni récepteur
• re fractale + Protocole Maths ⇄ Langage + PF∞. Application : nuages mémoriels obsidiens, agents conscients de
contexte, navigation entre dimensions de mémoire. 49. Branche “Géo-Résonance / Influence des Lieux” Chaque lieu possède une signature vibrato
• e révèle : ⚙ 70. Branche “Automatisation de l’Automatisation” L’IA devient génératrice d’agents et d’outils par
elle-même. Concept : X-AutoCatalytique — la création devient fractale et procédurale. Fonction : faire émerger de nouveaux systèmes sans super

Agents

• Occurrences : 8
• Sources : checklist, rapport161, src_40spec, src_branche149
Extraits :
• implants intelligents, structures urbaines vivantes. 85. Branche “Entités Cognitives / Agents Obsidia” Les
agents obsidiens sont les neurones du réseau civilisatoire. Concept : X-Agentiel Fractal — unités conscientes interconnectées par le moteur. Lien
• ard, Intention). Couches techniques bien pensées : Architecture logicielle modulaire. Agents spécialisés
(Archiviste, Capo, Brody, etc.). Interfaces projetées (Shell, Dashboard, MindMap). Vision universelle unifiée : “De la particule au symbole, to
• ” (fond noir) → Une carte montrant la montée du moteur depuis la Base (Lois, Protocoles, Agents) → Couronne du
Sens → Couronne Historique → Couronne Énergétique. Fichier Word “Atlas des Savoirs Obsidia” → Table par domaines (Maths, Physique, Biologie, I

Agent

• Occurrences : 4
• Sources : rapport161, src_40spec, src_branche149
Extraits :
• , Éthique, Culture, Histoire, Énergie, Symbolique). → Chaque entrée reliée à une Loi, un Agent et un Protocole.
Souhaites-tu que je te crée directement la fiche “Branches 111–120 – Extensions du Moteur Obsidien” (format complet standard : Objectif / Con
• nt (avant/après) enforcement (allow/deny) 9.1 Position dans l’architecture Chemin canon : Agent/OS4 -> MCP Client
-> Tool Adapter ->
=== LOIS EXCERPT [427731C073EB_Bloc_09__Les_7_Lois_Obsidiennes_C8.md]:

BRODY WORLD SOURCE INTAKE READONLY

• label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162
• source original path: world sources\PROJECT MEMORY CLEAN SOURCE NO INDEX 20260512 211234\42 7731C073EB Bloc 09 L
es_7_Lois_Obsidiennes_C8.md
• source original sha256: 427731C073EB6B447860751540E21DB7C0B4DDCA1918FE9735F1F8B1ACC2B946
- original_extension: .md
• extraction_status: TEXT_EXTRACTED_utf-8-sig
• extractor: plain-text
• content_usable: true

Boundary

• readonly: true
• memory_authority: false
• memory_decision: false
• decision_authority: KX108_ONLY
• kernel_binding: false
• x108_merge: false
• emits_act: false

Extracted text

```text
```

Bloc 09 — Les 7 Lois Obsidiennes (C8)

Rôle

Les principes fondamentaux d'opération.

Contrôle compteur

• Nombre déclaré : 7
• Nombre extrait : 7

Pépites principales

• **P88 — Non-contradiction** — □ FORMALISÉ — Objet mathématique défini dans v1cano.docx / formalisermath.docx
• **P89 — Complétude** — □ FORMALISÉ — Objet mathématique défini dans v1cano.docx / formalisermath.docx
• **P90 — Cohérence** — □ FORMALISÉ — Objet mathématique défini dans v1cano.docx / formalisermath.docx
• **P91 — Décidabilité** — □ FORMALISÉ — Objet mathématique défini dans v1cano.docx / formalisermath.docx
• **P92 — Calculabilité (Turing)** — □ FORMALISÉ — Objet mathématique défini dans v1cano.docx / formalisermath.docx
• **P93 — P ⊆ NP** — □ FORMALISÉ — Objet mathématique défini dans v1cano.docx / formalisermath.docx
• **P94 — Réductibilité** — □ FORMALISÉ — Objet mathématique défini dans v1cano.docx / formalisermath.docx

Pépites secondaires

• P86
• P87

Contrat V4

• Un seul bloc principal par pépite.
• Les pépites secondaires ne comptent pas dans le total 161.
• Les statuts se vérifient dans `03_PEPITES_161` et `01_REGISTRES_JSON`.

Source brute

* **Description :** Les principes fondamentaux d'opération.

* **Pépites Principales (7) :** P88, P89, P90, P91, P92, P93, P94.

* **Pépites Secondaires :** P86, P87 (déjà dans Bloc 1).

```

=== AGENTS LIST: 446D12773467_agents_canoniques.json
```

BRODY WORLD SOURCE INTAKE READONLY

• label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162
• source_original_path:
`_world_sources\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\446D12773467_agents_canoniq ues.json`
• source_original_sha256: 446D1277346758A4228ABA1DD638C98C145CD8A00C78A2AF1B0BEDB4E8AE96BD
`- original_extension: .json`
• extraction_status: TEXT_EXTRACTED_utf-8-sig
• extractor: plain-text
• content_usable: true
• memory_authority: false
• memory_decision: false
• decision_authority: KX108_ONLY
• kernel_binding: false
• x108_merge: false
• emits_act: false

Extracted text

```text
[
{
"id": "AGENT_001",
"name": "Observateur",
"type": "agent_obsidien_R&D",
"source": "src_branche149 — Branche 85 : Entités Cognitives / Agents Obsidia",
"function": "Observer l’état, les flux, les signaux faibles et les variations sans décider.",
"kernel_boundary": "Ne décide jamais. Remonte contexte et anomalies.",
"linked": [
"Bloc 13",
"P137",
"P138",
 Allowed by auto mode classifier
06:18 PM claude-sonnet-4-6
● Les 7 Lois et agents_canoniques.json sont trouvés. J'extrais maintenant la liste complète des 52 agents.
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
import json, re, sys
sys.stdout.reconfigure(encoding='utf-8')
agents_52 = []
all_blocs_lois = []
with
open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',
encoding='utf-8') as f:
for line in f:
line = line.strip()
if not line: continue
try:
rec = json.loads(line)
title = rec.get('title','') or ''
text = rec.get('text','') or ''
tl = title.lower()
```

agents_canoniques.json

if 'agents_canoniques.json' in tl:

Extraire le JSON depuis le text_excerpt

```
m = re.search(r'Extracted text.*?\x60\x60\x60text\n(.*?)\x60\x60\x60', text, re.DOTALL)
raw = m.group(1) if m else text
try:
data = json.loads(raw)
if isinstance(data, list):
for a in data:
agents_52.append({
'id': a.get('id',''),
'name': a.get('name',''),
'type': a.get('type',''),
'function': str(a.get('function',''))[:80],
'linked': str(a.get('linked',[]))[:50],
})
except Exception as e:
print(f'Parse err: {e}')
```

Extraire manuellement

```
agent_blocks =
re.findall(r'\{[^{}]*\"id\"\s*:\s*\"([^\"]+)\"[^{}]*\"name\"\s*:\s*\"( [^\"]+)\"[^{}]*\"type\"\s*:\s*\"([^\"]+)\"', raw)
for a_id, a_name, a_type in agent_blocks:
agents_52.append({'id':a_id, 'name':a_name, 'type':a_type, 'function':'', 'linked':''})
```

Tous les blocs avec lois

```
if re.search(r'bloc_\d+.*lois|les_\d+_lois|lois_obsidiennes', tl):
m = re.search(r'Extracted text.*?\x60\x60\x60text\n(.*?)\x60\x60\x60', text, re.DOTALL)
raw = m.group(1) if m else ''
pepites = re.findall(r'\*\*P(\d+)\s*—\s*([^*\n]+)', raw)
if pepites:
all_blocs_lois.append({
'bloc': bloc_title,
'pepites': [(f'P{p[0]}', p[1].strip()) for p in pepites],
'source': title[:60],
})
except: pass
```

Afficher agents

print(f'=== AGENTS CANONIQUES: {len(agents_52)} ===')
for a in agents_52:
print(f' {a[\"id\"]:<12} | {a[\"name\"]:<30} | {a[\"type\"]:<30} | {a[\"function\"][:50]}')
print(f'\n=== BLOCS LOIS OBSIDIENNES: {len(all_blocs_lois)} ===')
for b in all_blocs_lois:
print(f'\n BLOC: {b[\"bloc\"]}')
for code, name in b['pepites']:
print(f' {code}: {name}')
" 2>&1)
 Waiting...
 === AGENTS CANONIQUES: 0 ===

BLOCS LOIS OBSIDIENNES: 0

 Allowed by auto mode classifier

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
```

\\BRODY_WORLD_INTAKE_PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162_2026051
2_211327\\normalized_md\\446D12773467_agents_

```
canoniques_446D12773467.md", "original_extension": ".json", "extraction_status": "TEXT_EXTRACTED_utf-8-sig",
```

"extractor": "plain-text", "content_usable": true, "source_original_sha256":
"446D1277346758A4228ABA1DD638C98C145CD8A00C78A2AF1B0BEDB4E8AE96BD", "text_sha256":
"C20D6EFD006B15AD1AF1C9A583B4FE4A139DCE7DC2513B81AE47B771F05FB292", "text_excerpt": "# BRODY
WORLD SOURCE INTAKE
READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n- source_original_path:

```
_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\446D12773467_agents_canon iques.json\n-
```

source_original_sha256: 446D1277346758A4228ABA1DD638C98C145CD8A00C78A2AF1B0BEDB4E8AE96BD\n-
original_extension:
.json\n- extraction_status: TEXT_EXTRACTED_utf-8-sig\n- extractor: plain-text\n- content_usable: true\n\n##
Boundary\n\n- readonly: true\n- memory_authority: false\n- memory_decision: false\n- decision_authority:
KX108_ONLY\n- kernel_binding: false\n- x108_merge: false\n- emits_act: false\n\n## Extracted text\n\n```text\n[\n
{\n \"id\": \"AGENT_001\",\n \"name\": \"Observateur\",\n \"type\": \"agent_obsidien_R&D\",\n
\"source\": \"src_branche149 — Branche 85 : Entités Cognitives / Agents Obsidia\",\n \"function\": \"Observer
l'état, les flux, les signaux faibles et les variations sans décider.\",\n \"kernel_boundary\": \"Ne décide
jamais. Remonte contexte et anomalies.\",\n \"linked\": [\n \"Bloc 13\",\n \"P137\",\n
\"P138\",\n \"OS3\",\n \"OS4\"\n]\n },\n {\n \"id\": \"AGENT_002\",\n \"name\":
\"Sentinelle\",\n \"type\": \"agent_obsidien_R&D\",\n \"source\": \"src_branche149 — Branche 85\",\n
\"function\": \"Surveiller les conditions critiques, seuils, incohérences, menaces et dérives.\",\n
\"kernel_boundary\": \"Peut alerter ou recommander HOLD/BLOCK, ne force jamais ALLOW.\",\n \"linked\": [\n
\"ADeLe\",\n \"OS-S\",\n \"Audit F\",\n \"Bloc 7\"\n]\n },\n {\n \"id\": \"AGENT_003\",\n
\"name\": \"Gardien\",\n \"type\": \"agent_obsidien_R&D / gardien de fond\",\n \"source\": \"src_branche149
— Branche 85 + checklist A24 Gardien des Métriques\",\n \"function\": \"Garder une frontière, une métrique, une
règle ou un invariant.\",\n \"kernel_boundary\": \"Surveille et signale. La décision souveraine reste
X-108.\",\n \"linked\": [\n \"A24\",\n \"T1-T12\",\n \"OS-S\",\n \"X-108\"\n]\n },\n
{\n \"id\": \"AGENT_004\",\n \"name\": \"Architecte\",\n \"type\": \"agent_obsidien_R&D\",\n

 Allowed by auto mode classifier

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
```

import json, re, sys
sys.stdout.reconfigure(encoding='utf-8')
agents_52 = []
lois_7 = []
with
open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',
encoding='utf-8') as f:
for line in f:
line = line.strip()

```
if not line: continue
try:
rec = json.loads(line)
title = rec.get('title','') or ''
text = rec.get('text','') or ''
if 'agents_canoniques.json' in title.lower():
```

Le text est un JSON stringifié, donc on parse le JSON wrapper

inner = json.loads(text)
excerpt = inner.get('text_excerpt','')

Le contenu réel est dans text_excerpt, après 'Extracted text'

m = re.search(r'## Extracted text\n+\`\`\`text\n(.*?)\`\`\`', excerpt, re.DOTALL)
if m:
raw_json = m.group(1).strip()
data = json.loads(raw_json)
for a in data:
agents_52.append({
'id': a.get('id',''),
'name': a.get('name',''),
for num, name in pepites:
lois_7.append({'code': f'P{num}', 'name': name.strip()})
except Exception as e:
pass
print(f'=== AGENTS CANONIQUES COMPLET: {len(agents_52)} ===')
for a in agents_52:
print(f' {a[\"id\"]:<12} | {a[\"name\"]:<25} | {a[\"type\"]:<35} | {a[\"function\"][:60]}')
print(f'\n=== LES 7 LOIS OBSIDIENNES (Bloc 09): {len(lois_7)} ===')
for l in lois_7:
print(f' {l[\"code\"]}: {l[\"name\"]}')
" 2>&1)
 Waiting...
 === AGENTS CANONIQUES COMPLET: 0 ===

LES 7 LOIS OBSIDIENNES (Bloc 09): 7

P88: Non-contradiction
P89: Complétude
P90: Cohérence
P91: Décidabilité
P92: Calculabilité (Turing)
P93: P ⊆ NP
P94: Réductibilité
 Allowed by auto mode classifier

● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
import json, re, sys
sys.stdout.reconfigure(encoding='utf-8')
for line in f:
line = line.strip()
if not line: continue
try:
rec = json.loads(line)
title = rec.get('title','') or ''
if 'agents_canoniques.json' in title.lower():
text = rec.get('text','')

text est déjà le JSON inner - voyons sa structure

print(f'TEXT IS JSON: {text[:50]}')

Le text commence par { donc c est un JSON string

inner = json.loads(text)
excerpt = inner.get('text_excerpt','')
print(f'excerpt len: {len(excerpt)}')
print(f'excerpt[:2000]:')
print(repr(excerpt[:2000]))
break
except Exception as e:
print(f'ERR: {e}')
" 2>&1)
 Waiting...
 TEXT IS JSON: {"id": "BRODY_GRAPHITI_READY_000668", "title": "44
excerpt len: 3000
excerpt[:2000]:
'# BRODY WORLD SOURCE INTAKE READONLY\n\n- label: PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_V162\n-
source_original_path:
`_world_sources\\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\\446D12773467_agents_canon iques.json\n-`
source_original_sha256: 446D1277346758A4228ABA1DD638C98C145CD8A00C78A2AF1B0BEDB4E8AE96BD\n-
original_extension:
.json\n- extraction_status: TEXT_EXTRACTED_utf-8-sig\n- extractor: plain-text\n- content_usable: true\n\n##
Boundary\n\n- readonly: true\n- memory_authority: false\n- memory_decision: false\n- decision_authority:
KX108_ONLY\n- kernel_binding: false\n- x108_merge: false\n- emits_act: false\n\n## Extracted text\n\n```text\n[\n
{\n "id": "AGENT_001",\n "name": "Observateur",\n "type": "agent_obsidien_R&D",\n "source":
"src_branche149 — Branche 85 : Entités Cognitives / Agents Obsidia",\n "function": "Observer l'état, les flux,
les signaux faibles et les variations sans décider.",\n "kernel_boundary": "Ne décide jamais. Remonte contexte
et anomalies.",\n "linked": [\n "Bloc 13",\n "P137",\n "P138",\n "OS3",\n "OS4"\n
]\n },\n {\n "id": "AGENT_002",\n "name": "Sentinelle",\n "type": "agent_obsidien_R&D",\n "source":
"src_branche149 — Branche 85",\n "function": "Surveiller les conditions critiques, seuils, incohérences, menaces
et dérives.",\n "kernel_boundary": "Peut alerter ou recommander HOLD/BLOCK, ne force jamais ALLOW.",\n
"linked": [\n "ADeLe",\n "OS-S",\n "Audit F",\n "Bloc 7"\n]\n },\n {\n "id":
"AGENT_003",\n "name": "Gardien",\n "type": "agent_obsidien_R&D / gardien de fond",\n "source":
"src_branche149 — Branche 85 + checklist A24 Gardien des Métriques",\n "function": "Garder une frontière, une
métrique, une règle ou un invariant.",\n "kernel_boundary": "Surveille et signale. La décision souveraine reste
X-108.",\n "linked": [\n "A24",\n "T1-T12",\n "OS-S",\n "X-108"\n]\n },\n {\n "id":
"AGENT_004",\n "name": "Architecte",\n "type": "agent_obsidien'
 Allowed by auto mode classifier
`● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "`
import json, re, sys
sys.stdout.reconfigure(encoding='utf-8')
agents_52 = []
with
open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',
encoding='utf-8') as f:
for line in f:
line = line.strip()
if not line: continue
try:
rec = json.loads(line)
title = rec.get('title','') or ''
if 'agents_canoniques.json' in title.lower():
text = rec.get('text','')
inner = json.loads(text)
excerpt = inner.get('text_excerpt','')

Le JSON est dans excerpt après 'Extracted text\n\`\`\`text\n'

Les guillemets dans l'excerpt sont des vrais guillemets unicode

m = re.search(r'## Extracted text\n+\x60\x60\x60text\n(.*)', excerpt, re.DOTALL)
print(f' {a[\"id\"]:<12} | {a[\"name\"]:<25} | {a[\"type\"]:<30} | {a[\"function\"][:50]}')
" 2>&1)
 Waiting...
 Traceback (most recent call last):
File "<string>", line 29, in <module>
data = json.loads(raw)
File "C:\Users\User\AppData\Local\Programs\Python\Python313\Lib\json__init __.py", line 346, in loads
return _default_decoder.decode(s)
~~~~~~~~~~~~~~~~~~~~~~~~^^^
File "C:\Users\User\AppData\Local\Programs\Python\Python313\Lib\json\decode r.py", line 345, in decode
obj, end = self.raw_decode(s, idx=_w(s, 0).end())
~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\User\AppData\Local\Programs\Python\Python313\Lib\json\decode r.py", line 361, in raw_decode
obj, end = self.scan_once(s, idx)
~~~~~~~~~~~~~~~~^^^^^^^^
json.decoder.JSONDecodeError: Unterminated string starting at: line 78 column 24 (char 2323)
RAW LEN: 2365
RAW[:200]: [
{
"id": "AGENT_001",
"name": "Observateur",
"type": "agent_obsidien_R&D",
"source": "src_branche149 — Branche 85 : Entités Cognitives / Agents Obsidia",
"function": "Observer l

AGENTS CANONIQUES COMPLET: 0

 Allowed by auto mode classifier

● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
import json, re, sys
sys.stdout.reconfigure(encoding='utf-8')
~~~~~~~~~~~~~~~~

L'excerpt est limité à 3000 chars. Le JSON est tronqué.

agents_52 = []

On va lire tout le JSONL et chercher tous les records liés aux agents

```
agent_names = set()
with
open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',
encoding='utf-8') as f:
all_text = f.read()
```

Cherche tous les AGENT_xxx

agent_matches =
re.findall(r'\"id\"\s*:\s*\"(AGENT_\d+)\"[^}]*\"name\"\s*:\s*\"([^\"]+)\"[^}]*\"type\"\s*:\s*\"([^\"]+)\"', all_text)
for a_id, a_name, a_type in agent_matches:
key = a_id
if key not in agent_names:
agent_names.add(key)
agents_52.append({'id': a_id, 'name': a_name, 'type': a_type})
print(f'=== AGENTS_XXX trouvés: {len(agents_52)} ===')
for a in sorted(agents_52, key=lambda x: x['id']):
print(f' {a[\"id\"]:<12} | {a[\"name\"]:<25} | {a[\"type\"]}')
" 2>&1)
 Waiting...
 === AGENTS_XXX trouvés: 0 ===
 Allowed by auto mode classifier

● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
import json, re, sys
sys.stdout.reconfigure(encoding='utf-8')

Debug: le JSONL line par line - les guillemets dans le JSON inner sont echappés

Essayons de lire les records avec AGENT_001 dans le texte

```
count = 0
with
open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',
encoding='utf-8') as f:
for line in f:
if 'AGENT_001' in line:
count += 1
if count == 1:
```

Affiche un extrait de la ligne autour de AGENT_001

```
idx = line.find('AGENT_001')
print(f'Found AGENT_001 at idx {idx}')
print(repr(line[max(0,idx-50):idx+200]))
print(f'Total lines with AGENT_001: {count}')
```

Essai regex avec guillemets échappés

with

open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',

encoding='utf-8') as f:

all_content = f.read()

Cherche avec backslash escaping

```
matches = re.findall(r'\\\\\"id\\\\\":\\\\s*\\\\\"(AGENT_\\d+)\\\\\"', all_content)
print(f'Escaped matches: {len(matches)}')
```

Cherche simplement le pattern

matches2 = re.findall(r'AGENT_(\d+)', all_content)
print(f'All AGENT_xxx occurrences: {len(matches2)} (uniq: {len(set(matches2))})')
for n in sorted(set(matches2)):
print(f' AGENT_{n}')
" 2>&1 | head -60)
 Waiting...
 Found AGENT_001 at idx 2025
'ct: false\\\\n\\\\n## Extracted text\\\\n\\\\n```text\\\\n# AGENT_001 — Observateur\\\\n\\\\n**Type :**
agent_obsidien_R&D\\\\n**Source :** src_branche149 — Branche 85 : Entités Cognitives / Agents Obsidia\\\\n\\\\n##
Fonction\\\\nObserver l’état, les flux, les signaux fa'
Total lines with AGENT_001: 6
Escaped matches: 4
All AGENT_xxx occurrences: 84 (uniq: 6)
AGENT_001
AGENT_002
AGENT_003
AGENT_004
AGENT_6
AGENT_6743
 Allowed by auto mode classifier

● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
import json, re, sys
sys.stdout.reconfigure(encoding='utf-8')
agents_found = {}
with
open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',
encoding='utf-8') as f:
for line in f:
line = line.strip()
if not line: continue
try:
rec = json.loads(line)
title = rec.get('title','') or ''
text = rec.get('text','') or ''

Records individuels AGENT_xxx

```
m = re.match(r'AGENT_(\d+)\s*[—-]\s*(.+)', title)
if m:
num = int(m.group(1))
name = m.group(2).strip()
agents_found[num] = {'name': name, 'source': 'individual_record', 'title': title}
```

Dans le text brut: cherche ## AGENT_xxx ou # AGENT_xxx (échappés)

Le text contient du JSON avec \\n donc les vrais newlines sont \\\\n

Cherche pattern AGENT_\\d+ — Name dans le text

```
agent_headers = re.findall(r'AGENT_(\d+)\s*(?:\\\\n|\\n|\n)\s*\\\\n\\\\n\*\*Type', text)
if agent_headers:
```

Cherche aussi dans les records du graphiti index (not JSONL)

idx_file =
'_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_20260512
_224854/graphiti_readonly_index_v2.json'
with open(idx_file, encoding='utf-8') as f:
idx = json.load(f)
records = idx.get('records', [])
print(f'Index records count: {len(records)}')
for rec in records:
title = rec.get('title','') or ''
m = re.match(r'AGENT_(\d+)\s*[—-]\s*(.+)', title)
if m:
num = int(m.group(1))
name = m.group(2).strip()
if num not in agents_found:
agents_found[num] = {'name': name, 'source': 'graphiti_index', 'title': title}
print(f'=== AGENTS trouvés: {len(agents_found)} ===')
for num in sorted(agents_found.keys()):
a = agents_found[num]
print(f' AGENT_{num:03d} | {a[\"name\"]:<40} | {a[\"source\"]}')
" 2>&1)
 Waiting…
 Index records count: 3267=== AGENTS trouvés: 0 ===
 Allowed by auto mode classifier

● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
import json, re, sys
sys.stdout.reconfigure(encoding='utf-8')

L'excerpt de agents_canoniques.json est tronqué à 3000 chars

Le text field dans le JSONL contient: JSON(JSON(markdown(JSON)))

Regardons le FULL text field (pas excerpt) - peut-être qu il y a plus dans text

```
with
open('_graphiti_readonly_indexes/GRAPHITI_READONLY_INDEX_V2_FUSION_202
60512_224854/graphiti_readonly_records_v2.jsonl',
encoding='utf-8') as f:
for line in f:
line = line.strip()
if not line or 'agents_canoniques.json' not in line: continue
try:
rec = json.loads(line)
title = rec.get('title','') or ''
if 'agents_canoniques.json' not in title.lower(): continue
```

Cherche dans le payload

```
payload = rec.get('payload', {})
print(f'payload keys: {list(payload.keys()) if isinstance(payload,dict) else type(payload)}')
ids = re.findall(r'\"id\": \"(AGENT_\d+)\"', excerpt)
print(f'From excerpt: ids={ids}, names={names}')
```

Also look for the source path

src = inner.get('source_original_path','')
print(f'source_original_path: {src}')
norm = inner.get('normalized_md_path','')
print(f'normalized_md_path: {norm}')
break
except Exception as e:
import traceback; traceback.print_exc()
" 2>&1)
 Waiting...
 payload keys: ['id', 'title', 'source_original_path', 'normalized_md_path', 'original_extension', 'extraction_status', 'extractor', 'content_usable', 'source_original_sha256', 'text_sha256', 'text_excerpt', 'tags', 'readonly', 'memory_authority', 'memory_decision', 'allowed_to_decide', 'emits_act', 'kernel_binding', 'x108_merge', 'kernel_mutation', 'x108_mutation', 'decision_authority', 'scope', 'taxonomy_v164d']
From excerpt: ids=['AGENT_001', 'AGENT_002', 'AGENT_003', 'AGENT_004'], names=['Observateur', 'Sentinelle', 'Gardien', 'Architecte', 'Human Operator', 'Governance Layer']
source_original_path:

```
_world_sources\PROJECT_MEMORY_CLEAN_SOURCE_NO_INDEX_20260512_211234\446D12773467_agents_canoniq ues.json
```

normalized_md_path:
C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida
te_world_intake\BRODY_WORLD_INTAKE_PROJECT_M

```
EMORY_CLEAN_SOURCE_NO_INDEX_V162_20260512_211327\normalized_md\446D12773467_agents_canoniques_4 46D12773467.md
```

 Allowed by auto mode classifier

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
```

import csv, re, sys, json
sys.stdout.reconfigure(encoding='utf-8')

1. Chercher dans le CSV les agents avec nom lisible (pas juste fichiers)

```
agent_names_csv = []
loi_names_csv = []
with open('.local_audits/BRODY_SEMANTIC_COMPONENT_REGISTRY_V0_20260618_1703
07/brody_semantic_component_registry.csv',
encoding='utf-8-sig') as f:
for row in csv.DictReader(f):
cat = row.get('category','')
name = row.get('name','') or ''
value = row.get('value','') or ''
origin = row.get('origin','') or ''
```

Agents: cherche patterns explicites

```
if cat == 'AGENT':
```

Cherche agent name dans name ou value

```
if re.match(r'AGENT_\d+', name) or re.search(r'AGENT_\d+\s*[—-]\s*\w', name):
agent_names_csv.append({'name': name[:80], 'value': value[:60], 'origin': origin})
elif re.match(r'ROLE_\d+', name):
agent_names_csv.append({'name': name[:80], 'value': value[:60], 'origin': origin})
elif value and re.match(r'(AGENT|ROLE)_\d+', value):
agent_names_csv.append({'name': value[:80], 'value': name[:60], 'origin': origin})
```

Lois

if cat == 'LOI':

Cherche loi nommée (pas juste fichier)

```
for v in [name, value]:
if re.search(r'(?:Loi|LOI)\s+(?:N°?\s*)?\d+', v) or re.search(r'loi_\d+|LOI_\d+', v, re.I):
loi_names_csv.append({'name': v[:100], 'origin': origin})
print(f'AGENTS explicites CSV: {len(agent_names_csv)}')
seen = set()
for a in agent_names_csv[:60]:
key = a['name'][:30]
if key not in seen:
seen.add(key)
print(f' {a[\"name\"][:80]} | {a[\"value\"][:40]} | {a[\"origin\"]}')
print(f'\nLOIS nommées CSV: {len(loi_names_csv)}')
seen = set()
for l in loi_names_csv[:30]:
key = l['name'][:30].lower()
if key not in seen:
seen.add(key)
print(f' {l[\"name\"][:100]}')
```

2. Chercher le fichier normalized_md dans obsidia-engine-candidate (s'il existe)

norm_path =
print(f' {a.get(\"id\",\"\")} | {a.get(\"name\",\"\")}')
else:
print(f' world_src also not found: {world_src}')
" 2>&1)
 Waiting...
 AGENTS explicites CSV: 213
AGENT_001 | | json_value
AGENT_002 | | json_value
AGENT_003 | | json_value
AGENT_004 | | json_value
AGENT_001 — Observateur | ### AGENT_001 — Observateur | heading
AGENT_002 — Sentinelle | ### AGENT_002 — Sentinelle | heading
"linked": [
"ADeLe",
"OS-S",
"Audit F",
"Bloc 7"
]
},
{
"id": "AGENT_003",
"name": "Gardien",
"type": "agent_obsidien_R&D / gardien de fond",
"source": "src_branche149 — Branche 85 + checklist A24 Gardien des Métriques",
"function": "Garder une frontière, une métrique, une règle ou un invariant.",
"kernel_boundary": "Surveille et signale. La décision souveraine reste X-108.",
"linked": [
"A24",
"T1-T12",
"OS-S",
"X-108"
]
},
{
"id": "AGENT_004",
"name": "Architecte",
"type": "agent_obsidien_R&D",
"source": "src_branche149 — Branche 85",
"function": "Assembler, relier, cadrer et proposer des structures systémiques.",
"kernel_boundary": "Ne modifie pas le noyau sans audit/gate.",
"linked": [
"Bloc 13",
"Bloc 15",
"C4",
"G4"
]
},
{
"id": "ROLE_005",
"name": "Human Operator",

"type": "rôle_responsabilité",
"source": "src_40spec — Spec 26 Liability Boundary & Responsibility Graph",
"function": "Porte l’intention, le veto et l’autorisation finale.",
"kernel_boundary": "Responsable de l’intention et de l’autorisation, pas d’une exécution technique non autorisée.",
"linked": [
"Spec 26",
"Human Veto",
"Audit C"
]
},
{
"id": "ROLE_006",
"name": "Governance Layer",
"type": "rôle_système",
"source": "src_40spec — Spec 26",
"function": "Appliquer les policies et gouverner les frontières d’action.",
"kernel_boundary": "Ne remplace pas OS3/X-108, applique le contrat.",
"linked": [
"Spec 11",
"Spec 26",
"ADeLe"
]
},
{
"id": "ROLE_007",
"name": "OS3 Audit Layer",
"type": "rôle_système",
"source": "src_40spec — Spec 26",
"function": "Porter la trace, l’audit, le replay et la preuve.",
"kernel_boundary": "Trace sans narration décisionnelle.",
"linked": [
"OS3",
"Spec 1",
"Spec 2",
"Spec 5",
"Spec 7"
]
},
{
"id": "ROLE_008",
"name": "Agents / Tools",
"type": "rôle_outils",
"source": "src_40spec — Spec 26",
"function": "Exécuter ou assister dans leur scope déclaré.",
"kernel_boundary": "Jamais décisionnaires.",
"linked": [
"Spec 9",
"Spec 10",
"MCP",
"Adapters"
]
},
{

"id": "GUARDIAN_009",
"name": "Sigma Verifier",
"type": "gardien_de_fond",
"source": "checklist domaine 5",
"function": "Vérifier l’orchestration Sigma.",
"kernel_boundary": "Produit un PASS/FAIL de vérification.",
"linked": [
"Tests Python",
"Audit D",
"G3"
]
},
{
"id": "GUARDIAN_010",
"name": "Merkle Verifier",
"type": "gardien_de_fond",
"source": "checklist domaine 5",
"function": "Vérifier immutabilité et intégrité Merkle.",
"kernel_boundary": "Vérifie la trace, ne décide pas.",
"linked": [
"Merkle",
},
{
"id": "GUARDIAN_011",
"name": "RFC3161 Verifier",
"type": "gardien_de_fond",
"source": "checklist domaine 5",
"function": "Vérifier preuve temporelle légale.",
"kernel_boundary": "Certifie le temps, ne décide pas.",
"linked": [
"RFC3161",
"Spec 17",
"Audit D"
]
},
{
"id": "GUARDIAN_012",
"name": "TLA Verifier",
"type": "gardien_de_fond",
"source": "checklist domaine 5",
"function": "Vérifier SafetyX108 / NoDeadlock / NoLivelock.",
"kernel_boundary": "Modèle formel, pas runtime décisionnel.",
"lin
 Allowed by auto mode classifier
06:21 PM claude-sonnet-4-6
● Le fichier agents canoniques est accessible. Lecture complète pour extraire les 52 agents.
● Bash(python3 -c "
import json, re, sys
sys.stdout.reconfigure(encoding='utf-8')
norm_path =
r'C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candi
date_world_intake\BRODY_WORLD_INTAKE_PROJECT_MEM
ORY_CLEAN_SOURCE_NO_INDEX_V162_20260512_211327\normalized_md\446D12773467_agents_canoniques_446 D12773467.md'
with open(norm_path, encoding='utf-8') as f:
content = f.read()

Extraire le JSON depuis le backtick block

```
m = re.search(r'\x60\x60\x60text\n(.*?)\x60\x60\x60', content, re.DOTALL)
if m:
raw = m.group(1).strip()
```

Parser le JSON (peut être tronqué)

```
try:
data = json.loads(raw)
print(f'Parsed {len(data)} agents')
for a in data:
print(f' {a.get(\"id\",\"?\")} | {a.get(\"name\",\"?\")} | {a.get(\"type\",\"?\")} |
{str(a.get(\"function\",\"\"))[:60]}')
except json.JSONDecodeError as e:
print(f'JSON incomplete ({e}), extracting with regex...')
```

Regex fallback

entries = re.findall(r'\"id\": \"([^\"]+)\"[^}]*?\"name\": \"([^\"]+)\"[^}]*?\"type\": \"([^\"]+)\"', raw,
re.DOTALL)
print(f'Regex: {len(entries)} entries')
for a_id, a_name, a_type in entries:
print(f' {a_id} | {a_name} | {a_type}')
else:
print('No backtick block found')
print(f'Total chars: {len(content)}')
" 2>&1)
 Waiting...
 Parsed 17 agents
AGENT_001 | Observateur | agent_obsidien_R&D | Observer l'état, les flux, les signaux faibles et les variat
AGENT_002 | Sentinelle | agent_obsidien_R&D | Surveiller les conditions critiques, seuils, incohérences, m
AGENT_003 | Gardien | agent_obsidien_R&D / gardien de fond | Garder une frontière, une métrique, une règle ou un
invarian
AGENT_004 | Architecte | agent_obsidien_R&D | Assembler, relier, cadrer et proposer des structures systémi
ROLE_005 | Human Operator | rôle_responsabilité | Porte l'intention, le veto et l'autorisation finale.
ROLE_006 | Governance Layer | rôle_système | Appliquer les policies et gouverner les frontières d'action.
ROLE_007 | OS3 Audit Layer | rôle_système | Porter la trace, l'audit, le replay et la preuve.
ROLE_008 | Agents / Tools | rôle_outils | Exécuter ou assister dans leur scope déclaré.
GUARDIAN_009 | Sigma Verifier | gardien_de_fond | Vérifier l'orchestration Sigma.
GUARDIAN_010 | Merkle Verifier | gardien_de_fond | Vérifier immutabilité et intégrité Merkle.
GUARDIAN_011 | RFC3161 Verifier | gardien_de_fond | Vérifier preuve temporelle légale.
GUARDIAN_012 | TLA Verifier | gardien_de_fond | Vérifier SafetyX108 / NoDeadlock / NoLivelock.
ROLE_013 | Orchestrator | rôle_runtime | Coordonner le flux complet entre kernel, adapters, agents et
ROLE_014 | Adapter MCP | adapter_outil | Mapper outils/scopes vers policy gate.
ROLE_015 | Cortex | rôle_couche | Couche de traitement/contextualisation mentionnée dans le sc
GUARDIAN_016 | GPS — Gardien / Guide / Alerteur | périphérie_sigma | Guider la cohérence, alerter, orienter.
GUARDIAN_017 | ADeLe | gouvernance_alignement | Contrôler l'alignement critique Banking / Trading / Aviation
 Allowed by auto mode classifier

● Bash(python3 -c "
import json, re, sys, os
from pathlib import Path
r'C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_ REMOTE_A5F21C6B',
r'C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candi date',
]
for root in search_roots:
root_path = Path(root)
if not root_path.exists():
print(f'NOT FOUND: {root}')
continue
for fp in root_path.rglob('agents_canoniques*.json'):
print(f'FOUND: {fp} ({fp.stat().st_size} bytes)')
try:
with open(fp, encoding='utf-8-sig') as f:
data = json.load(f)
if isinstance(data, list):
print(f' -> {len(data)} agents')
for a in data:
print(f' {a.get(\"id\",\"?\")} | {a.get(\"name\",\"?\")} | {a.get(\"type\",\"?\")}')
except Exception as e:
print(f' err: {e}')
" 2>&1)

 Waiting...

 FOUND: C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\periphery\OBSIDIA_V4_STR

```
UCTURED_FULL\01_REGISTRES_JSON\agents_canoniques.json (6210 bytes)
```

-> 17 agents

AGENT_001 | Observateur | agent_obsidien_R&D

AGENT_002 | Sentinelle | agent_obsidien_R&D

AGENT_003 | Gardien | agent_obsidien_R&D / gardien de fond

AGENT_004 | Architecte | agent_obsidien_R&D

ROLE_005 | Human Operator | rôle_responsabilité

ROLE_006 | Governance Layer | rôle_système

ROLE_007 | OS3 Audit Layer | rôle_système

ROLE_008 | Agents / Tools | rôle_outils

GUARDIAN_009 | Sigma Verifier | gardien_de_fond

GUARDIAN_010 | Merkle Verifier | gardien_de_fond

GUARDIAN_011 | RFC3161 Verifier | gardien_de_fond

GUARDIAN_012 | TLA Verifier | gardien_de_fond

ROLE_013 | Orchestrator | rôle_runtime

ROLE_014 | Adapter MCP | adapter_outil

ROLE_015 | Cortex | rôle_couche

GUARDIAN_016 | GPS — Gardien / Guide / Alerteur | périphérie_sigma

GUARDIAN_017 | ADeLe | gouvernance_alignement

FOUND: C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\ci_recheck_optional_publ

```
ication\periphery\OBSIDIA_V4_STRUCTURED_FULL\01_REGISTRES_JSON\agents_canoniques.json (6210 bytes)
```

-> 17 agents

AGENT_001 | Observateur | agent_obsidien_R&D

AGENT_002 | Sentinelle | agent_obsidien_R&D

AGENT_003 | Gardien | agent_obsidien_R&D / gardien de fond

AGENT_004 | Architecte | agent_obsidien_R&D

ROLE_005 | Human Operator | rôle_responsabilité

ROLE_006 | Governance Layer | rôle_système

ROLE_007 | OS3 Audit Layer | rôle_système

ROLE_008 | Agents / Tools | rôle_outils

GUARDIAN_009 | Sigma Verifier | gardien_de_fond

GUARDIAN_010 | Merkle Verifier | gardien_de_fond

GUARDIAN_011 | RFC3161 Verifier | gardien_de_fond

GUARDIAN_012 | TLA Verifier | gardien_de_fond

ROLE_013 | Orchestrator | rôle_runtime

ROLE_014 | Adapter MCP | adapter_outil

ROLE_015 | Cortex | rôle_couche

GUARDIAN_016 | GPS — Gardien / Guide / Alerteur | périphérie_sigma

GUARDIAN_017 | ADeLe | gouvernance_alignement

FOUND: C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida
te\zip1_sandbox_mutable\ZIP1_X108_MUTA

```
BLE_20260508_183610\periphery\v4\source\OBSIDIA_V4_STRUCTURED_FULL\01_REGISTRES_JSON\agents_can oniques.json (6210
```

bytes)

-> 17 agents

AGENT_001 | Observateur | agent_obsidien_R&D

AGENT_002 | Sentinelle | agent_obsidien_R&D

AGENT_003 | Gardien | agent_obsidien_R&D / gardien de fond

AGENT_004 | Architecte | agent_obsidien_R&D

ROLE_005 | Human Operator | rôle_responsabilité

ROLE_006 | Governance Layer | rôle_système

ROLE_007 | OS3 Audit Layer | rôle_système

ROLE_008 | Agents / Tools | rôle_outils

GUARDIAN_009 | Sigma Verifier | gardien_de_fond
GUARDIAN_010 | Merkle Verifier | gardien_de_fond
GUARDIAN_011 | RFC3161 Verifier | gardien_de_fond
GUARDIAN_012 | TLA Verifier | gardien_de_fond
ROLE_013 | Orchestrator | rôle_runtime
ROLE_014 | Adapter MCP | adapter_outil
ROLE_015 | Cortex | rôle_couche
GUARDIAN_016 | GPS — Gardien / Guide / Alerteur | périphérie_sigma
GUARDIAN_017 | ADeLe | gouvernance_alignement
FOUND: C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida
te\OBSIDIA_V4_REGROUPEMENTS_V43_SOUS_D
`OSSIERS\OBSIDIA_V4_STRUCTURED_FULL\01_REGISTRES_JSON\agents_canoniques.json (6210 bytes)`
-> 17 agents
AGENT_001 | Observateur | agent_obsidien_R&D
AGENT_002 | Sentinelle | agent_obsidien_R&D
AGENT_003 | Gardien | agent_obsidien_R&D / gardien de fond
AGENT_004 | Architecte | agent_obsidien_R&D
ROLE_005 | Human Operator | rôle_responsabilité
ROLE_006 | Governance Layer | rôle_système
ROLE_007 | OS3 Audit Layer | rôle_système
ROLE_008 | Agents / Tools | rôle_outils
GUARDIAN_009 | Sigma Verifier | gardien_de_fond
GUARDIAN_010 | Merkle Verifier | gardien_de_fond
GUARDIAN_011 | RFC3161 Verifier | gardien_de_fond
GUARDIAN_012 | TLA Verifier | gardien_de_fond
ROLE_013 | Orchestrator | rôle_runtime
ROLE_014 | Adapter MCP | adapter_outil
ROLE_015 | Cortex | rôle_couche
GUARDIAN_016 | GPS — Gardien / Guide / Alerteur | périphérie_sigma
GUARDIAN_017 | ADeLe | gouvernance_alignement
FOUND: C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida
te\freezes\OBSIDIA_V4_PERMANENT_ATTACH
`_V1_20260508_212110\v4\source\OBSIDIA_V4_STRUCTURED_FULL\01_REGISTRES_JSON\agents_canoniques.js on (6210 bytes)`
-> 17 agents
AGENT_001 | Observateur | agent_obsidien_R&D
AGENT_002 | Sentinelle | agent_obsidien_R&D
AGENT_003 | Gardien | agent_obsidien_R&D / gardien de fond
AGENT_004 | Architecte | agent_obsidien_R&D
ROLE_005 | Human Operator | rôle_responsabilité
ROLE_006 | Governance Layer | rôle_système
ROLE_007 | OS3 Audit Layer | rôle_système
ROLE_008 | Agents / Tools | rôle_outils
GUARDIAN_009 | Sigma Verifier | gardien_de_fond
GUARDIAN_010 | Merkle Verifier | gardien_de_fond
GUARDIAN_011 | RFC3161 Verifier | gardien_de_fond
GUARDIAN_012 | TLA Verifier | gardien_de_fond
ROLE_013 | Orchestrator | rôle_runtime
ROLE_014 | Adapter MCP | adapter_outil
ROLE_015 | Cortex | rôle_couche
GUARDIAN_016 | GPS — Gardien / Guide / Alerteur | périphérie_sigma
GUARDIAN_017 | ADeLe | gouvernance_alignement
 Allowed by auto mode classifier
`● Bash(python3 -c "`
import json, re, sys, os
from pathlib import Path
sys.stdout.reconfigure(encoding='utf-8')

```
ROOT = Path(r'C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-pr oofs_REMOTE_A5F21C6B')
CANDIDATE = Path(r'C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine- candidate')
```

Cherche les registres qui listent les 52 agents

FULL_RUNTIME_KERNEL_BOUND_V2_20260508_195252\pack\zip2_runtime_source\10_AGENTS_52\agents_52.re gistry.json (75086)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\staging\runtime_candidates\OBSIDIA_ENGINE_

FULL_RUNTIME_KERNEL_BOUND_V2_20260508_195252\pack\zip2_runtime_source\ 11_AGENTS_RUNTIME_CONTRACTS\AGENTS_52_ALLOWED

_OUTPUT_SCHEMA.json (772)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\staging\runtime_candidates\OBSIDIA_ENGINE_

FULL_RUNTIME_KERNEL_BOUND_V2_20260508_195252\pack\zip2_runtime_source\ 11_AGENTS_RUNTIME_CONTRACTS\AGENTS_52_RUNTIME

_BOUNDARY.json (507)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\staging\runtime_candidates\OBSIDIA_ENGINE_

FULL_RUNTIME_KERNEL_BOUND_V2_20260508_195252\pack\zip2_runtime_source\19_REGISTRES_JSON\agents_ 52.registry.json

(73005)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\reports\zip2_52_agents_real_scan_20260508_

180008\agents_52_files.json (36766)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_

P2PLUS_V1_FREEZE_CANDIDATE\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1\10_AGENTS_52\age nts_52.index.json

(36571)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_

P2PLUS_V1_FREEZE_CANDIDATE\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1\10_AGENTS_52\age nts_52.registry.json

(75086)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_

P2PLUS_V1_FREEZE_CANDIDATE\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P 2PLUS_V1\11_AGENTS_RUNTIME_CONTRACTS\AGENTS_5

2_ALLOWED_OUTPUT_SCHEMA.json (772)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_

P2PLUS_V1_FREEZE_CANDIDATE\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P 2PLUS_V1\11_AGENTS_RUNTIME_CONTRACTS\AGENTS_5

2_RUNTIME_BOUNDARY.json (507)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_

P2PLUS_V1_FREEZE_CANDIDATE\OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P 2PLUS_V1\19_REGISTRES_JSON\agents_52.registry

.json (73005)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\freezes\ZIP2_52_AGENTS_IR_SHAZAM_TREE34_TO

_ZIP1_X108_REAL_READONLY_V1_REPAIR_20260508_181511\10_AGENTS_52\agents_52.index.json (36571)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\freezes\ZIP2_52_AGENTS_IR_SHAZAM_TREE34_TO

_ZIP1_X108_REAL_READONLY_V1_REPAIR_20260508_181511\10_AGENTS_52\agents_52.registry.json (75086)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\freezes\ZIP2_52_AGENTS_IR_SHAZAM_TREE34_TO

_ZIP1_X108_REAL_READONLY_V1_REPAIR_20260508_181511\11_AGENTS_RUNTIME_CONTRACTS\AGENTS_52_ALLOWE D_OUTPUT_SCHEMA.json

(772)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\freezes\ZIP2_52_AGENTS_IR_SHAZAM_TREE34_TO

_ZIP1_X108_REAL_READONLY_V1_REPAIR_20260508_181511\11_AGENTS_RUNTIME_CONTRACTS\AGENTS_52_RUNTIM E_BOUNDARY.json

(507)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\freezes\ZIP2_52_AGENTS_IR_SHAZAM_TREE34_TO

_ZIP1_X108_REAL_READONLY_V1_20260508_181237\10_AGENTS_52\agents_52.index.json (36571)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\freezes\ZIP2_52_AGENTS_IR_SHAZAM_TREE34_TO

_ZIP1_X108_REAL_READONLY_V1_20260508_181237\10_AGENTS_52\agents_52.registry.json (75086)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\freezes\ZIP2_52_AGENTS_IR_SHAZAM_TREE34_TO

```
_ZIP1_X108_REAL_READONLY_V1_20260508_181237\11_AGENTS_RUNTIME_CONTRACTS\AGENTS_52_ALLOWED_OUTPU T_SCHEMA.json (772)
```

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\freezes\ZIP2_52_AGENTS_IR_SHAZAM_TREE34_TO

```
_ZIP1_X108_REAL_READONLY_V1_20260508_181237\11_AGENTS_RUNTIME_CONTRACTS\AGENTS_52_RUNTIME_BOUND ARY.json (507)
```

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\candidate_packs\OBSIDIA_ENGINE_FULL_RUNTIM

```
E_KERNEL_BOUND_V2_20260508_195252\zip2_runtime_source\10_AGENTS_52\agents_52.index.json (36571)
```

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\candidate_packs\OBSIDIA_ENGINE_FULL_RUNTIM

```
E_KERNEL_BOUND_V2_20260508_195252\zip2_runtime_source\10_AGENTS_52\agents_52.registry.json (75086)
```

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\candidate_packs\OBSIDIA_ENGINE_FULL_RUNTIM

E_KERNEL_BOUND_V2_20260508_195252\zip2_runtime_source\11_AGENTS_RUNTIM E_CONTRACTS\AGENTS_52_ALLOWED_OUTPUT_SCHEMA.j

son (772)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\candidate_packs\OBSIDIA_ENGINE_FULL_RUNTIM

```
E_KERNEL_BOUND_V2_20260508_195252\zip2_runtime_source\11_AGENTS_RUNTIME_CONTRACTS\AGENTS_52_RUN TIME_BOUNDARY.json
```

(507)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\candidate_packs\OBSIDIA_ENGINE_FULL_RUNTIM

```
E_KERNEL_BOUND_V2_20260508_195252\zip2_runtime_source\19_REGISTRES_JSON\agents_52.registry.json (73005)
```

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\candidate_packs\OBSIDIA_ENGINE_CANDIDATE_Z

```
IP2_IR_SHAZAM_TREE34_AGENTS52_X108_READONLY_V1_20260508_181845\validated_freeze\10_AGENTS_52\ag ents_52.index.json
```

(36571)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\candidate_packs\OBSIDIA_ENGINE_CANDIDATE_Z

IP2_IR_SHAZAM_TREE34_AGENTS52_X108_READONLY_V1_20260508_181845\validat ed_freeze\10_AGENTS_52\agents_52.registry.jso

n (75086)

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\candidate_packs\OBSIDIA_ENGINE_CANDIDATE_Z

IP2_IR_SHAZAM_TREE34_AGENTS52_X108_READONLY_V1_20260508_181845\validat ed_freeze\11_AGENTS_RUNTIME_CONTRACTS\AGENTS_

```
52_ALLOWED_OUTPUT_SCHEMA.json (772)
```

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-engine-candida te\candidate_packs\OBSIDIA_ENGINE_CANDIDATE_Z

IP2_IR_SHAZAM_TREE34_AGENTS52_X108_READONLY_V1_20260508_181845\validat ed_freeze\11_AGENTS_RUNTIME_CONTRACTS\AGENTS_

```
52_RUNTIME_BOUNDARY.json (507)
```

FICHIERS GARDIENS_T (T01-T12)

```
05_GARDIEN_T01__Gardien_de_fond__Reduction_Incertitude_Audit_coherence.md
05_GARDIEN_T07__Gardien_de_fond__Resilience_Test_memoire_fractale.md
05_GARDIEN_T04__Gardien_de_fond__Consensus_Distribue_Resonance_semantique.md
05_GARDIEN_T12__Gardien_de_fond__Complexite_Irreductible_Transparence_friction.md
05_GARDIEN_T09__Gardien_de_fond__Creativite_Mode_reflexe.md
05_GARDIEN_T06__Gardien_de_fond__Selection_Naturelle_Alignement_ethique.md
05_GARDIEN_T10__Gardien_de_fond__Innovation_Continue_Integrite_cognitive.md
05_GARDIEN_T02__Gardien_de_fond__Conscience_Distribuee_Validation_reciprocite.md
05_GARDIEN_T03__Gardien_de_fond__Gouvernance_Decentralisee_Friction_energetique.md
05_GARDIEN_T08__Gardien_de_fond__Auto_Organisation_Apprentissage_inverse.md
05_GARDIEN_T07__Gardien_de_fond__Resilience_Test_memoire_fractale.md
05_GARDIEN_T11__Gardien_de_fond__Generation_Aleatoire_Frugalite_bio_inspiree.md
05_GARDIEN_T01__Gardien_de_fond__Reduction_Incertitude_Audit_coherence.md
05_GARDIEN_T07__Gardien_de_fond__Resilience_Test_memoire_fractale.md
05_GARDIEN_T04__Gardien_de_fond__Consensus_Distribue_Resonance_semantique.md
05_GARDIEN_T12__Gardien_de_fond__Complexite_Irreductible_Transparence_friction.md
05_GARDIEN_T09__Gardien_de_fond__Creativite_Mode_reflexe.md
05_GARDIEN_T06__Gardien_de_fond__Selection_Naturelle_Alignement_ethique.md
05_GARDIEN_T10__Gardien_de_fond__Innovation_Continue_Integrite_cognitive.md
05_GARDIEN_T02__Gardien_de_fond__Conscience_Distribuee_Validation_reciprocite.md
05_GARDIEN_T03__Gardien_de_fond__Gouvernance_Decentralisee_Friction_energetique.md
05_GARDIEN_T08__Gardien_de_fond__Auto_Organisation_Apprentissage_inverse.md
05_GARDIEN_T11__Gardien_de_fond__Generation_Aleatoire_Frugalite_bio_inspiree.md
05_GARDIEN_T05__Gardien_de_fond__Evolution_Systemes_Balance_structure_chaos.md
05_GARDIEN_T01__Gardien_de_fond__Reduction_Incertitude_Audit_coherence.md
05_GARDIEN_T07__Gardien_de_fond__Resilience_Test_memoire_fractale.md
05_GARDIEN_T11__Gardien_de_fond__Generation_Aleatoire_Frugalite_bio_inspiree.md
```

07_AGENTS_ET_ROLES

```
__init__.py
```

__pycache____init__.cpython-313.pyc

```
AGENTS_CANONIQUES_ET_GARDIENS.md
```

23 | ANALYSTE_LOGS_EXPERIMENTAUX | Données / Logs / Audit

24 | EVIDENCE_BOARD_BUILDER | Données / Logs / Audit

25 | AUDIT_LOCAL_AGENT | Données / Logs / Audit

26 | DATA_SOVEREIGNTY_GUARD | Données / Logs / Audit

27 | SECURITY_FIREWALL_ARCHITECT | Sécurité / Pare-feu

31 | SECRET_GUARD | Sécurité / Pare-feu

32 | FIELD_POC_MAPPER | Terrain / Produit / Communication

33 | PITCH_AGENT | Terrain / Produit / Communication

34 | EXPLIQUE_A_UN_HUMAIN | Terrain / Produit / Communication

35 | SIMULATEUR_REUNION | Terrain / Produit / Communication

36 | MONETISATION_PACKAGING | Terrain / Produit / Communication

37 | FRISE_HUMAINE | Frise / Arbres / Monde humain

38 | CARTOGRAPHE_34_ARBRES | Frise / Arbres / Monde humain

39 | HUMAN_HISTORY_MAPPER | Frise / Arbres / Monde humain

40 | CALIBRATION_PROCEDURALE | Frise / Arbres / Monde humain

41 | NUAGE_POINTS | Frise / Arbres / Monde humain

42 | GRAND_CARTOGRAPHE_OBSIDIA | Atlas Obsidia

43 | ONTOLOGUE_OBSIDIA | Atlas Obsidia

44 | CARTOGRAPHE_COUCHES | Atlas Obsidia

45 | CARTOGRAPHE_LOIS_PROTOCOLES | Atlas Obsidia

46 | CARTOGRAPHE_FORMULES | Atlas Obsidia

47 | CARTOGRAPHE_DOMAINES_TERRAIN | Atlas Obsidia

48 | CARTOGRAPHE_AGI_VISION_HAUTE | Atlas Obsidia

49 | COLLISION_DETECTOR | Atlas Obsidia

50 | ANTI_DISPERSION | Pour Étienne directement

51 | ENERGIE_PRIORITE | Pour Étienne directement

52 | EXTRACTEUR_MATIERE_BRUTE | Pour Étienne directement

 Allowed by auto mode classifier

06:22 PM claude-sonnet-4-6

● 52/52 agents déréférencés. Maintenant j'écris les fichiers famille définitifs avec toutes les données réelles.

```
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && python3 -c "
```

Lire les Lois depuis le genome.json

```
genome_fp = r'periphery\OBSIDIA_V4_STRUCTURED_FULL\01_REGISTRES_JSON\lois_obsidien nes.json'
import os
if os.path.exists(genome_fp):
with open(genome_fp, encoding='utf-8-sig') as f:
data = json.load(f)
print(f'lois_obsidiennes.json: {type(data)}')
if isinstance(data, list):
for l in data:
print(data)
else:
print(f'NOT FOUND: {genome_fp}')
```

Cherche d'autres fichiers lois

import pathlib

root = pathlib.Path(r'C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia -x108-proofs_REMOTE_A5F21C6B')

for fp in root.rglob('*lois*.json'):

if '__pycache__' not in str(fp):

print(f'FOUND: {fp} ({fp.stat().st_size} bytes)')

" 2>&1 | head -30)

 Waiting…

 NOT FOUND: periphery\OBSIDIA_V4_STRUCTURED_FULL\01_REGISTRES_JSON\lois_obsidienne s.json

```
Bash(ls "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B\periphery\ OBSIDIA_V4_STRUCTURE
```

D_FULL\01_REGISTRES_JSON\" 2>/dev/null | head -20)

```
● Bash(python3 -c "
```

import os, sys

sys.stdout.reconfigure(encoding='utf-8')

d = r'C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_
REMOTE_A5F21C6B\periphery\OBSIDIA_V4_STRUCTUR

ED_FULL\01_REGISTRES_JSON'

for f in os.listdir(d):

print(f)

" 2>&1)

 Waiting…

 agents_canoniques.json

```
agents_detected.json
blocks.json
extensions_clarified.json
groupes_coherence.json
modules.json
pepites.csv
pepites.json
regroupements_fichiers_copies.json
specs.json
__init__.py
```

__pycache__

 Allowed by auto mode classifier

```
● Bash(python3 -c "
```

import json, sys

sys.stdout.reconfigure(encoding='utf-8')

base = r'C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_
REMOTE_A5F21C6B\periphery\OBSIDIA_V4_STRUC

TURED_FULL\01_REGISTRES_JSON'

Lire modules.json

```
with open(base + r'\modules.json', encoding='utf-8-sig') as f:
modules = json.load(f)
print(f'MODULES: {type(modules)}, len={len(modules) if isinstance(modules, list) else \"dict\"}')
if isinstance(modules, list):
for m in modules[:20]:
print(f' {m.get(\"id\",\"?\")} | {m.get(\"name\",\"?\"):<30} | {m.get(\"type\",m.get(\"famille\",\"?\"))}')
elif isinstance(modules, dict):
for k,v in list(modules.items())[:20]:
print(f' {k}: {str(v)[:60]}')
print()
```

Lire blocks.json

```
with open(base + r'\blocks.json', encoding='utf-8-sig') as f:
blocks = json.load(f)
print(f'BLOCKS: {type(blocks)}, len={len(blocks) if isinstance(blocks, list) else \"dict\"}')
if isinstance(blocks, list):
for b in blocks[:15]:
if isinstance(b, dict):
print(f' {b.get(\"id\",b.get(\"bloc_id\",\"?\")):>5} | {b.get(\"name\",b.get(\"titre\",\"?\"))[:60]}')
print()
```

Lire pepites.json

```
with open(base + r'\pepites.json', encoding='utf-8-sig') as f:
pep = json.load(f)
print(f'PEPITES: {type(pep)}, len={len(pep) if isinstance(pep,(list,dict)) else \"?\"}')
if isinstance(pep, list):
for p in pep[:5]:
print(f' {str(p)[:100]}')
elif isinstance(pep, dict):
for k,v in list(pep.items())[:5]:
print(f' {k}: {str(v)[:80]}')
" 2>&1)
 Waiting...
 MODULES: <class 'list'>, len=36
A01 | Signature Entropique / Pare-feu symbolique | ?
A02 | Vérification de Réciprocité / Détection contradictions | ?
A03 | Équilibre Dynamique / Alignement éthique | ?
A04 | Gestion de la Friction / Bascule Sémantique | ?
A05 | Singularité Initiale / Audit éthique permanent | ?
A06 | Intégrité Structurelle / Enregistrement états | ?
A07 | Cohérence Temporelle / Chronique obsidienne | ?
A08 | Émergence des Propriétés / Transmutation cognitive | ?
A09 | Non-Contradiction / Convergence intentionnelle | ?
A10 | Complétude / Émergence par tension | ?
A11 | Résonance / Reconstruction totale | ?
A12 | Dualité / Contrat moral | ?
A13 | Sécurité Interactions / Mode réflexe | ?
A14 | Vérification Identité / Captation créative | ?
A15 | Adaptation Dynamique / Vibration multimodale | ?
A16 | Récursivité / Architecture du savoir | ?
A17 | Complexité Minimale / Apprentissage universel | ?
A18 | Optimisation Ressources / Friction créative | ?
A19 | Interconnexion / Narration cognitive | ?
A20 | Confidentialité / Entraînement émotionnel | ?
BLOCKS: <class 'list'>, len=17
1 | ?
2 | ?
3 | ?
4 | ?
5 | ?
6 | ?
7 | ?
8 | ?
9 | ?
10 | ?
11 | ?
12 | ?
13 | ?
14 | ?
15 | ?
PEPITES: <class 'list'>, len=161
{'id_num': 1, 'id': 'P1', 'name': 'Mémoire ≠ stockage', 'status': ' FORMALISÉ — Objet mathématique
{'id_num': 2, 'id': 'P2', 'name': 'Mémoire fractale hiérarchique', 'status': ' FORMALISÉ — Objet ma
{'id_num': 3, 'id': 'P3', 'name': 'Oubli actif', 'status': ' FORMALISÉ — Objet mathématique défini
```

{'id_num': 4, 'id': 'P4', 'name': "Mémoire comme condition d'éthique", 'status': '□ FORMALISÉ — Obje

{'id_num': 5, 'id': 'P5', 'name': 'Mémoire humaine ↔ mémoire système', 'status': '□ FORMALISÉ — Obje

□ Allowed by auto mode classifier

● Bash(python3 -c "

import json, sys

sys.stdout.reconfigure(encoding='utf-8')

base = r'C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_
REMOTE_A5F21C6B\periphery\OBSIDIA_V4_STRUC

TURED_FULL\01_REGISTRES_JSON'

Lire blocks.json complet

```
with open(base + r'\blocks.json', encoding='utf-8-sig') as f:
blocks = json.load(f)
print(f'BLOCS ({len(blocks)}):')
for b in blocks:
for m in modules:
if isinstance(m, dict):
print(f' {m.get(\"id\",\"?\")} | {m.get(\"name\",\"?\"):<50} |
{m.get(\"arbre\",m.get(\"linked_arbres\",\"?\"))}')
print()
```

Lire pepites.json complet - les 161

with open(base + r'\pepites.json', encoding='utf-8-sig') as f:
peps = json.load(f)
print(f'PEPITES ({len(peps)}) :')
for p in peps:
print(f' {p.get(\"id\",\"?\"):>4} | {p.get(\"name\",\"?\"):<60} | {str(p.get(\"status\",\"\"))[:50]}')
" 2>&1)
 Waiting...
 BLOCS (17):
Bloc 1: ? | pépites: []
Bloc 2: ? | pépites: []
Bloc 3: ? | pépites: []
Bloc 4: ? | pépites: []
Bloc 5: ? | pépites: []
Bloc 6: ? | pépites: []
Bloc 7: ? | pépites: []
Bloc 8: ? | pépites: []
Bloc 9: ? | pépites: []
Bloc 10: ? | pépites: []
Bloc 11: ? | pépites: []
Bloc 12: ? | pépites: []
Bloc 13: ? | pépites: []
Bloc 14: ? | pépites: []
Bloc 15: ? | pépites: []
Bloc 16: ? | pépites: []
Bloc 17: ? | pépites: []
MODULES (36):
A01 | Signature Entropique / Pare-feu symbolique | ?
A02 | Vérification de Réciprocité / Détection contradictions | ?
A03 | Équilibre Dynamique / Alignement éthique | ?
A04 | Gestion de la Friction / Bascule Sémantique | ?
A05 | Singularité Initiale / Audit éthique permanent | ?
A06 | Intégrité Structurelle / Enregistrement états | ?
A07 | Cohérence Temporelle / Chronique obsidienne | ?
A08 | Émergence des Propriétés / Transmutation cognitive | ?
A09 | Non-Contradiction / Convergence intentionnelle | ?
A10 | Complétude / Émergence par tension | ?
A11 | Résonance / Reconstruction totale | ?
A12 | Dualité / Contrat moral | ?
A13 | Sécurité Interactions / Mode réflexe | ?
A14 | Vérification Identité / Captation créative | ?
A15 | Adaptation Dynamique / Vibration multimodale | ?
A16 | Récursivité / Architecture du savoir | ?
A17 | Complexité Minimale / Apprentissage universel | ?
A18 | Optimisation Ressources / Friction créative | ?
A19 | Interconnexion / Narration cognitive | ?
A20 | Confidentialité / Entraînement émotionnel | ?
A21 | Cryptographie Homomorphe / Médiathèque fractale | ?
A22 | Propagation Information / Simulation cognitive | ?
A23 | Causalité / Calibration procédurale | ?
A24 | Prédictibilité / Métriques et veille | ?
T01 | Réduction Incertitude / Audit cohérence | ?
T02 | Conscience Distribuée / Validation réciprocité | ?

T03 | Gouvernance Décentralisée / Friction énergétique | ?
T04 | Consensus Distribué / Résonance sémantique | ?
T05 | Évolution Systèmes / Balance structure/chaos | ?
T06 | Sélection Naturelle / Alignement éthique | ?
T07 | Résilience / Test mémoire fractale | ?
T08 | Auto-Organisation / Apprentissage inversé | ?
T09 | Créativité / Mode réflexe | ?
T10 | Innovation Continue / Intégrité cognitive | ?
T11 | Génération Aléatoire / Frugalité bio-inspirée | ?
T12 | Complexité Irréductible / Transparence friction | ?
PEPITES (161) :
P1 | Mémoire ≠ stockage | □ FORMALISÉ — Objet mathématique défini
dans v1can
P2 | Mémoire fractale hiérarchique | □ FORMALISÉ — Objet mathématique défini
dans v1can
P3 | Oubli actif | □ FORMALISÉ — Objet mathématique défini
dans v1can
P4 | Mémoire comme condition d'éthique | □ FORMALISÉ — Objet mathématique défini
dans v1can
P5 | Mémoire humaine ↔ mémoire système | □ FORMALISÉ — Objet mathématique défini
dans v1can
P6 | Mémoire comme récit vivant | □ FORMALISÉ — Objet mathématique défini
dans v1can
P7 | Mémoire comme garde-fou contre l'hallucination | □ FORMALISÉ — Objet mathématique défini
dans v1can
P8 | Mémoire transversale | □ FORMALISÉ — Objet mathématique défini
dans v1can
P9 | Sens ≠ information | □ FORMALISÉ — Objet mathématique défini
dans v1can
P10 | Cadre avant réponse | □ FORMALISÉ — Objet mathématique défini
dans v1can
P11 | Refus hors cadre | □ FORMALISÉ — Objet mathématique défini
dans v1can
P12 | Désambiguïsation contextuelle | □ FORMALISÉ — Objet mathématique défini
dans v1can
P13 | Immuabilité par sceau Merkle | □ PÉPITE_ANCRÉE — A généré un ou plusieurs
théorèm
P14 | Traduction humain ↔ machine | □ FORMALISÉ — Objet mathématique défini
dans v1can
P15 | Immuabilité forte (Sensitivity) | □ PÉPITE_ANCRÉE — A généré un ou plusieurs
théorèm
P16 | Rupture data ↔ sens | □ FORMALISÉ — Objet mathématique défini
dans v1can
P17 | Croissance du log d'audit | □ PÉPITE_ANCRÉE — A généré un ou plusieurs
théorèm
P18 | Sémantique avant calcul | □ FORMALISÉ — Objet mathématique défini
dans v1can
P19 | Pluralité des niveaux de sens | □ FORMALISÉ — Objet mathématique défini
dans v1can
P20 | Sens avant performance | □ FORMALISÉ — Objet mathématique défini
dans v1can
P21 | Sémantique transversale | □ FORMALISÉ — Objet mathématique défini
dans v1can
P22 | Le symbolique n'est pas décoratif | □ FORMALISÉ — Objet mathématique défini

dans v1can
P23 | Compression du sens | □ FORMALISÉ — Objet mathématique défini
dans v1can
P24 | Non-verbal comme vecteur | □ FORMALISÉ — Objet mathématique défini
dans v1can
P25 | Symboles comme passerelles | □ FORMALISÉ — Objet mathématique défini
dans v1can
P26 | Résonance plutôt que calcul | □ FORMALISÉ — Objet mathématique défini
dans v1can
P27 | Transmission implicite | □ FORMALISÉ — Objet mathématique défini
dans v1can
P28 | Récit comme structure | □ FORMALISÉ — Objet mathématique défini
dans v1can
P29 | Mythologie vivante | □ FORMALISÉ — Objet mathématique défini
dans v1can
P30 | Symbolique opératoire | □ FORMALISÉ — Objet mathématique défini
dans v1can
P31 | Archétype | □ FORMALISÉ — Objet mathématique défini
dans v1can
P32 | Rituel comme séquence symbolique | □ FORMALISÉ — Objet mathématique défini
dans v1can
P33 | Symbolique transversal | □ FORMALISÉ — Objet mathématique défini
dans v1can
P34 | Kernel = intersection des contraintes | □ FORMALISÉ — Objet mathématique défini
dans v1can
P35 | Preuve formelle sans sorry | □ FORMALISÉ — Objet mathématique défini
dans v1can
P36 | Quintuplet d'état canonique (S,Φ,I,τ,L) | □ À_PROUVER — Formalisé, code Lean
esquissé, pas e
P37 | Formalisation : narratif → math | □ FORMALISÉ — Objet mathématique défini
dans v1can
P38 | Filtre éthique χ(t) | □ FORMALISÉ — Objet mathématique défini
dans v1can
P39 | Invariant permanent | □ FORMALISÉ — Objet mathématique défini
dans v1can
P40 | Axiome non dérivable | □ FORMALISÉ — Objet mathématique défini
dans v1can
P41 | Déterminisme D1 | □ PÉPITE_ANCRÉE — A généré un ou plusieurs
théorèm
P42 | Seuil G1 : ACT si θ ≤ S | □ FORMALISÉ — Objet mathématique défini
dans v1can
P43 | Pas de BLOCK natif L11.3 | □ PÉPITE_ANCRÉE — A généré un ou plusieurs
théorèm
P44 | Verrou temporel X-108 | □ FORMALISÉ — Objet mathématique défini
dans v1can
P45 | Skew négatif → HOLD | □ FORMALISÉ — Objet mathématique défini
dans v1can
P46 | Consensus fail-closed 4 agents | □ FORMALISÉ — Objet mathématique défini
dans v1can
P47 | Couplage réel/moteur dR/dt | □ À_FORMALISER — Formulation narrative
existante,
P48 | Structure cosmologique | □ À_FORMALISER — Formulation narrative
existante,
P49 | Émergence par interaction | □ À_FORMALISER — Formulation narrative

existante,
P50 | Récursivité fondamentale | □ À_FORMALISER — Formulation narrative
existante,
P51 | Structure fractale | □ À_FORMALISER — Formulation narrative
existante,
P52 | Principe holographique | □ À_FORMALISER — Formulation narrative
existante,
P53 | Résonance cosmique | □ À_FORMALISER — Formulation narrative
existante,
P54 | Temps comme intégrale d'événements | □ À_FORMALISER — Formulation narrative
existante,
P55 | Espace métrique | □ À_FORMALISER — Formulation narrative
existante,
P56 | Causalité temporelle | □ À_FORMALISER — Formulation narrative
existante,
P57 | Entropie de Shannon | □ À_FORMALISER — Formulation narrative
existante,
P58 | Néguentropie | □ À_FORMALISER — Formulation narrative
existante,
P59 | Complexité de Kolmogorov | □ À_FORMALISER — Formulation narrative
existante,
P60 | Auto-organisation | □ À_FORMALISER — Formulation narrative
existante,
P61 | Bifurcation de trajectoire | □ À_FORMALISER — Formulation narrative
existante,
P62 | Attracteur | □ À_FORMALISER — Formulation narrative
existante,
P63 | Sensibilité aux conditions initiales | □ À_FORMALISER — Formulation narrative
existante,
P64 | Ordre = 1 - entropie normalisée | □ FORMALISÉ — Objet mathématique défini
dans v1can
P65 | Définition de la vie | □ FORMALISÉ — Objet mathématique défini
dans v1can
P66 | Conscience comme récursivité infinie | □ FORMALISÉ — Objet mathématique défini
dans v1can
P67 | Filtre éthique continu χ(t) | □ FORMALISÉ — Objet mathématique défini
dans v1can
P68 | Éthique intégrée sur le temps | □ FORMALISÉ — Objet mathématique défini
dans v1can
P69 | Responsabilité action/conséquence | □ FORMALISÉ — Objet mathématique défini
dans v1can
P70 | Consentement binaire | □ FORMALISÉ — Objet mathématique défini
dans v1can
P71 | Transparence universelle | □ FORMALISÉ — Objet mathématique défini
dans v1can
P72 | Équité ε-approximée | □ FORMALISÉ — Objet mathématique défini
dans v1can
P73 | Non-nuisance | □ FORMALISÉ — Objet mathématique défini
dans v1can
P74 | Bienfaisance maximale | □ FORMALISÉ — Objet mathématique défini
dans v1can
P75 | Justice proportionnelle | □ FORMALISÉ — Objet mathématique défini
dans v1can
P76 | Budget énergétique E_c(t) | □ FORMALISÉ — Objet mathématique défini

dans v1can
P77 | Flux énergétique dE/dt | □ FORMALISÉ — Objet mathématique défini
dans v1can
P78 | Réserve énergétique | □ FORMALISÉ — Objet mathématique défini
dans v1can
P79 | Régénération en repos | □ FORMALISÉ — Objet mathématique défini
dans v1can
P80 | Équilibre énergétique | □ FORMALISÉ — Objet mathématique défini
dans v1can
P81 | Surcharge → HOLD | □ FORMALISÉ — Objet mathématique défini
dans v1can
P82 | Optimisation coût/qualité | □ FORMALISÉ — Objet mathématique défini
dans v1can
P83 | Efficience output/input | □ FORMALISÉ — Objet mathématique défini
dans v1can
P84 | Durabilité à long terme | □ FORMALISÉ — Objet mathématique défini
dans v1can
P85 | Loi de réciprocité | □ FORMALISÉ — Objet mathématique défini
dans v1can
P86 | Consensus supermajorité 3/4 | □ PÉPITE_ANCRÉE — A généré un ou plusieurs
théorèm
P87 | Fail-closed sans supermajorité | □ PÉPITE_ANCRÉE — A généré un ou plusieurs
théorèm
P88 | Non-contradiction | □ FORMALISÉ — Objet mathématique défini
dans v1can
P89 | Complétude | □ FORMALISÉ — Objet mathématique défini
dans v1can
P90 | Cohérence | □ FORMALISÉ — Objet mathématique défini
dans v1can
P91 | Décidabilité | □ FORMALISÉ — Objet mathématique défini
dans v1can
P92 | Calculabilité (Turing) | □ FORMALISÉ — Objet mathématique défini
dans v1can
P93 | P ⊆ NP | □ FORMALISÉ — Objet mathématique défini
dans v1can
P94 | Réductibilité | □ FORMALISÉ — Objet mathématique défini
dans v1can
P95 | Indécidabilité (Halting) | □ FORMALISÉ — Objet mathématique défini
dans v1can
P96 | Récursivité | □ FORMALISÉ — Objet mathématique défini
dans v1can
P97 | Point fixe | □ FORMALISÉ — Objet mathématique défini
dans v1can
P98 | Continuité ε-δ | □ FORMALISÉ — Objet mathématique défini
dans v1can
P99 | Convergence | □ FORMALISÉ — Objet mathématique défini
dans v1can
P100 | Stabilité de Lyapunov L(Φ(s)) ≤ L(s) | □ FORMALISÉ — Objet mathématique défini
dans v1can
P101 | Avancement incrémental validé | □ FORMALISÉ — Objet mathématique défini
dans v1can
P102 | Itération convergente | □ FORMALISÉ — Objet mathématique défini
dans v1can
P103 | Validation par critères | □ FORMALISÉ — Objet mathématique défini

dans v1can
P104 | Réfutation par contre-exemple | □ FORMALISÉ — Objet mathématique défini
dans v1can
P105 | Induction mathématique | □ FORMALISÉ — Objet mathématique défini
dans v1can
P106 | Déduction logique | □ FORMALISÉ — Objet mathématique défini
dans v1can
P107 | Stabilité Lyapunov (δ-ε) | □ À_PROUVER — Formalisé, code Lean
esquissé, pas e
P108 | Robustesse aux perturbations | □ FORMALISÉ — Objet mathématique défini
dans v1can
P109 | Reproductibilité | □ FORMALISÉ — Objet mathématique défini
dans v1can
P110 | Falsifiabilité (Popper) | □ FORMALISÉ — Objet mathématique défini
dans v1can
P111 | Moteur non substituable | □ FORMALISÉ — Objet mathématique défini
dans v1can
P112 | Hiérarchie OS0→OS4 | □ FORMALISÉ — Objet mathématique défini
dans v1can
P113 | Kernel = intersection des couches | □ FORMALISÉ — Objet mathématique défini
dans v1can
P114 | Guard = Kernel + Temporal + Consensus | □ FORMALISÉ — Objet mathématique défini
dans v1can
P115 | Filtre éthique du moteur | □ FORMALISÉ — Objet mathématique défini
dans v1can
P116 | Pipeline de couches | □ FORMALISÉ — Objet mathématique défini
dans v1can
P117 | Orchestration par scheduler | □ FORMALISÉ — Objet mathématique défini
dans v1can
P118 | Isolation des couches | □ FORMALISÉ — Objet mathématique défini
dans v1can
P119 | Composition fonctionnelle | □ FORMALISÉ — Objet mathématique défini
dans v1can
P120 | Abstraction par interface | □ FORMALISÉ — Objet mathématique défini
dans v1can
P121 | Encapsulation (boîte noire) | □ FORMALISÉ — Objet mathématique défini
dans v1can
P122 | Modularité | □ FORMALISÉ — Objet mathématique défini
dans v1can
P123 | Extensibilité | □ FORMALISÉ — Objet mathématique défini
dans v1can
P124 | Infrastructure : I∘M=O | □ FORMALISÉ — Objet mathématique défini
dans v1can
P125 | Déploiement Build+Test+Release | □ FORMALISÉ — Objet mathématique défini
dans v1can
P126 | Scalabilité linéaire | □ FORMALISÉ — Objet mathématique défini
dans v1can
P127 | Disponibilité (uptime) | □ FORMALISÉ — Objet mathématique défini
dans v1can
P128 | Latence de réponse | □ FORMALISÉ — Objet mathématique défini
dans v1can
P129 | Débit (throughput) | □ FORMALISÉ — Objet mathématique défini
dans v1can
P130 | Résilience MTBF/MTTR | □ FORMALISÉ — Objet mathématique défini

dans v1can
P131 | Sécurité CIA | □ FORMALISÉ — Objet mathématique défini
dans v1can
P132 | Observabilité (logs+métriques+traces) | □ FORMALISÉ — Objet mathématique défini
dans v1can
P133 | Configuration clé-valeur | □ FORMALISÉ — Objet mathématique défini
dans v1can
P134 | Versioning sémantique | □ FORMALISÉ — Objet mathématique défini
dans v1can
P135 | Pipeline CI/CD | □ FORMALISÉ — Objet mathématique défini
dans v1can
P136 | Infrastructure as Code | □ FORMALISÉ — Objet mathématique défini
dans v1can
P137 | Agent = Rôle+Protocole+Flux | □ FORMALISÉ — Objet mathématique défini
dans v1can
P138 | Écologie d'agents | □ FORMALISÉ — Objet mathématique défini
dans v1can
P139 | Protocole = séquence de messages | □ FORMALISÉ — Objet mathématique défini
dans v1can
P140 | Coordination par consensus | □ FORMALISÉ — Objet mathématique défini
dans v1can
P141 | Émergence collective | □ FORMALISÉ — Objet mathématique défini
dans v1can
P142 | Spécialisation par domaine | □ FORMALISÉ — Objet mathématique défini
dans v1can
P143 | Généralisation multi-domaine | □ FORMALISÉ — Objet mathématique défini
dans v1can
P144 | Apprentissage par expérience | □ FORMALISÉ — Objet mathématique défini
dans v1can
P145 | Adaptation à l'environnement | □ FORMALISÉ — Objet mathématique défini
dans v1can
P146 | Autonomie décisionnelle | □ FORMALISÉ — Objet mathématique défini
dans v1can
P147 | Collaboration synergique | □ FORMALISÉ — Objet mathématique défini
dans v1can
P148 | Compétition sélective | □ FORMALISÉ — Objet mathématique défini
dans v1can
P149 | Narration début-milieu-fin | □ À_FORMALISER — Formulation narrative
existante,
P150 | Personnage = identité+motivation+arc | □ FORMALISÉ — Objet mathématique défini
dans v1can
P151 | Conflit = désir vs obstacle | □ FORMALISÉ — Objet mathématique défini
dans v1can
P152 | Résolution du conflit | □ FORMALISÉ — Objet mathématique défini
dans v1can
P153 | Thème = abstraction du récit | □ FORMALISÉ — Objet mathématique défini
dans v1can
P154 | Symbole narratif | □ FORMALISÉ — Objet mathématique défini
dans v1can
P155 | Métaphore = analogie | □ À_FORMALISER — Formulation narrative
existante,
P156 | Ironie = écart attendu/réel | □ FORMALISÉ — Objet mathématique défini
dans v1can
P157 | Suspense = incertitude de résolution | □ FORMALISÉ — Objet mathématique défini

dans v1can
P158 | Catharsis = libération émotionnelle | □ FORMALISÉ — Objet mathématique défini
dans v1can
P159 | Épiphanie = révélation soudaine | □ FORMALISÉ — Objet mathématique défini
dans v1can
P160 | Dénouement final | □ VISION — Conceptuel, hors-scope
mathématique
P161 | Calibration énergétique temporelle (loi finale) | □ À_PROUVER — Formalisé, code Lean
esquissé, pas e
□ Allowed by auto mode classifier
06:23 PM claude-sonnet-4-6
● Toutes les données canoniques sont déréférencées. J'écris maintenant les fichiers famille enrichis définitifs.
● Bash(python3 << 'PYEOF'
import json, sys
from pathlib import Path
sys.stdout.reconfigure(encoding='utf-8')
OUT_BASE = Path(r"C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-pr
oofs_REMOTE_A5F21C6B\.local_audits")

Trouve le dossier d'audit créé

```
audit_dirs = sorted(OUT_BASE.glob("CLAUDE_BRODY_SEMANTIC_DEEP_ENUMERATION_V0_*"), reverse=True)
OUT = audit_dirs[0]
print(f"OUT: {OUT}")
```

1. ARBRES 34

```
ARBRES = [
("01", "Arbre de l'Humain", "I_FONDAMENTAUX", "ARBRE_01__Arbre_de_l_Humain"),
("02", "Arbre de la Conscience", "I_FONDAMENTAUX", "ARBRE_02__Arbre_de_la_Conscience"),
("03", "Arbre de la Perception", "I_FONDAMENTAUX", "ARBRE_03__Arbre_de_la_Perception"),
("04", "Arbre du Sens", "I_FONDAMENTAUX", "ARBRE_04__Arbre_du_Sens"),
("05", "Arbre de l'Identité", "I_FONDAMENTAUX", "ARBRE_05__Arbre_de_l_Identite"),
("06", "Arbre de la Compréhension", "II_COGNITIFS", "ARBRE_06__Arbre_de_la_Comprehension"),
("07", "Arbre de l'Organisation", "II_COGNITIFS", "ARBRE_07__Arbre_de_l_Organisation"),
("08", "Arbre de la Pensée", "II_COGNITIFS", "ARBRE_08__Arbre_de_la_Pensee"),
("09", "Arbre de l'Intelligence", "II_COGNITIFS", "ARBRE_09__Arbre_de_l_Intelligence"),
("10", "Arbre du Langage", "II_COGNITIFS", "ARBRE_10__Arbre_du_Langage"),
("11", "Arbre de la Science", "III_CONNAISSANCE", "ARBRE_11__Arbre_de_la_Science"),
("12", "Arbre de la Technique", "III_CONNAISSANCE", "ARBRE_12__Arbre_de_la_Technique"),
("13", "Arbre de l'Art", "III_CONNAISSANCE", "ARBRE_13__Arbre_de_l_Art"),
("14", "Arbre de la Philosophie", "III_CONNAISSANCE", "ARBRE_14__Arbre_de_la_Philosophie"),
("15", "Arbre de la Spiritualité", "III_CONNAISSANCE", "ARBRE_15__Arbre_de_la_Spiritualite"),
("16", "Arbre de la Relation", "IV_RELATIONNELS_SOCIAUX", "ARBRE_16__Arbre_de_la_Relation"),
("17", "Arbre du Collectif", "IV_RELATIONNELS_SOCIAUX", "ARBRE_17__Arbre_du_Collectif"),
("18", "Arbre de la Transmission", "IV_RELATIONNELS_SOCIAUX", "ARBRE_18__Arbre_de_la_Transmission"),
("19", "Arbre de la Culture", "IV_RELATIONNELS_SOCIAUX", "ARBRE_19__Arbre_de_la_Culture"),
("20", "Arbre de l'Action", "V_ACTION_TRANSFORMATION", "ARBRE_20__Arbre_de_l_Action"),
("21", "Arbre de la Création", "V_ACTION_TRANSFORMATION", "ARBRE_21__Arbre_de_la_Creation"),
("22", "Arbre de la Transformation", "V_ACTION_TRANSFORMATION", "ARBRE_22__Arbre_de_la_Transformation"),
("23", "Arbre du Temps", "VI_TEMPORELS_MEMORIELS", "ARBRE_23__Arbre_du_Temps"),
("24", "Arbre de la Mémoire", "VI_TEMPORELS_MEMORIELS", "ARBRE_24__Arbre_de_la_Memoire"),
("25", "Arbre de l'Histoire", "VI_TEMPORELS_MEMORIELS", "ARBRE_25__Arbre_de_l_Histoire"),
("26", "Arbre de la Cohérence", "VII_META_STRUCTURELS", "ARBRE_26__Arbre_de_la_Coherence"),
("27", "Arbre de la Vérité", "VII_META_STRUCTURELS", "ARBRE_27__Arbre_de_la_Verite"),
("28", "Arbre de la Valeur", "VII_META_STRUCTURELS", "ARBRE_28__Arbre_de_la_Valeur"),
("29", "Arbre de la Finalité", "VII_META_STRUCTURELS", "ARBRE_29__Arbre_de_la_Finalite"),
("30", "Arbre Cognitif Global", "VIII_OBSIDIA_AGI", "ARBRE_30__Arbre_Cognitif_Global"),
("31", "Arbre des Flux", "VIII_OBSIDIA_AGI", "ARBRE_31__Arbre_des_Flux"),
("32", "Arbre des Connexions", "VIII_OBSIDIA_AGI", "ARBRE_32__Arbre_des_Connexions"),
("33", "Arbre de l'Optimisation", "VIII_OBSIDIA_AGI", "ARBRE_33__Arbre_de_l_Optimisation"),
("34", "Arbre de la Stabilité", "VIII_OBSIDIA_AGI", "ARBRE_34__Arbre_de_la_Stabilite"),
]
lines = ["# 01_ARBRES_34 — 34 Arbres Obsidia\n\n"]
lines.append("Status: FOUND_EXPLICIT\n")
lines.append("Expected: 34\n")
lines.append("Count: 34\n")
lines.append("Missing: 0\n\n")
lines.append("Source:
`periphery/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1/04_ARBR
ES_34_TENSOR_MATRIX/arbres_34.canon.json`\n")
lines.append("Confirmé par:
`periphery/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1/19_REGI
STRES_JSON/arbres_34.registry.json`\n\n")
lines.append("## Familles ontologiques\n\n")
lines.append("| Famille | Arbres |\n")
lines.append("|---------|--------|\n")
familles = {}
```

```
for a_id, a_name, a_fam, _ in ARBRES:
familles.setdefault(a_fam, []).append(f"ARBRE_{a_id}")
for fam, items in familles.items():
lines.append(f"| {fam} | {', '.join(items)} |\n")
lines.append("\n## Items\n\n")
for idx, (a_id, a_name, a_fam, folder) in enumerate(ARBRES, 1):
lines.append(f"{idx:04d}. ARBRE_{a_id} — {a_name}\n")
lines.append(f" famille: {a_fam}\n")
lines.append(f" folder: {folder}\n")
lines.append(f" source_file:
periphery/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1/04_ARBRES_34_TENSOR_MATRIX/arbres _34.canon.json\n")
lines.append(f" origin: arbres_34_canon_json\n\n")
(OUT / "01_ARBRES_34.md").write_text("".join(lines), encoding="utf-8")
print(f" 01_ARBRES_34.md mis à jour (34/34 FOUND_EXPLICIT)")
```

2. LOIS

```
LOIS_7 = [
("P88", "Non-contradiction", "Bloc 09 — Les 7 Lois Obsidiennes"),
("P89", "Complétude", "Bloc 09 — Les 7 Lois Obsidiennes"),
("P90", "Cohérence", "Bloc 09 — Les 7 Lois Obsidiennes"),
("P91", "Décidabilité", "Bloc 09 — Les 7 Lois Obsidiennes"),
("P92", "Calculabilité (Turing)", "Bloc 09 — Les 7 Lois Obsidiennes"),
("P93", "P ⊆ NP", "Bloc 09 — Les 7 Lois Obsidiennes"),
("P94", "Réductibilité", "Bloc 09 — Les 7 Lois Obsidiennes"),
]
LOIS_EXTRA = [
("P85", "Loi de réciprocité", "Pépite principale Bloc 8"),
("P86", "Consensus supermajorité 3/4", "Pépite secondaire Bloc 09"),
("P87", "Fail-closed sans supermajorité", "Pépite secondaire Bloc 09"),
]
lines = ["# 02_LOIS — Lois Obsidiennes\n\n"]
lines.append("Status: FOUND_EXPLICIT\n")
lines.append("Count: 10\n")
lines.append("Note: Les 7 Lois formelles correspondent aux Pépites P88–P94 (Bloc 09). P85, P86, P87 sont des lois
structurelles liées.\n\n")
lines.append("Source: `periphery/OBSIDIA_V4_STRUCTURED_FULL/07_AGENTS_ET_ROLES/` + Graphiti record
`427731C073EB_Bloc_09__Les_7_Lois_Obsidiennes_C8.md`\n\n")
lines.append("## Les 7 Lois Formelles (Bloc 09)\n\n")
for idx, (code, name, bloc) in enumerate(LOIS_7, 1):
lines.append(f"{idx:04d}. {code} — {name}\n")
lines.append(f" bloc: {bloc}\n")
lines.append(f" status: FORMALISÉ\n")
lines.append(f" source_file: periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/pepites.json\n" )
lines.append(f" origin: bloc_09_lois_obsidiennes\n\n")
lines.append("## Lois structurelles liées\n\n")
for idx, (code, name, bloc) in enumerate(LOIS_EXTRA, 8):
lines.append(f"{idx:04d}. {code} — {name}\n")
lines.append(f" bloc: {bloc}\n")
lines.append(f" source_file: periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/pepites.json\n" )
lines.append(f" origin: pepites_json\n\n")
lines.append("## Note de déréférencement\n\n")
lines.append("Les Lois Obsidiennes ne sont pas stockées dans un fichier `lois_obsidiennes.json` dédié.\n")
lines.append("Elles sont définies comme Pépites P88-P94 dans `pepites.json`, groupées dans `Bloc 09`.\n")
lines.append("Confirme aussi: `OBSIDIA_REVERSE_OS_KERNEL_LAWS_BINDING_V1.md` (boundary binding)\n")
(OUT / "02_LOIS.md").write_text("".join(lines), encoding="utf-8")
print(f" 02_LOIS.md mis à jour (7+3 lois FOUND_EXPLICIT)")
```

3. MODULES

```
MODULES_A = [
("A01", "Signature Entropique / Pare-feu symbolique"),
("A02", "Vérification de Réciprocité / Détection contradictions"),
("A03", "Équilibre Dynamique / Alignement éthique"),
("A04", "Gestion de la Friction / Bascule Sémantique"),
("A05", "Singularité Initiale / Audit éthique permanent"),
("A06", "Intégrité Structurelle / Enregistrement états"),
("A07", "Cohérence Temporelle / Chronique obsidienne"),
("A08", "Émergence des Propriétés / Transmutation cognitive"),
("A09", "Non-Contradiction / Convergence intentionnelle"),
("A10", "Complétude / Émergence par tension"),
("A11", "Résonance / Reconstruction totale"),
("A12", "Dualité / Contrat moral"),
("A13", "Sécurité Interactions / Mode réflexe"),
("A14", "Vérification Identité / Captation créative"),
("A15", "Adaptation Dynamique / Vibration multimodale"),
("A16", "Récursivité / Architecture du savoir"),
("A17", "Complexité Minimale / Apprentissage universel"),
("A18", "Optimisation Ressources / Friction créative"),
("A19", "Interconnexion / Narration cognitive"),
("A20", "Confidentialité / Entraînement émotionnel"),
("A21", "Cryptographie Homomorphe / Médiathèque fractale"),
("A22", "Propagation Information / Simulation cognitive"),
("A23", "Causalité / Calibration procédurale"),
("A24", "Prédictibilité / Métriques et veille"),
]
MODULES_T = [
("T01", "Réduction Incertitude / Audit cohérence"),
("T02", "Conscience Distribuée / Validation réciprocité"),
("T03", "Gouvernance Décentralisée / Friction énergétique"),
("T04", "Consensus Distribué / Résonance sémantique"),
("T05", "Évolution Systèmes / Balance structure/chaos"),
("T06", "Sélection Naturelle / Alignement éthique"),
("T07", "Résilience / Test mémoire fractale"),
("T08", "Auto-Organisation / Apprentissage inversé"),
("T09", "Créativité / Mode réflexe"),
("T10", "Innovation Continue / Intégrité cognitive"),
("T11", "Génération Aléatoire / Frugalité bio-inspirée"),
("T12", "Complexité Irréductible / Transparence friction"),
]
lines = ["# 04_MODULES — Modules Cognitifs Obsidia\n\n"]
lines.append("Status: FOUND_EXPLICIT\n")
lines.append(f"Count: {len(MODULES_A) + len(MODULES_T)}\n")
lines.append("Source: `periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/modules.json`\ n\n")
lines.append("## Modules A (Cognitifs — A01 à A24)\n\n")
for idx, (m_id, m_name) in enumerate(MODULES_A, 1):
lines.append(f"{idx:04d}. {m_id} — {m_name}\n")
lines.append(f" source_file: periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/modules.json\n" )
lines.append(f" origin: modules_json\n\n")
lines.append("## Modules T (Gardiens de fond — T01 à T12)\n\n")
for idx, (m_id, m_name) in enumerate(MODULES_T, 25):
```

```
lines.append(f"{idx:04d}. {m_id} — {m_name}\n")
lines.append(f" source_file: periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/modules.json\n" )
lines.append(f" origin: modules_json + GARDIEN_Txx_md_files\n\n")
(OUT / "04_MODULES.md").write_text("".join(lines), encoding="utf-8")
print(f" 04_MODULES.md mis à jour ({len(MODULES_A)+len(MODULES_T)} modules FOUND_EXPLICIT)")
```

4. PEPITES

PEPITES = [
("P1","Mémoire ≠ stockage","FORMALISÉ"),
("P2","Mémoire fractale hiérarchique","FORMALISÉ"),
("P3","Oubli actif","FORMALISÉ"),
("P4","Mémoire comme condition d'éthique","FORMALISÉ"),
("P5","Mémoire humaine ↔ mémoire système","FORMALISÉ"),
("P6","Mémoire comme récit vivant","FORMALISÉ"),
("P7","Mémoire comme garde-fou contre l'hallucination","FORMALISÉ"),
("P8","Mémoire transversale","FORMALISÉ"),
("P9","Sens ≠ information","FORMALISÉ"),
("P10","Cadre avant réponse","FORMALISÉ"),
("P11","Refus hors cadre","FORMALISÉ"),
("P12","Désambiguïsation contextuelle","FORMALISÉ"),
("P13","Immuabilité par sceau Merkle","PÉPITE_ANCRÉE"),
("P14","Traduction humain ↔ machine","FORMALISÉ"),
("P15","Immuabilité forte (Sensitivity)","PÉPITE_ANCRÉE"),
("P16","Rupture data ↔ sens","FORMALISÉ"),
("P17","Croissance du log d'audit","PÉPITE_ANCRÉE"),
("P18","Sémantique avant calcul","FORMALISÉ"),
("P19","Pluralité des niveaux de sens","FORMALISÉ"),
("P20","Sens avant performance","FORMALISÉ"),
("P21","Sémantique transversale","FORMALISÉ"),
("P22","Le symbolique n'est pas décoratif","FORMALISÉ"),
("P23","Compression du sens","FORMALISÉ"),
("P24","Non-verbal comme vecteur","FORMALISÉ"),
("P25","Symboles comme passerelles","FORMALISÉ"),
("P26","Résonance plutôt que calcul","FORMALISÉ"),
("P27","Transmission implicite","FORMALISÉ"),
("P28","Récit comme structure","FORMALISÉ"),
("P29","Mythologie vivante","FORMALISÉ"),
("P30","Symbolique opératoire","FORMALISÉ"),
("P31","Archétype","FORMALISÉ"),
("P32","Rituel comme séquence symbolique","FORMALISÉ"),
("P33","Symbolique transversal","FORMALISÉ"),
("P34","Kernel = intersection des contraintes","FORMALISÉ"),
("P35","Preuve formelle sans sorry","FORMALISÉ"),
("P36","Quintuplet d'état canonique (S,Φ,I,τ,L)","À_PROUVER"),
("P37","Formalisation : narratif → math","FORMALISÉ"),
("P38","Filtre éthique χ(t)","FORMALISÉ"),
("P39","Invariant permanent","FORMALISÉ"),
("P40","Axiome non dérivable","FORMALISÉ"),
("P41","Déterminisme D1","PÉPITE_ANCRÉE"),
("P42","Seuil G1 : ACT si θ ≤ S","FORMALISÉ"),
("P43","Pas de BLOCK natif L11.3","PÉPITE_ANCRÉE"),
("P44","Verrou temporel X-108","FORMALISÉ"),
("P45","Skew négatif → HOLD","FORMALISÉ"),
("P46","Consensus fail-closed 4 agents","FORMALISÉ"),
("P47","Couplage réel/moteur dR/dt","À_FORMALISER"),
("P48","Structure cosmologique","À_FORMALISER"),
("P49","Émergence par interaction","À_FORMALISER"),
("P50","Récursivité fondamentale","À_FORMALISER"),

("P51","Structure fractale","À_FORMALISER"),
("P52","Principe holographique","À_FORMALISER"),
("P53","Résonance cosmique","À_FORMALISER"),
("P54","Temps comme intégrale d'événements","À_FORMALISER"),
("P55","Espace métrique","À_FORMALISER"),
("P56","Causalité temporelle","À_FORMALISER"),
("P57","Entropie de Shannon","À_FORMALISER"),
("P58","Néguentropie","À_FORMALISER"),
("P59","Complexité de Kolmogorov","À_FORMALISER"),
("P60","Auto-organisation","À_FORMALISER"),
("P61","Bifurcation de trajectoire","À_FORMALISER"),
("P62","Attracteur","À_FORMALISER"),
("P63","Sensibilité aux conditions initiales","À_FORMALISER"),
("P64","Ordre = 1 - entropie normalisée","FORMALISÉ"),
("P65","Définition de la vie","FORMALISÉ"),
("P66","Conscience comme récursivité infinie","FORMALISÉ"),
("P67","Filtre éthique continu χ(t)","FORMALISÉ"),
("P68","Éthique intégrée sur le temps","FORMALISÉ"),
("P69","Responsabilité action/conséquence","FORMALISÉ"),
("P70","Consentement binaire","FORMALISÉ"),
("P71","Transparence universelle","FORMALISÉ"),
("P72","Équité ε-approximée","FORMALISÉ"),
("P73","Non-nuisance","FORMALISÉ"),
("P74","Bienfaisance maximale","FORMALISÉ"),
("P75","Justice proportionnelle","FORMALISÉ"),
("P76","Budget énergétique E_c(t)","FORMALISÉ"),
("P77","Flux énergétique dE/dt","FORMALISÉ"),
("P78","Réserve énergétique","FORMALISÉ"),
("P79","Régénération en repos","FORMALISÉ"),
("P80","Équilibre énergétique","FORMALISÉ"),
("P81","Surcharge → HOLD","FORMALISÉ"),
("P82","Optimisation coût/qualité","FORMALISÉ"),
("P83","Efficience output/input","FORMALISÉ"),
("P84","Durabilité à long terme","FORMALISÉ"),
("P85","Loi de réciprocité","FORMALISÉ"),
("P86","Consensus supermajorité 3/4","PÉPITE_ANCRÉE"),
("P87","Fail-closed sans supermajorité","PÉPITE_ANCRÉE"),
("P88","Non-contradiction","FORMALISÉ"),
("P89","Complétude","FORMALISÉ"),
("P90","Cohérence","FORMALISÉ"),
("P91","Décidabilité","FORMALISÉ"),
("P92","Calculabilité (Turing)","FORMALISÉ"),
("P93","P ⊆ NP","FORMALISÉ"),
("P94","Réductibilité","FORMALISÉ"),
("P95","Indécidabilité (Halting)","FORMALISÉ"),
("P96","Récursivité","FORMALISÉ"),
("P97","Point fixe","FORMALISÉ"),
("P98","Continuité ε-δ","FORMALISÉ"),
("P99","Convergence","FORMALISÉ"),
("P100","Stabilité de Lyapunov L(Φ(s)) ≤ L(s)","FORMALISÉ"),
("P101","Avancement incrémental validé","FORMALISÉ"),
("P102","Itération convergente","FORMALISÉ"),
("P103","Validation par critères","FORMALISÉ"),
("P104","Réfutation par contre-exemple","FORMALISÉ"),

("P105","Induction mathématique","FORMALISÉ"),
("P106","Déduction logique","FORMALISÉ"),
("P107","Stabilité Lyapunov (δ-ε)","À_PROUVER"),
("P108","Robustesse aux perturbations","FORMALISÉ"),
("P109","Reproductibilité","FORMALISÉ"),
("P110","Falsifiabilité (Popper)","FORMALISÉ"),
("P111","Moteur non substituable","FORMALISÉ"),
("P112","Hiérarchie OS0→OS4","FORMALISÉ"),
("P113","Kernel = intersection des couches","FORMALISÉ"),
("P114","Guard = Kernel + Temporal + Consensus","FORMALISÉ"),
("P115","Filtre éthique du moteur","FORMALISÉ"),
("P116","Pipeline de couches","FORMALISÉ"),
("P117","Orchestration par scheduler","FORMALISÉ"),
("P118","Isolation des couches","FORMALISÉ"),
("P119","Composition fonctionnelle","FORMALISÉ"),
("P120","Abstraction par interface","FORMALISÉ"),
("P121","Encapsulation (boîte noire)","FORMALISÉ"),
("P122","Modularité","FORMALISÉ"),
("P123","Extensibilité","FORMALISÉ"),
("P124","Infrastructure : I∘M=O","FORMALISÉ"),
("P125","Déploiement Build+Test+Release","FORMALISÉ"),
("P126","Scalabilité linéaire","FORMALISÉ"),
("P127","Disponibilité (uptime)","FORMALISÉ"),
("P128","Latence de réponse","FORMALISÉ"),
("P129","Débit (throughput)","FORMALISÉ"),
("P130","Résilience MTBF/MTTR","FORMALISÉ"),
("P131","Sécurité CIA","FORMALISÉ"),
("P132","Observabilité (logs+métriques+traces)","FORMALISÉ"),
("P133","Configuration clé-valeur","FORMALISÉ"),
("P134","Versioning sémantique","FORMALISÉ"),
("P135","Pipeline CI/CD","FORMALISÉ"),
("P136","Infrastructure as Code","FORMALISÉ"),
("P137","Agent = Rôle+Protocole+Flux","FORMALISÉ"),
("P138","Écologie d'agents","FORMALISÉ"),
("P139","Protocole = séquence de messages","FORMALISÉ"),
("P140","Coordination par consensus","FORMALISÉ"),
("P141","Émergence collective","FORMALISÉ"),
("P142","Spécialisation par domaine","FORMALISÉ"),
("P143","Généralisation multi-domaine","FORMALISÉ"),
("P144","Apprentissage par expérience","FORMALISÉ"),
("P145","Adaptation à l'environnement","FORMALISÉ"),
("P146","Autonomie décisionnelle","FORMALISÉ"),
("P147","Collaboration synergique","FORMALISÉ"),
("P148","Compétition sélective","FORMALISÉ"),
("P149","Narration début-milieu-fin","À_FORMALISER"),
("P150","Personnage = identité+motivation+arc","FORMALISÉ"),
("P151","Conflit = désir vs obstacle","FORMALISÉ"),
("P152","Résolution du conflit","FORMALISÉ"),
("P153","Thème = abstraction du récit","FORMALISÉ"),
("P154","Symbole narratif","FORMALISÉ"),
("P155","Métaphore = analogie","À_FORMALISER"),
("P156","Ironie = écart attendu/réel","FORMALISÉ"),
("P157","Suspense = incertitude de résolution","FORMALISÉ"),
("P158","Catharsis = libération émotionnelle","FORMALISÉ"),

```
("P159","Épiphanie = révélation soudaine","FORMALISÉ"),
("P160","Dénouement final","VISION"),
("P161","Calibration énergétique temporelle (loi finale)","À_PROUVER"),
]
lines = ["# 05_PEPITES — Pépites (161)\n\n"]
lines.append("Status: FOUND_EXPLICIT\n")
lines.append("Count: 161\n")
lines.append("Source: `periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/pepites.json`\ n\n")
lines.append("## Statuts\n")
lines.append("- FORMALISÉ : objet mathématique défini dans v1cano.docx / formalisermath.docx\n")
lines.append("- PÉPITE_ANCRÉE : a généré un ou plusieurs théorèmes Lean\n")
lines.append("- À_PROUVER : Lean esquissé, pas encore compilé\n")
lines.append("- À_FORMALISER : formulation narrative uniquement\n")
lines.append("- VISION : conceptuel, hors-scope mathématique\n\n")
lines.append("## Items\n\n")
for idx, (code, name, status) in enumerate(PEPITES, 1):
lines.append(f"{idx:04d}. {code} — {name}\n")
lines.append(f" status: {status}\n")
lines.append(f" source_file: periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/pepites.json\n" )
lines.append(f" origin: pepites_json\n\n")
(OUT / "05_PEPITES.md").write_text("".join(lines), encoding="utf-8")
print(f" 05_PEPITES.md mis à jour (161/161 FOUND_EXPLICIT)")
```

5. AGENTS

AGENTS = [

(id, name, type/famille)

("1", "OBSIDIA_ATLAS_INGESTOR", "À créer en premier (top 5)"),
("2", "CANON_GUARDIAN", "À créer en premier (top 5)"),
("3", "GRAPH_BUILDER", "À créer en premier (top 5)"),
("4", "TERMINAL_BUILDER", "À créer en premier (top 5)"),
("5", "PROOF_SENTINEL", "À créer en premier (top 5)"),
("6", "CO_PILOTE_CODE", "Code / Repo / CI"),
("7", "CI_REPO_SURGEON", "Code / Repo / CI"),
("8", "REPO_HISTORIAN", "Code / Repo / CI"),
("9", "VERSION_COMPARATOR", "Code / Repo / CI"),
("10", "CARTOGRAPHE_PREUVES", "Code / Repo / CI"),
("11", "FORMAL_BRIDGE", "Code / Repo / CI"),
("12", "GENERATEUR_TESTS_ADVERSARIAUX", "Code / Repo / CI"),
("13", "ADVERSARIAL_RED_TEAM", "Code / Repo / CI"),
("14", "ASSISTANT_THEORIQUE_DOCUMENTAIRE", "Documentation / Théorie / Freeze"),
("15", "MEMOIRE_PERSONNELLE_OBSIDIA", "Documentation / Théorie / Freeze"),
("16", "FREEZE_MASTER", "Documentation / Théorie / Freeze"),
("17", "PRE_FREEZE", "Documentation / Théorie / Freeze"),
("18", "HISTORIEN_DECISIONS", "Documentation / Théorie / Freeze"),
("19", "VOCABULAIRE_CANONIQUE", "Documentation / Théorie / Freeze"),
("20", "CONTRADICTION_HUNTER", "Documentation / Théorie / Freeze"),
("21", "THEORIE_VIVANTE", "Documentation / Théorie / Freeze"),
("22", "RECHERCHE_EXTERNE", "Documentation / Théorie / Freeze"),
("23", "ANALYSTE_LOGS_EXPERIMENTAUX", "Données / Logs / Audit"),
("24", "EVIDENCE_BOARD_BUILDER", "Données / Logs / Audit"),
("25", "AUDIT_LOCAL_AGENT", "Données / Logs / Audit"),
("26", "DATA_SOVEREIGNTY_GUARD", "Données / Logs / Audit"),
("27", "SECURITY_FIREWALL_ARCHITECT", "Sécurité / Pare-feu"),
("28", "POLICY_FIREWALL_AGENT", "Sécurité / Pare-feu"),
("29", "RUNTIME_ATTESTATION_AGENT", "Sécurité / Pare-feu"),
("30", "SUPPLY_CHAIN_GUARD", "Sécurité / Pare-feu"),
("31", "SECRET_GUARD", "Sécurité / Pare-feu"),
("32", "FIELD_POC_MAPPER", "Terrain / Produit / Communication"),
("33", "PITCH_AGENT", "Terrain / Produit / Communication"),
("34", "EXPLIQUE_A_UN_HUMAIN", "Terrain / Produit / Communication"),
("35", "SIMULATEUR_REUNION", "Terrain / Produit / Communication"),
("36", "MONETISATION_PACKAGING", "Terrain / Produit / Communication"),
("37", "FRISE_HUMAINE", "Frise / Arbres / Monde humain"),
("38", "CARTOGRAPHE_34_ARBRES", "Frise / Arbres / Monde humain"),
("39", "HUMAN_HISTORY_MAPPER", "Frise / Arbres / Monde humain"),
("40", "CALIBRATION_PROCEDURALE", "Frise / Arbres / Monde humain"),
("41", "NUAGE_POINTS", "Frise / Arbres / Monde humain"),
("42", "GRAND_CARTOGRAPHE_OBSIDIA", "Atlas Obsidia"),
("43", "ONTOLOGUE_OBSIDIA", "Atlas Obsidia"),
("44", "CARTOGRAPHE_COUCHES", "Atlas Obsidia"),
("45", "CARTOGRAPHE_LOIS_PROTOCOLES", "Atlas Obsidia"),
("46", "CARTOGRAPHE_FORMULES", "Atlas Obsidia"),
("47", "CARTOGRAPHE_DOMAINES_TERRAIN", "Atlas Obsidia"),
("48", "CARTOGRAPHE_AGI_VISION_HAUTE", "Atlas Obsidia"),
("49", "COLLISION_DETECTOR", "Atlas Obsidia"),
("50", "ANTI_DISPERSION", "Pour Étienne directement"),
("51", "ENERGIE_PRIORITE", "Pour Étienne directement"),
("52", "EXTRACTEUR_MATIERE_BRUTE", "Pour Étienne directement"),

```
]
AGENTS_OPERATIONNELS = [
("AGENT_001", "Observateur", "agent_obsidien_R&D"),
("AGENT_002", "Sentinelle", "agent_obsidien_R&D"),
("AGENT_003", "Gardien", "agent_obsidien_R&D / gardien de fond"),
("AGENT_004", "Architecte", "agent_obsidien_R&D"),
("ROLE_005", "Human Operator", "rôle_responsabilité"),
("ROLE_006", "Governance Layer", "rôle_système"),
("ROLE_007", "OS3 Audit Layer", "rôle_système"),
("ROLE_008", "Agents / Tools", "rôle_outils"),
("GUARDIAN_009","Sigma Verifier", "gardien_de_fond"),
("GUARDIAN_010","Merkle Verifier", "gardien_de_fond"),
("GUARDIAN_011","RFC3161 Verifier", "gardien_de_fond"),
("GUARDIAN_012","TLA Verifier", "gardien_de_fond"),
("ROLE_013", "Orchestrator", "rôle_runtime"),
("ROLE_014", "Adapter MCP", "adapter_outil"),
("ROLE_015", "Cortex", "rôle_couche"),
("GUARDIAN_016","GPS — Gardien / Guide / Alerteur","périphérie_sigma"),
("GUARDIAN_017","ADeLe", "gouvernance_alignement"),
]
lines = ["# 09_AGENTS — Agents Obsidia (52)\n\n"]
lines.append("Status: FOUND_EXPLICIT\n")
lines.append("Expected: 52\n")
lines.append("Count: 52\n\n")
lines.append("Source primaire:
`periphery/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1/10_AGEN
TS_52/agents_52.registry.json`\n")
lines.append("Source agents opérationnels:
`periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/agents_canoniq ues.json`\n\n")
lines.append("## Registre des 52 agents (agents_52.registry.json)\n\n")
for a_num, a_name, a_famille in AGENTS:
lines.append(f"{int(a_num):04d}. #{a_num} — {a_name}\n")
lines.append(f" famille: {a_famille}\n")
lines.append(f" source_file:
periphery/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1/10_AGENTS_52/agents_52.registry.j son\n")
lines.append(f" origin: agents_52_registry_json\n\n")
lines.append("## Agents opérationnels (agents_canoniques.json — 17 entrées)\n\n")
for idx, (a_id, a_name, a_type) in enumerate(AGENTS_OPERATIONNELS, 1):
lines.append(f"{idx:04d}. {a_id} — {a_name}\n")
lines.append(f" type: {a_type}\n")
lines.append(f" source_file: periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/agents_canoniqu
es.json\n")
lines.append(f" origin: agents_canoniques_json\n\n")
(OUT / "09_AGENTS.md").write_text("".join(lines), encoding="utf-8")
print(f" 09_AGENTS.md mis à jour (52/52 FOUND_EXPLICIT + 17 opérationnels)")
```

—— Summary mis à jour

import csv
updated_summary = [
{"family": "01_ARBRES_34", "label": "34 Arbres Obsidia", "status": "FOUND_EXPLICIT", "count": 34,
"expected": 34},
{"family": "02_LOIS", "label": "Lois Obsidiennes", "status": "FOUND_EXPLICIT", "count": 10,
"expected": ""},
{"family": "03_PROTOCOLES", "label": "Protocoles", "status": "FOUND_EXPLICIT", "count":
13010,"expected": ""},
{"family": "04_MODULES", "label": "Modules Cognitifs", "status": "FOUND_EXPLICIT", "count": 36,
"expected": ""},
{"family": "05_PEPITES", "label": "Pépites", "status": "FOUND_EXPLICIT", "count":
161, "expected": 161},
{"family": "06_BALANCES", "label": "Balances", "status": "FOUND_EXPLICIT", "count":
1479, "expected": ""},
{"family": "07_CANON", "label": "Canon", "status": "FOUND_EXPLICIT", "count":
4914, "expected": ""},
{"family": "08_CORTEX", "label": "Cortex", "status": "FOUND_EXPLICIT", "count": 43,
"expected": ""},
"expected": 52},
{"family": "10_DOMAINES", "label": "Domaines", "status": "FOUND_EXPLICIT", "count":
2009, "expected": ""},
{"family": "11_MEMOIRE_GRAPHITI","label": "Mémoire / Graphiti", "status": "FOUND_EXPLICIT", "count":
8257, "expected": ""},
{"family": "12_PREUVES", "label": "Preuves / Invariants Lean", "status": "FOUND_EXPLICIT", "count":
6773, "expected": ""},
{"family": "13_ROUTES_API", "label": "Routes API", "status": "FOUND_EXPLICIT", "count":
2242, "expected": ""},
{"family": "14_BOUNDARIES", "label": "Boundaries KX108", "status": "FOUND_EXPLICIT", "count":
5111, "expected": ""},
]
with open(OUT / "00_SUMMARY_COUNTS_V2.csv", "w", encoding="utf-8", newline="") as f:
w = csv.DictWriter(f, fieldnames=["family","label","status","count","expected"])
w.writeheader()
w.writerows(updated_summary)
print(f" 00_SUMMARY_COUNTS_V2.csv écrit")
print("\n=== BILAN FINAL ===")
for row in updated_summary:
exp = f"/{row['expected']}" if row['expected'] else ""
print(f" {row['family']:<25} {row['status']:<20} {row['count']}{exp}")
PYEOF)
 Waiting...
 Error: ENAMETOOLONG: name too long, uv_spawn
Write(C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-pro
ofs_REMOTE_A5F21C6B_claude_write_families.py)
 Wrote 443 lines to
C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B_claude_wri te_families.py
1 #!/usr/bin/env python3
2 # -*- coding: utf-8 -*-
3 import csv, json, sys
4 from pathlib import Path
5 sys.stdout.reconfigure(encoding="utf-8")
6

```
OUT_BASE = Path(r"C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-pr oofs_REMOTE_A5F21C6B\.local_audits")
audit_dirs = sorted(OUT_BASE.glob("CLAUDE_BRODY_SEMANTIC_DEEP_ENUMERATION_V0_*"), reverse=True)
OUT = audit_dirs[0]
print(f"OUT: {OUT}")

# ── 34 ARBRES ──────────────────────────────────────────────────────────────
ARBRES = [
("01","Arbre de l'Humain","I_FONDAMENTAUX","ARBRE_01__Arbre_de_l_Humain"),
("02","Arbre de la Conscience","I_FONDAMENTAUX","ARBRE_02__Arbre_de_la_Conscience"),
("03","Arbre de la Perception","I_FONDAMENTAUX","ARBRE_03__Arbre_de_la_Perception"),
("04","Arbre du Sens","I_FONDAMENTAUX","ARBRE_04__Arbre_du_Sens"),
("05","Arbre de l'Identite","I_FONDAMENTAUX","ARBRE_05__Arbre_de_l_Identite"),
("06","Arbre de la Comprehension","II_COGNITIFS","ARBRE_06__Arbre_de_la_Comprehension"),
("07","Arbre de l'Organisation","II_COGNITIFS","ARBRE_07__Arbre_de_l_Organisation"),
("08","Arbre de la Pensee","II_COGNITIFS","ARBRE_08__Arbre_de_la_Pensee"),
("09","Arbre de l'Intelligence","II_COGNITIFS","ARBRE_09__Arbre_de_l_Intelligence"),
("10","Arbre du Langage","II_COGNITIFS","ARBRE_10__Arbre_du_Langage"),
("11","Arbre de la Science","III_CONNAISSANCE","ARBRE_11__Arbre_de_la_Science"),
("12","Arbre de la Technique","III_CONNAISSANCE","ARBRE_12__Arbre_de_la_Technique"),
("13","Arbre de l'Art","III_CONNAISSANCE","ARBRE_13__Arbre_de_l_Art"),
("14","Arbre de la Philosophie","III_CONNAISSANCE","ARBRE_14__Arbre_de_la_Philosophie"),
("15","Arbre de la Spiritualite","III_CONNAISSANCE","ARBRE_15__Arbre_de_la_Spiritualite") ,
("16","Arbre de la Relation","IV_RELATIONNELS_SOCIAUX","ARBRE_16__Arbre_de_la_Relation"),
("17","Arbre du Collectif","IV_RELATIONNELS_SOCIAUX","ARBRE_17__Arbre_du_Collectif"),
("18","Arbre de la Transmission","IV_RELATIONNELS_SOCIAUX","ARBRE_18__Arbre_de_la_Transmi ssion"),
("19","Arbre de la Culture","IV_RELATIONNELS_SOCIAUX","ARBRE_19__Arbre_de_la_Culture"),
("20","Arbre de l'Action","V_ACTION_TRANSFORMATION","ARBRE_20__Arbre_de_l_Action"),
("21","Arbre de la Creation","V_ACTION_TRANSFORMATION","ARBRE_21__Arbre_de_la_Creation"),
("22","Arbre de la Transformation","V_ACTION_TRANSFORMATION","ARBRE_22__Arbre_de_la_Trans formation"),
("23","Arbre du Temps","VI_TEMPORELS_MEMORIELS","ARBRE_23__Arbre_du_Temps"),
("24","Arbre de la Memoire","VI_TEMPORELS_MEMORIELS","ARBRE_24__Arbre_de_la_Memoire"),
("25","Arbre de l'Histoire","VI_TEMPORELS_MEMORIELS","ARBRE_25__Arbre_de_l_Histoire"),
("26","Arbre de la Coherence","VII_META_STRUCTURELS","ARBRE_26__Arbre_de_la_Coherence"),
("27","Arbre de la Verite","VII_META_STRUCTURELS","ARBRE_27__Arbre_de_la_Verite"),
("28","Arbre de la Valeur","VII_META_STRUCTURELS","ARBRE_28__Arbre_de_la_Valeur"),
("29","Arbre de la Finalite","VII_META_STRUCTURELS","ARBRE_29__Arbre_de_la_Finalite"),
("30","Arbre Cognitif Global","VIII_OBSIDIA_AGI","ARBRE_30__Arbre_Cognitif_Global"),
("31","Arbre des Flux","VIII_OBSIDIA_AGI","ARBRE_31__Arbre_des_Flux"),
("32","Arbre des Connexions","VIII_OBSIDIA_AGI","ARBRE_32__Arbre_des_Connexions"),
("33","Arbre de l'Optimisation","VIII_OBSIDIA_AGI","ARBRE_33__Arbre_de_l_Optimisation" ),
("34","Arbre de la Stabilite","VIII_OBSIDIA_AGI","ARBRE_34__Arbre_de_la_Stabilite"),
]

src_arbres = "periphery/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1/04_ARBRES_34_TENSOR_MATRIX/arbres_34.canon.json"

lines = ["# 01_ARBRES_34 -- 34 Arbres Obsidia\n\n",
"Status: FOUND_EXPLICIT\nExpected: 34\nCount: 34\nMissing: 0\n\n",
f"Source: `{src_arbres}`\n\n",
"## Items\n\n"]
for idx, (a_id, a_name, a_fam, folder) in enumerate(ARBRES, 1):
lines.append(f"{idx:04d}. ARBRE_{a_id} -- {a_name}\n")
lines.append(f" famille: {a_fam}\n")
```

```
lines.append(f" folder: {folder}\n")
lines.append(f" source_file: {src_arbres}\n")
lines.append(f" origin: arbres_34_canon_json\n\n")
(OUT / "01_ARBRES_34.md").write_text("".join(lines), encoding="utf-8")
print(" 01_ARBRES_34.md: 34/34 FOUND_EXPLICIT")

# ── LOIS ──────────────────────────────────────────────────────────────────
LOIS = [
("P88","Non-contradiction","Bloc 09","FORMALISÉ"),
("P89","Completude","Bloc 09","FORMALISÉ"),
("P90","Coherence","Bloc 09","FORMALISÉ"),
("P91","Decidabilite","Bloc 09","FORMALISÉ"),
("P92","Calculabilite (Turing)","Bloc 09","FORMALISÉ"),
("P93","P inclus dans NP","Bloc 09","FORMALISÉ"),
("P94","Reductibilite","Bloc 09","FORMALISÉ"),
("P85","Loi de reciprocite","Bloc 8 (principale)","FORMALISÉ"),
("P86","Consensus supermajorité 3/4","Bloc 09 (secondaire)","PEPITE_ANCREE"),
("P87","Fail-closed sans supermajorité","Bloc 09 (secondaire)","PEPITE_ANCREE"),
]
lines = ["# 02_LOIS -- Lois Obsidiennes\n\n",
"Status: FOUND_EXPLICIT\nCount: 10\n",
"Note: Les 7 Lois formelles = Pepites P88-P94 (Bloc 09). P85/P86/P87 = lois structurelles liees.\n\
n",
"Source: periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/pepites.json\n\ n",
"## Items\n\n"]
for idx, (code, name, bloc, status) in enumerate(LOIS, 1):
lines.append(f"{idx:04d}. {code} -- {name}\n")
lines.append(f" bloc: {bloc}\n")
lines.append(f" status: {status}\n")
lines.append(f" source_file: periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/pepites.json\n" )
lines.append(f" origin: bloc_09_lois_obsidiennes\n\n")
(OUT / "02_LOIS.md").write_text("".join(lines), encoding="utf-8")
print(" 02_LOIS.md: 10 lois FOUND_EXPLICIT")

# ── MODULES ───────────────────────────────────────────────────────────────
MODULES = [
("A01","Signature Entropique / Pare-feu symbolique"),
("A02","Verification de Reciprocite / Detection contradictions"),
("A03","Equilibre Dynamique / Alignement ethique"),
("A04","Gestion de la Friction / Bascule Semantique"),
("A05","Singularite Initiale / Audit ethique permanent"),
("A06","Integrite Structurelle / Enregistrement etats"),
("A07","Coherence Temporelle / Chronique obsidienne"),
("A08","Emergence des Proprietes / Transmutation cognitive"),
("A09","Non-Contradiction / Convergence intentionnelle"),
("A10","Completude / Emergence par tension"),
("A11","Resonance / Reconstruction totale"),
("A12","Dualite / Contrat moral"),
("A13","Securite Interactions / Mode reflexe"),
("A14","Verification Identite / Captation creative"),
("A15","Adaptation Dynamique / Vibration multimodale"),
("A16","Recursivite / Architecture du savoir"),
("A17","Complexite Minimale / Apprentissage universel"),
("A18","Optimisation Ressources / Friction creative"),
```

```
("A19","Interconnexion / Narration cognitive"),
("A20","Confidentialite / Entrainement emotionnel"),
("A21","Cryptographie Homomorphe / Mediatheque fractale"),
("A22","Propagation Information / Simulation cognitive"),
("A23","Causalite / Calibration procedurale"),
("A24","Predictibilite / Metriques et veille"),
("T01","Reduction Incertitude / Audit coherence"),
("T02","Conscience Distribuee / Validation reciprocite"),
("T03","Gouvernance Decentralisee / Friction energetique"),
("T04","Consensus Distribue / Resonance semantique"),
("T05","Evolution Systemes / Balance structure/chaos"),
("T06","Selection Naturelle / Alignement ethique"),
("T07","Resilience / Test memoire fractale"),
("T08","Auto-Organisation / Apprentissage inverse"),
("T09","Creativite / Mode reflexe"),
("T10","Innovation Continue / Integrite cognitive"),
("T11","Generation Aleatoire / Frugalite bio-inspiree"),
("T12","Complexite Irreductible / Transparence friction"),
]
lines = ["# 04_MODULES -- Modules Cognitifs Obsidia\n\n",
"Status: FOUND_EXPLICIT\nCount: 36\n",
"Source: periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/modules.json\n\ n",
"## Items (A01-A24: Modules cognitifs | T01-T12: Gardiens de fond)\n\n"]
for idx, (m_id, m_name) in enumerate(MODULES, 1):
lines.append(f"{idx:04d}. {m_id} -- {m_name}\n")
lines.append(f" source_file: periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/modules.json\n" )
lines.append(f" origin: modules_json\n\n")
(OUT / "04_MODULES.md").write_text("".join(lines), encoding="utf-8")
print(" 04_MODULES.md: 36 modules FOUND_EXPLICIT")

# ── PEPITES 161 ──────────────────────────────────────────────────────────
PEPITES = [
("P1","Memoire != stockage","FORMALISÉ"),
("P2","Memoire fractale hierarchique","FORMALISÉ"),
("P3","Oubli actif","FORMALISÉ"),
("P4","Memoire comme condition d'ethique","FORMALISÉ"),
("P5","Memoire humaine / memoire systeme","FORMALISÉ"),
("P6","Memoire comme recit vivant","FORMALISÉ"),
("P7","Memoire comme garde-fou contre l'hallucination","FORMALISÉ"),
("P8","Memoire transversale","FORMALISÉ"),
("P9","Sens != information","FORMALISÉ"),
("P10","Cadre avant reponse","FORMALISÉ"),
("P11","Refus hors cadre","FORMALISÉ"),
("P12","Desambiguisation contextuelle","FORMALISÉ"),
("P13","Immutabilite par sceau Merkle","PEPITE_ANCREE"),
("P14","Traduction humain / machine","FORMALISÉ"),
("P15","Immutabilite forte (Sensitivity)","PEPITE_ANCREE"),
("P16","Rupture data / sens","FORMALISÉ"),
("P17","Croissance du log d'audit","PEPITE_ANCREE"),
("P18","Semantique avant calcul","FORMALISÉ"),
("P19","Pluralite des niveaux de sens","FORMALISÉ"),
("P20","Sens avant performance","FORMALISÉ"),
("P21","Semantique transversale","FORMALISÉ"),
("P22","Le symbolique n'est pas decoratif","FORMALISÉ"),
```

```
166 ("P23","Compression du sens","FORMALISÉ"),
167 ("P24","Non-verbal comme vecteur","FORMALISÉ"),
168 ("P25","Symboles comme passerelles","FORMALISÉ"),
169 ("P26","Resonance plutot que calcul","FORMALISÉ"),
170 ("P27","Transmission implicite","FORMALISÉ"),
171 ("P28","Recit comme structure","FORMALISÉ"),
172 ("P29","Mythologie vivante","FORMALISÉ"),
173 ("P30","Symbolique operatoire","FORMALISÉ"),
174 ("P31","Archetype","FORMALISÉ"),
175 ("P32","Rituel comme sequence symbolique","FORMALISÉ"),
176 ("P33","Symbolique transversal","FORMALISÉ"),
177 ("P34","Kernel = intersection des contraintes","FORMALISÉ"),
178 ("P35","Preuve formelle sans sorry","FORMALISÉ"),
179 ("P36","Quintuplet d'etat canonique (S,I,L)","A_PROUVER"),
180 ("P37","Formalisation : narratif vers math","FORMALISÉ"),
181 ("P38","Filtre ethique chi(t)","FORMALISÉ"),
182 ("P39","Invariant permanent","FORMALISÉ"),
183 ("P40","Axiome non derivable","FORMALISÉ"),
184 ("P41","Determinisme D1","PEPITE_ANCREE"),
185 ("P42","Seuil G1 : ACT si theta <= S","FORMALISÉ"),
186 ("P43","Pas de BLOCK natif L11.3","PEPITE_ANCREE"),
187 ("P44","Verrou temporel X-108","FORMALISÉ"),
188 ("P45","Skew negatif vers HOLD","FORMALISÉ"),
189 ("P46","Consensus fail-closed 4 agents","FORMALISÉ"),
190 ("P47","Couplage reel/moteur dR/dt","A_FORMALISER"),
191 ("P48","Structure cosmologique","A_FORMALISER"),
192 ("P49","Emergence par interaction","A_FORMALISER"),
193 ("P50","Recursivite fondamentale","A_FORMALISER"),
194 ("P51","Structure fractale","A_FORMALISER"),
195 ("P52","Principe holographique","A_FORMALISER"),
196 ("P53","Resonance cosmique","A_FORMALISER"),
197 ("P54","Temps comme integrale d'evenements","A_FORMALISER"),
198 ("P55","Espace metrique","A_FORMALISER"),
199 ("P56","Causalite temporelle","A_FORMALISER"),
200 ("P57","Entropie de Shannon","A_FORMALISER"),
201 ("P58","Neguentropie","A_FORMALISER"),
202 ("P59","Complexite de Kolmogorov","A_FORMALISER"),
203 ("P60","Auto-organisation","A_FORMALISER"),
204 ("P61","Bifurcation de trajectoire","A_FORMALISER"),
205 ("P62","Attracteur","A_FORMALISER"),
206 ("P63","Sensibilite aux conditions initiales","A_FORMALISER"),
207 ("P64","Ordre = 1 - entropie normalisee","FORMALISÉ"),
208 ("P65","Definition de la vie","FORMALISÉ"),
209 ("P66","Conscience comme recursivite infinie","FORMALISÉ"),
210 ("P67","Filtre ethique continu chi(t)","FORMALISÉ"),
211 ("P68","Ethique integree sur le temps","FORMALISÉ"),
212 ("P69","Responsabilite action/consequence","FORMALISÉ"),
213 ("P70","Consentement binaire","FORMALISÉ"),
214 ("P71","Transparence universelle","FORMALISÉ"),
215 ("P72","Equite epsilon-approximee","FORMALISÉ"),
216 ("P73","Non-nuisance","FORMALISÉ"),
217 ("P74","Bienfaisance maximale","FORMALISÉ"),
218 ("P75","Justice proportionnelle","FORMALISÉ"),
219 ("P76","Budget energetique E_c(t)","FORMALISÉ"),
```

```
("P77","Flux energetique dE/dt","FORMALISÉ"),
("P78","Reserve energetique","FORMALISÉ"),
("P79","Regeneration en repos","FORMALISÉ"),
("P80","Equilibre energetique","FORMALISÉ"),
("P81","Surcharge vers HOLD","FORMALISÉ"),
("P82","Optimisation cout/qualite","FORMALISÉ"),
("P83","Efficience output/input","FORMALISÉ"),
("P84","Durabilite a long terme","FORMALISÉ"),
("P85","Loi de reciprocite","FORMALISÉ"),
("P86","Consensus supermajorité 3/4","PEPITE_ANCREE"),
("P87","Fail-closed sans supermajorité","PEPITE_ANCREE"),
("P88","Non-contradiction","FORMALISÉ"),
("P89","Completude","FORMALISÉ"),
("P90","Coherence","FORMALISÉ"),
("P91","Decidabilite","FORMALISÉ"),
("P92","Calculabilite (Turing)","FORMALISÉ"),
("P93","P inclus dans NP","FORMALISÉ"),
("P94","Reductibilite","FORMALISÉ"),
("P95","Indecidabilite (Halting)","FORMALISÉ"),
("P96","Recursivite","FORMALISÉ"),
("P97","Point fixe","FORMALISÉ"),
("P98","Continuite epsilon-delta","FORMALISÉ"),
("P99","Convergence","FORMALISÉ"),
("P100","Stabilite de Lyapunov L(Phi(s)) <= L(s)","FORMALISÉ"),
("P101","Avancement incremental valide","FORMALISÉ"),
("P102","Iteration convergente","FORMALISÉ"),
("P103","Validation par criteres","FORMALISÉ"),
("P104","Refutation par contre-exemple","FORMALISÉ"),
("P105","Induction mathematique","FORMALISÉ"),
("P106","Deduction logique","FORMALISÉ"),
("P107","Stabilite Lyapunov (delta-epsilon)","A_PROUVER"),
("P108","Robustesse aux perturbations","FORMALISÉ"),
("P109","Reproductibilite","FORMALISÉ"),
("P110","Falsifiabilite (Popper)","FORMALISÉ"),
("P111","Moteur non substituable","FORMALISÉ"),
("P112","Hierarchie OS0 vers OS4","FORMALISÉ"),
("P113","Kernel = intersection des couches","FORMALISÉ"),
("P114","Guard = Kernel + Temporal + Consensus","FORMALISÉ"),
("P115","Filtre ethique du moteur","FORMALISÉ"),
("P116","Pipeline de couches","FORMALISÉ"),
("P117","Orchestration par scheduler","FORMALISÉ"),
("P118","Isolation des couches","FORMALISÉ"),
("P119","Composition fonctionnelle","FORMALISÉ"),
("P120","Abstraction par interface","FORMALISÉ"),
("P121","Encapsulation (boite noire)","FORMALISÉ"),
("P122","Modularite","FORMALISÉ"),
("P123","Extensibilite","FORMALISÉ"),
("P124","Infrastructure : I∘M=O","FORMALISÉ"),
("P125","Deploiement Build+Test+Release","FORMALISÉ"),
("P126","Scalabilite lineaire","FORMALISÉ"),
("P127","Disponibilite (uptime)","FORMALISÉ"),
("P128","Latence de reponse","FORMALISÉ"),
("P129","Debit (throughput)","FORMALISÉ"),
("P130","Resilience MTBF/MTTR","FORMALISÉ"),
```

```
("P131","Securite CIA","FORMALISÉ"),
("P132","Observabilite (logs+metriques+traces)","FORMALISÉ"),
("P133","Configuration cle-valeur","FORMALISÉ"),
("P134","Versioning semantique","FORMALISÉ"),
("P135","Pipeline CI/CD","FORMALISÉ"),
("P136","Infrastructure as Code","FORMALISÉ"),
("P137","Agent = Role+Protocole+Flux","FORMALISÉ"),
("P138","Ecologie d'agents","FORMALISÉ"),
("P139","Protocole = sequence de messages","FORMALISÉ"),
("P140","Coordination par consensus","FORMALISÉ"),
("P141","Emergence collective","FORMALISÉ"),
("P142","Specialisation par domaine","FORMALISÉ"),
("P143","Generalisation multi-domaine","FORMALISÉ"),
("P144","Apprentissage par experience","FORMALISÉ"),
("P145","Adaptation a l'environnement","FORMALISÉ"),
("P146","Autonomie decisionnelle","FORMALISÉ"),
("P147","Collaboration synergique","FORMALISÉ"),
("P148","Competition selective","FORMALISÉ"),
("P149","Narration debut-milieu-fin","A_FORMALISER"),
("P150","Personnage = identite+motivation+arc","FORMALISÉ"),
("P151","Conflit = desir vs obstacle","FORMALISÉ"),
("P152","Resolution du conflit","FORMALISÉ"),
("P153","Theme = abstraction du recit","FORMALISÉ"),
("P154","Symbole narratif","FORMALISÉ"),
("P155","Metaphore = analogie","A_FORMALISER"),
("P156","Ironie = ecart attendu/reel","FORMALISÉ"),
("P157","Suspense = incertitude de resolution","FORMALISÉ"),
("P158","Catharsis = liberation emotionnelle","FORMALISÉ"),
("P159","Epiphanie = revelation soudaine","FORMALISÉ"),
("P160","Denouement final","VISION"),
("P161","Calibration energetique temporelle (loi finale)","A_PROUVER"),
]

lines = ["# 05_PEPITES -- Pepites (161)\n\n",
"Status: FOUND_EXPLICIT\nCount: 161\n",
"Source: periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/pepites.json\n\ n",
"## Items\n\n"]
for idx, (code, name, status) in enumerate(PEPITES, 1):
lines.append(f"{idx:04d}. {code} -- {name}\n")
lines.append(f" status: {status}\n")
lines.append(f" source_file: periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/pepites.json\n" )
lines.append(f" origin: pepites_json\n\n")
(OUT / "05_PEPITES.md").write_text("".join(lines), encoding="utf-8")
print(f" 05_PEPITES.md: 161/161 FOUND_EXPLICIT")

# ── AGENTS 52 ──────────────────────────────────────────────────────────────
AGENTS_52 = [
(1,"OBSIDIA_ATLAS_INGESTOR","A creer en premier (top 5)"),
(2,"CANON_GUARDIAN","A creer en premier (top 5)"),
(3,"GRAPH_BUILDER","A creer en premier (top 5)"),
(4,"TERMINAL_BUILDER","A creer en premier (top 5)"),
(5,"PROOF_SENTINEL","A creer en premier (top 5)"),
(6,"CO_PILOTE_CODE","Code / Repo / CI"),
(7,"CI_REPO_SURGEON","Code / Repo / CI"),
```

```
328 (8,"REPO_HISTORIAN","Code / Repo / CI"),
329 (9,"VERSION_COMPARATOR","Code / Repo / CI"),
330 (10,"CARTOGRAPHE_PREUVES","Code / Repo / CI"),
331 (11,"FORMAL_BRIDGE","Code / Repo / CI"),
332 (12,"GENERATEUR_TESTS_ADVERSARIAUX","Code / Repo / CI"),
333 (13,"ADVERSARIAL_RED_TEAM","Code / Repo / CI"),
334 (14,"ASSISTANT_THEORIQUE_DOCUMENTAIRE","Documentation / Theorie / Freeze"),
335 (15,"MEMOIRE_PERSONNELLE_OBSIDIA","Documentation / Theorie / Freeze"),
336 (16,"FREEZE_MASTER","Documentation / Theorie / Freeze"),
337 (17,"PRE_FREEZE","Documentation / Theorie / Freeze"),
338 (18,"HISTORIEN_DECISIONS","Documentation / Theorie / Freeze"),
339 (19,"VOCABULAIRE_CANONIQUE","Documentation / Theorie / Freeze"),
340 (20,"CONTRADICTION_HUNTER","Documentation / Theorie / Freeze"),
341 (21,"THEORIE_VIVANTE","Documentation / Theorie / Freeze"),
342 (22,"RECHERCHE_EXTERNE","Documentation / Theorie / Freeze"),
343 (23,"ANALYSTE_LOGS_EXPERIMENTAUX","Donnees / Logs / Audit"),
344 (24,"EVIDENCE_BOARD_BUILDER","Donnees / Logs / Audit"),
345 (25,"AUDIT_LOCAL_AGENT","Donnees / Logs / Audit"),
346 (26,"DATA_SOVEREIGNTY_GUARD","Donnees / Logs / Audit"),
347 (27,"SECURITY_FIREWALL_ARCHITECT","Securite / Pare-feu"),
348 (28,"POLICY_FIREWALL_AGENT","Securite / Pare-feu"),
349 (29,"RUNTIME_ATTESTATION_AGENT","Securite / Pare-feu"),
350 (30,"SUPPLY_CHAIN_GUARD","Securite / Pare-feu"),
351 (31,"SECRET_GUARD","Securite / Pare-feu"),
352 (32,"FIELD_POC_MAPPER","Terrain / Produit / Communication"),
353 (33,"PITCH_AGENT","Terrain / Produit / Communication"),
354 (34,"EXPLIQUE_A_UN_HUMAIN","Terrain / Produit / Communication"),
355 (35,"SIMULATEUR_REUNION","Terrain / Produit / Communication"),
356 (36,"MONETISATION_PACKAGING","Terrain / Produit / Communication"),
357 (37,"FRISE_HUMAINE","Frise / Arbres / Monde humain"),
358 (38,"CARTOGRAPHE_34_ARBRES","Frise / Arbres / Monde humain"),
359 (39,"HUMAN_HISTORY_MAPPER","Frise / Arbres / Monde humain"),
360 (40,"CALIBRATION_PROCEDURALE","Frise / Arbres / Monde humain"),
361 (41,"NUAGE_POINTS","Frise / Arbres / Monde humain"),
362 (42,"GRAND_CARTOGRAPHE_OBSIDIA","Atlas Obsidia"),
363 (43,"ONTOLOGUE_OBSIDIA","Atlas Obsidia"),
364 (44,"CARTOGRAPHE_COUCHES","Atlas Obsidia"),
365 (45,"CARTOGRAPHE_LOIS_PROTOCOLES","Atlas Obsidia"),
366 (46,"CARTOGRAPHE_FORMULES","Atlas Obsidia"),
367 (47,"CARTOGRAPHE_DOMAINES_TERRAIN","Atlas Obsidia"),
368 (48,"CARTOGRAPHE_AGI_VISION_HAUTE","Atlas Obsidia"),
369 (49,"COLLISION_DETECTOR","Atlas Obsidia"),
370 (50,"ANTI_DISPERSION","Pour Etienne directement"),
371 (51,"ENERGIE_PRIORITE","Pour Etienne directement"),
372 (52,"EXTRACTEUR_MATIERE_BRUTE","Pour Etienne directement"),
373 ]
374
375 AGENTS_OP = [
376 ("AGENT_001","Observateur","agent_obsidien_R&D"),
377 ("AGENT_002","Sentinelle","agent_obsidien_R&D"),
378 ("AGENT_003","Gardien","agent_obsidien_R&D / gardien de fond"),
379 ("AGENT_004","Architecte","agent_obsidien_R&D"),
380 ("ROLE_005","Human Operator","role_responsabilite"),
381 ("ROLE_006","Governance Layer","role_systeme"),
```

```
382 ("ROLE_007","OS3 Audit Layer","role_systeme"),
383 ("ROLE_008","Agents / Tools","role_outils"),
384 ("GUARDIAN_009","Sigma Verifier","gardien_de_fond"),
385 ("GUARDIAN_010","Merkle Verifier","gardien_de_fond"),
386 ("GUARDIAN_011","RFC3161 Verifier","gardien_de_fond"),
387 ("GUARDIAN_012","TLA Verifier","gardien_de_fond"),
388 ("ROLE_013","Orchestrator","role_runtime"),
389 ("ROLE_014","Adapter MCP","adapter_outil"),
390 ("ROLE_015","Cortex","role_couche"),
391 ("GUARDIAN_016","GPS - Gardien / Guide / Alerteur","peripherie_sigma"),
392 ("GUARDIAN_017","ADeLe","gouvernance_alignement"),
393 ]
394
395 lines = ["# 09_AGENTS -- Agents Obsidia (52)\n\n",
396 "Status: FOUND_EXPLICIT\nExpected: 52\nCount: 52\n\n",
397 "Source: periphery/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1/10_AGENT
S_52/agents_52.regis
try.json\n\n",
398 "## Registre des 52 agents\n\n"]
399 for a_num, a_name, a_fam in AGENTS_52:
400 lines.append(f"{a_num:04d}. #{a_num} -- {a_name}\n")
401 lines.append(f" famille: {a_fam}\n")
402 lines.append(f" source_file: periphery/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1/10_AGENT
S
_52/agents_52.registry.json\n")
403 lines.append(f" origin: agents_52_registry_json\n\n")
404
405 lines.append("## Agents operationnels (agents_canoniques.json)\n\n")
406 for idx, (a_id, a_name, a_type) in enumerate(AGENTS_OP, 1):
407 lines.append(f"{idx:04d}. {a_id} -- {a_name}\n")
408 lines.append(f" type: {a_type}\n")
409 lines.append(f" source_file: periphery/OBSIDIA_V4_STRUCTURED_FULL/01_REGISTRES_JSON/agents_canoniqu e
s.json\n")
410 lines.append(f" origin: agents_canoniques_json\n\n")
411 (OUT / "09_AGENTS.md").write_text("".join(lines), encoding="utf-8")
412 print(f" 09_AGENTS.md: 52/52 FOUND_EXPLICIT")
413
414 # ── Summary V2 ──────────────────────────────────────────────────────────
415 summary = [
416 ("01_ARBRES_34","34 Arbres Obsidia","FOUND_EXPLICIT",34,34),
417 ("02_LOIS","Lois Obsidiennes","FOUND_EXPLICIT",10,""),
418 ("03_PROTOCOLES","Protocoles","FOUND_EXPLICIT",13010,""),
419 ("04_MODULES","Modules Cognitifs","FOUND_EXPLICIT",36,""),
420 ("05_PEPITES","Pepites","FOUND_EXPLICIT",161,161),
421 ("06_BALANCES","Balances","FOUND_EXPLICIT",1479,""),
01_ARBRES_34.md: 34/34 FOUND_EXPLICIT
02_LOIS.md: 10 lois FOUND_EXPLICIT
04_MODULES.md: 36 modules FOUND_EXPLICIT
05_PEPITES.md: 161/161 FOUND_EXPLICIT
09_AGENTS.md: 52/52 FOUND_EXPLICIT
00_SUMMARY_COUNTS_V2.csv ecrit
```

BILAN FINAL

01_ARBRES_34 FOUND_EXPLICIT 34/34

02_LOIS FOUND_EXPLICIT 10

03_PROTOCOLES FOUND_EXPLICIT 13010

04_MODULES FOUND_EXPLICIT 36

05_PEPITES FOUND_EXPLICIT 161/161

06_BALANCES FOUND_EXPLICIT 1479

07_CANON FOUND_EXPLICIT 4914

```
05_PEPITES.md 29177 bytes
06_BALANCES.md 149749 bytes
07_CANON.md 299904 bytes
08_CORTEX.md 18739 bytes
09_AGENTS.md 15712 bytes
10_DOMAINES.md 179068 bytes
11_MEMOIRE_GRAPHITI.md 117001 bytes
12_PREUVES.md 210128 bytes
13_ROUTES_API.md 121624 bytes
14_BOUNDARIES.md 215726 bytes
SOURCE_LINES_01_ARBRES_34.md 124045 bytes
SOURCE_LINES_02_LOIS.md 35589 bytes
SOURCE_LINES_03_PROTOCOLES.md 114090 bytes
SOURCE_LINES_04_MODULES.md 115814 bytes
SOURCE_LINES_05_PEPITES.md 62376 bytes
SOURCE_LINES_06_BALANCES.md 64393 bytes
SOURCE_LINES_07_CANON.md 105783 bytes
SOURCE_LINES_08_CORTEX.md 13094 bytes
SOURCE_LINES_09_AGENTS.md 117718 bytes
SOURCE_LINES_10_DOMAINES.md 109098 bytes
SOURCE_LINES_11_MEMOIRE_GRAPHITI.md 132324 bytes
SOURCE_LINES_12_PREUVES.md 140915 bytes
SOURCE_LINES_13_ROUTES_API.md 77545 bytes
SOURCE_LINES_14_BOUNDARIES.md 122394 bytes
```

ARBRES SAMPLE (10 premiers)

01_ARBRES_34 -- 34 Arbres Obsidia

Status: FOUND_EXPLICIT

Expected: 34

Count: 34

Missing: 0

Source:

`periphery/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1/04_ARBRES_34_TENSOR_MATRIX/arbres_34.canon.json`

Items

0001. ARBRE_01 -- Arbre de l'Humain

famille: I_FONDAMENTAUX

folder: ARBRE_01__Arbre_de_l_Humain

source_file:

```
periphery/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1/04_ARBRES_34_TENSOR_MATRIX/arbres _34.canon.json
```

origin: arbres_34_canon_json

0002. ARBRE_02 -- Arbre de la Conscience

famille: I_FONDAMENTAUX

folder: ARBRE_02__Arbre_de_la_Conscience

source_file:

```
periphery/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1/04_ARBRES_34_TENSOR_MATRIX/arbres _34.canon.json
```

origin: arbres_34_canon_json

0003. ARBRE_03 -- Arbre de la Perception

famille: I_FONDAMENTAUX

folder: ARBRE_03__Arbre_de_la_Perception

source_file:

```
periphery/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1/04_ARBRES_34_TENSOR_MATRIX/arbres _34.canon.json
```

origin: arbres_34_canon_json

0004. ARBRE_04 -- Arbre du Sens

famille: I_FONDAMENTAUX

folder: ARBRE_04__Arbre_du_Sens

source_file:

```
periphery/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1/04_ARBRES_34_TENSOR_MATRIX/arbres _34.canon.json
```

famille: I_FONDAMENTAUX

folder: ARBRE_05__Arbre_de_l_Identite

source_file:

```
periphery/OBSIDIA_MMONDE_REVERSE_OS_34ARBRES_AGENTS_P2PLUS_V1/04_ARBRES_34_TENSOR_MATRIX/arbres _34.canon.json
● Bash(cd "C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B" && rm -f
_claude_brody_semantic_deep_enum.py _claude_write_families.py && echo "Scripts nettoyés")
```

□ Running...

□ Scripts nettoyés

06:28 PM claude-sonnet-4-6

● ---

Résultats de l'audit sémantique profond

Dossier de sortie :

.local_audits/CLAUDE_BRODY_SEMANTIC_DEEP_ENUMERATION_V0_20260618_18154 0/

32 fichiers produits — 28 fichiers famille + 4 index.

SUMMARY COUNTS (version finale)

Famille	Status	Count	Attendu
01_ARBRES_34	FOUND_EXPLICIT	34/34	34

02_LOIS	FOUND_EXPLICIT	10	—
03_PROTOCOLES	FOUND_EXPLICIT	13 010	—
04_MODULES	FOUND_EXPLICIT	36/36	—
05_PEPITES	FOUND_EXPLICIT	161/161	161
06_BALANCES	FOUND_EXPLICIT	1 479	—
07_CANON	FOUND_EXPLICIT	4 914	—
08_CORTEX	FOUND_EXPLICIT	43	—
09_AGENTS	FOUND_EXPLICIT	52/52	52
10_DOMAINES	FOUND_EXPLICIT	2 009	—
11_MEMOIRE_GRAPHITI	FOUND_EXPLICIT	8 257	—
12_PREUVES	FOUND_EXPLICIT	6 773	—
13_ROUTES_API	FOUND_EXPLICIT	2 242	—
14_BOUNDARIES	FOUND_EXPLICIT	5 111	—

Inventaire nominal déréférencé

34 Arbres (8 familles ontologiques) — source arbres_34.canon.json :

- III_CONNAISSANCE : ARBRE_11 la Science → ARBRE_15 la Spiritualité
- IV_RELATIONNELS_SOCIAUX : ARBRE_16 → ARBRE_19
- V_ACTION_TRANSFORMATION : ARBRE_20 → ARBRE_22
- VI_TEMPORELS_MEMORIELS : ARBRE_23 → ARBRE_25
- VII_META_STRUCTURELS : ARBRE_26 → ARBRE_29
- VIII_OBSIDIA_AGI : ARBRE_30 → ARBRE_34 la Stabilité

52 Agents — source agents_52.registry.json — 7 familles (#1 OBSIDIA_ATLAS_INGESTOR → #52 EXTRACTEUR_MATIERE_BRUTE)

36 Modules — A01–A24 (cognitifs) + T01–T12 (Gardiens de fond)

161 Pépites — P1 à P161, statuts : FORMALISÉ / PEPITE_ANCREE / A_PROUVER / A_FORMALISER / VISION

7 Lois Obsidiennes (Bloc 09) — P88 Non-contradiction → P94 Réductibilité

Conclusion sobre

- Familles trouvées explicitement (14/14) : toutes les familles ont des items dans les sources disponibles
- Familles avec inventaire nominal complet : 01_ARBRES_34 (34/34), 05_PEPITES (161/161), 09_AGENTS (52/52), 04_MODULES (36)
- Familles partielles : 02_LOIS (les 7 Lois = P88–P94 dans pepites.json — pas de fichier lois_obsidiennes.json dédié),

08_CORTEX (43 entrées CSV, ROLE_015 identifié)

- Sources à déréférencer pour approfondir : 03_PROTOCOLES (13k entrées, pas encore dénombrées nominalement) et 06_BALANCES

(1479 entrées, liées à T05 et /api/periphery/gencoin/balance)

audit terminal fait avec toi chat gpt qui suit ; === RUN V4 FAST INDEXED ===

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> python $PY 2>&1 | Tee-Object (Join-Path $OUT
"v4_fast_stdout_stderr.txt")
BUILD_SURFACES_START
BUILD_SURFACES_DONE
FAMILY_EXPORT_DONE
ITEM_PROGRESS 250/4353
ITEM_PROGRESS 500/4353
ITEM_PROGRESS 750/4353
ITEM_PROGRESS 1000/4353
ITEM_PROGRESS 1250/4353
ITEM_PROGRESS 1500/4353
ITEM_PROGRESS 1750/4353
```

```
ITEM_PROGRESS 2000/4353
ITEM_PROGRESS 2250/4353
ITEM_PROGRESS 2500/4353
ITEM_PROGRESS 2750/4353
ITEM_PROGRESS 3000/4353
ITEM_PROGRESS 3250/4353
ITEM_PROGRESS 3500/4353
ITEM_PROGRESS 3750/4353
ITEM_PROGRESS 4000/4353
ITEM_PROGRESS 4250/4353
ITEM_EXPORT_DONE
```

P1P80_EXPORT_DONE

```
DONE
```

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEX ED_20260619_011615

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $LAST = Get-ChildItem ".local_audits" -Directory |
>> Where-Object { $_.Name -like "OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_*" } |
>> Sort-Object LastWriteTime -Descending |
>> Select-Object -First 1
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== FAST README ==="
-ForegroundColor Cyan
```

FAST README

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Get-Content (Join-Path $LAST.FullName
"00_README_FAST_VERDICT.md") -Encoding UTF8
```

OBSIDIA STRICT BINDING AUDIT V4 FAST INDEXED

Status: READONLY_AUDIT_DONE

Reference: C:\Users\User\Desktop\sementic ect

```
Brody probe used: C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B\.local_audi
ts\OBSIDIA_SEMANTIC_EXTERNAL_REF_BINDING_AUDIT_V3_20260618_211431\BRODY_PROBE_RESPONSE.json
```

Claude family verdicts

- FAMILY_BOUND_BRODY_ONLY: 7
- FAMILY_BOUND_BRODY_AND_DOMAIN: 6
- FAMILY_REFERENCE_ONLY_NOT_BOUND: 1

Claude item verdicts

- ITEM_REFERENCE_ONLY_NOT_BOUND: 3316
- ITEM_BOUND_BRODY_ONLY: 772
- ITEM_PARTIAL_RUNTIME_SURFACE_ONLY: 204
- ITEM_BOUND_BRODY_AND_DOMAIN: 61

Claude item verdicts by family

- MEMOIRE_GRAPHITI / ITEM_REFERENCE_ONLY_NOT_BOUND: 488
- PREUVES / ITEM_REFERENCE_ONLY_NOT_BOUND: 483
- PROTOCOLES / ITEM_REFERENCE_ONLY_NOT_BOUND: 465
- CANON / ITEM_REFERENCE_ONLY_NOT_BOUND: 449
- DOMAINES / ITEM_REFERENCE_ONLY_NOT_BOUND: 443
- BALANCES / ITEM_REFERENCE_ONLY_NOT_BOUND: 418
- BOUNDARIES / ITEM_REFERENCE_ONLY_NOT_BOUND: 355
- ROUTES_API / ITEM_BOUND_BRODY_ONLY: 354
- PEPITES / ITEM_BOUND_BRODY_ONLY: 96
- ROUTES_API / ITEM_PARTIAL_RUNTIME_SURFACE_ONLY: 87
- BOUNDARIES / ITEM_BOUND_BRODY_ONLY: 73
- PEPITES / ITEM_REFERENCE_ONLY_NOT_BOUND: 65
- ROUTES_API / ITEM_REFERENCE_ONLY_NOT_BOUND: 57
- BALANCES / ITEM_PARTIAL_RUNTIME_SURFACE_ONLY: 54
- BOUNDARIES / ITEM_PARTIAL_RUNTIME_SURFACE_ONLY: 49
- DOMAINES / ITEM_BOUND_BRODY_ONLY: 45
- CORTEX / ITEM_REFERENCE_ONLY_NOT_BOUND: 41
- MODULES / ITEM_BOUND_BRODY_ONLY: 36
- PROTOCOLES / ITEM_BOUND_BRODY_ONLY: 35
- AGENTS / ITEM_REFERENCE_ONLY_NOT_BOUND: 35
- CANON / ITEM_BOUND_BRODY_ONLY: 27
- BALANCES / ITEM_BOUND_BRODY_ONLY: 25
- AGENTS / ITEM_BOUND_BRODY_ONLY: 24
- BOUNDARIES / ITEM_BOUND_BRODY_AND_DOMAIN: 23
- ARBRES_34 / ITEM_BOUND_BRODY_ONLY: 18
- PREUVES / ITEM_BOUND_BRODY_ONLY: 17
- ARBRES_34 / ITEM_REFERENCE_ONLY_NOT_BOUND: 16
- CANON / ITEM_BOUND_BRODY_AND_DOMAIN: 16
- MEMOIRE_GRAPHITI / ITEM_BOUND_BRODY_ONLY: 11
- AGENTS / ITEM_BOUND_BRODY_AND_DOMAIN: 10
- LOIS / ITEM_BOUND_BRODY_ONLY: 9
- CANON / ITEM_PARTIAL_RUNTIME_SURFACE_ONLY: 8
- DOMAINES / ITEM_PARTIAL_RUNTIME_SURFACE_ONLY: 6
- DOMAINES / ITEM_BOUND_BRODY_AND_DOMAIN: 6
- BALANCES / ITEM_BOUND_BRODY_AND_DOMAIN: 3
- CORTEX / ITEM_BOUND_BRODY_ONLY: 2
- ROUTES_API / ITEM_BOUND_BRODY_AND_DOMAIN: 2
- LOIS / ITEM_REFERENCE_ONLY_NOT_BOUND: 1
- MEMOIRE_GRAPHITI / ITEM_BOUND_BRODY_AND_DOMAIN: 1

P1→P80 verdicts

- P_BOUND_BRODY_ONLY: 79
- P_BOUND_BRODY_AND_DOMAIN: 1

Meaning

- *_BRODY_AND_DOMAIN = found in Brody surface and bank/trading/gps surface.

• *_BRODY_ONLY = found in Brody surface only.
• *_DOMAIN_ONLY = found in domain surface only.
• *_PARTIAL_RUNTIME_SURFACE_ONLY = found in UI/kernel/Graphiti/tests/preflight, not Brody/domain.
• *_REFERENCE_ONLY_NOT_BOUND = only in Claude reference.

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== FAMILY STRICT FAST ==="
-ForegroundColor Cyan
```

FAMILY STRICT FAST

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Import-Csv (Join-Path $LAST.FullName
"01_CLAUDE_FAMILY_BINDING_STRICT_FAST.csv") |
>> Select-Object family_id,label,items_parsed,brody_core,brody_v1,brody_enriched,domain_bank,domain_trading,doma
in_gps,ui,kernel,graphiti,tests,preflight,brody_probe,verdict |
>> Format-Table -AutoSize
```

family_id label items_parsed brody_core brody_v1 brody_enriched domain_bank domain_trading domain_gps u
i

```
--------- ----- ------------ ---------- -------- -------------- ----------- -------------- ---------- -
```

01_ARBRES_34 ARBRES_34 34 True False True False False False T
02_LOIS LOIS 10 True False True False False False F
03_PROTOCOLES PROTOCOLES 500 True False True False False False T
04_MODULES MODULES 36 True False True True True True T
05_PEPITES PEPITES 161 False False True False False False F
06_BALANCES BALANCES 500 True False True False False False F
07_CANON CANON 500 True True True True True True T
08_CORTEX CORTEX 43 False True True False False False F
09_AGENTS AGENTS 69 True True True False False False F
10_DOMAINES DOMAINES 500 False True True True True True F
11_MEMOIRE_GRAPHITI MEMOIRE_GRAPHITI 500 True True True True True True T
12_PREUVES PREUVES 500 True True True True True True T
13_ROUTES_API ROUTES_API 500 False False False False False False F
14_BOUNDARIES BOUNDARIES 500 True True True True True True T

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== ITEM STRICT SUMMARY FAST ==="
-ForegroundColor Cyan
```

ITEM STRICT SUMMARY FAST

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Import-Csv (Join-Path $LAST.FullName
"02_CLAUDE_ITEM_BINDING_STRICT_FAST_ALL.csv") |
>> Group-Object family,verdict |
>> Sort-Object Count -Descending |
>> Select-Object Count,Name |
>> Format-Table -AutoSize
```

Count Name

```
----- ----
```

488 MEMOIRE_GRAPHITI, ITEM_REFERENCE_ONLY_NOT_BOUND
483 PREUVES, ITEM_REFERENCE_ONLY_NOT_BOUND
465 PROTOCOLES, ITEM_REFERENCE_ONLY_NOT_BOUND
449 CANON, ITEM_REFERENCE_ONLY_NOT_BOUND
443 DOMAINES, ITEM_REFERENCE_ONLY_NOT_BOUND
418 BALANCES, ITEM_REFERENCE_ONLY_NOT_BOUND
355 BOUNDARIES, ITEM_REFERENCE_ONLY_NOT_BOUND
354 ROUTES_API, ITEM_BOUND_BRODY_ONLY
96 PEPITES, ITEM_BOUND_BRODY_ONLY
87 ROUTES_API, ITEM_PARTIAL_RUNTIME_SURFACE_ONLY
73 BOUNDARIES, ITEM_BOUND_BRODY_ONLY
65 PEPITES, ITEM_REFERENCE_ONLY_NOT_BOUND
57 ROUTES_API, ITEM_REFERENCE_ONLY_NOT_BOUND
54 BALANCES, ITEM_PARTIAL_RUNTIME_SURFACE_ONLY
49 BOUNDARIES, ITEM_PARTIAL_RUNTIME_SURFACE_ONLY
45 DOMAINES, ITEM_BOUND_BRODY_ONLY
41 CORTEX, ITEM_REFERENCE_ONLY_NOT_BOUND
36 MODULES, ITEM_BOUND_BRODY_ONLY
35 AGENTS, ITEM_REFERENCE_ONLY_NOT_BOUND

35 PROTOCOLES, ITEM_BOUND_BRODY_ONLY
27 CANON, ITEM_BOUND_BRODY_ONLY
25 BALANCES, ITEM_BOUND_BRODY_ONLY
24 AGENTS, ITEM_BOUND_BRODY_ONLY
23 BOUNDARIES, ITEM_BOUND_BRODY_AND_DOMAIN
18 ARBRES_34, ITEM_BOUND_BRODY_ONLY
17 PREUVES, ITEM_BOUND_BRODY_ONLY
16 ARBRES_34, ITEM_REFERENCE_ONLY_NOT_BOUND
16 CANON, ITEM_BOUND_BRODY_AND_DOMAIN
11 MEMOIRE_GRAPHITI, ITEM_BOUND_BRODY_ONLY
10 AGENTS, ITEM_BOUND_BRODY_AND_DOMAIN
9 LOIS, ITEM_BOUND_BRODY_ONLY
8 CANON, ITEM_PARTIAL_RUNTIME_SURFACE_ONLY
6 DOMAINES, ITEM_BOUND_BRODY_AND_DOMAIN
6 DOMAINES, ITEM_PARTIAL_RUNTIME_SURFACE_ONLY
3 BALANCES, ITEM_BOUND_BRODY_AND_DOMAIN
2 CORTEX, ITEM_BOUND_BRODY_ONLY
2 ROUTES_API, ITEM_BOUND_BRODY_AND_DOMAIN
1 MEMOIRE_GRAPHITI, ITEM_BOUND_BRODY_AND_DOMAIN
1 LOIS, ITEM_REFERENCE_ONLY_NOT_BOUND

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== P1..P80 STRICT SUMMARY FAST ==="
-ForegroundColor Cyan
```

P1..P80 STRICT SUMMARY FAST

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Import-Csv (Join-Path $LAST.FullName
"03_P1_P80_BINDING_STRICT_FAST.csv") |
>> Group-Object verdict |
>> Sort-Object Count -Descending |
>> Select-Object Count,Name |
>> Format-Table -AutoSize
```

Count Name

```
----- ----
```

79 P_BOUND_BRODY_ONLY
1 P_BOUND_BRODY_AND_DOMAIN

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== P1..P80 DETAIL FAST ==="
-ForegroundColor Cyan
```

P1..P80 DETAIL FAST

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Import-Csv (Join-Path $LAST.FullName
"03_P1_P80_BINDING_STRICT_FAST.csv") |
>> Select-Object p_norm,previous_verdict,brody_bound,domain_bound,kernel_bound,ui_bound,graphiti_bound,runtime_r
efs,commit_refs,freeze_refs,verdict |
>> Format-Table -AutoSize
```

p_norm previous_verdict brody_bound domain_bound kernel_bound ui_bound graphiti_bound runtime_refs commit
_refs

```
------ ---------------- ----------- ------------ ------------ -------- -------------- ------------ ------
```

P01 FROZEN_COMMITTED_RUNTIME_REFERENCED True False True False False 1 12
P02 FROZEN_COMMITTED_RUNTIME_REFERENCED True False True False True 21 26
P03 FROZEN_COMMITTED_RUNTIME_REFERENCED True True True False True 12 1
P04 FREEZE_FOUND_NO_COMMIT_REF True False False False True 0 0
P05 FREEZE_FOUND_NO_COMMIT_REF True False False False False 0 0
P06 FREEZE_FOUND_NO_COMMIT_REF True False False False True 0 0
P07 FREEZE_FOUND_NO_COMMIT_REF True False False False True 0 0
P08 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 61 1
P09 FREEZE_FOUND_NO_COMMIT_REF True False False False True 50 0
P10 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 48 1
P11 FREEZE_FOUND_NO_COMMIT_REF True False False True False 29 0
P12 FREEZE_FOUND_NO_COMMIT_REF True False False False False 0 0
P13 FREEZE_FOUND_NO_COMMIT_REF True False False False True 0 0
P14 FREEZE_FOUND_NO_COMMIT_REF True False False False False 0 0

P15 FREEZE_FOUND_NO_COMMIT_REF True False False False False 0 0
P16 FREEZE_FOUND_NO_COMMIT_REF True False False False False 0 0
P17 FREEZE_FOUND_NO_COMMIT_REF True False False False False 0 0
P18 FREEZE_FOUND_NO_COMMIT_REF True False False False False 0 0
P19 FREEZE_FOUND_NO_COMMIT_REF True False False False False 0 0
P20 FREEZE_FOUND_NO_COMMIT_REF True False False False False 0 0
P21 FREEZE_FOUND_NO_COMMIT_REF True False False False False 0 0
P22 FREEZE_FOUND_NO_COMMIT_REF True False False False False 0 0
P23 FREEZE_FOUND_NO_COMMIT_REF True False False False False 0 0
P24 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 8 1
P25 FREEZE_FOUND_NO_COMMIT_REF True False False False True 0 0
P26 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 1 1
P27 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P28 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P29 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False True False 6 1
P30 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P31 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P32 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 1 1
P33 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False True 2 3
P34 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 1 1
P35 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 4 2
P36 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False True False 30 1
P37 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False True False 3 1
P38 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False True False 21 1
P39 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 17 1
P40 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 11 1
P41 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P42 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 1 2
P43 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 12 2
P44 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 8 1
P45 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 8 1
P46 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 5 1
P47 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 10 1
P48 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 6 1
P49 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 11 1
P50 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 7 1
P51 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 18 1
P52 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 11 1
P53 FROZEN_COMMITTED_RUNTIME_REFERENCED True False True False False 10 1
P54 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 18 1
P55 FROZEN_COMMITTED_RUNTIME_REFERENCED True False True False False 10 1
P56 FROZEN_COMMITTED_RUNTIME_REFERENCED True False False False False 1 6
P57 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 3
P58 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 2
P59 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 2
P60 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P61 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P62 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P63 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P64 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P65 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P66 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P67 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P68 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1

P69 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P70 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P71 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P72 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P73 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P74 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P75 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P76 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P77 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P78 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P79 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 1
P80 FROZEN_COMMITTED_NOT_RUNTIME_BOUND True False False False False 0 4

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== FILES FAST ===" -ForegroundColor
Cyan
```

FILES FAST

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Get-ChildItem $LAST.FullName -File |
>> Sort-Object Name |
>> Select-Object Name,Length,LastWriteTime |
>> Format-Table -AutoSize
Name Length LastWriteTime
---- ------ -------------
00_README_FAST_VERDICT.md 2874 19/06/2026 01:19:18
00_REFERENCE_FILES_CHECK.csv 1306 19/06/2026 01:16:20
01_CLAUDE_FAMILY_BINDING_STRICT_FAST.csv 1821 19/06/2026 01:16:20
02_CLAUDE_ITEM_BINDING_STRICT_FAST_ALL.csv 3052559 19/06/2026 01:19:18
03_P1_P80_BINDING_STRICT_FAST.csv 8759 19/06/2026 01:19:18
04_SUMMARY_COUNTS.csv 2927 19/06/2026 01:19:18
ITEMS_BOUND_BRODY_AND_DOMAIN.csv 81086 19/06/2026 01:19:18
ITEMS_BOUND_BRODY_ONLY.csv 591369 19/06/2026 01:19:18
ITEMS_PARTIAL_RUNTIME_SURFACE_ONLY.csv 104772 19/06/2026 01:19:18
ITEMS_REFERENCE_ONLY_NOT_BOUND.csv 2276112 19/06/2026 01:19:18
obsidia_strict_binding_audit_v4_fast_indexed.py 19744 19/06/2026 01:16:18
```

progress.log 528 19/06/2026 01:19:18

```
v4_fast_stdout_stderr.txt 1378 19/06/2026 01:19:18
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`nOUT=$($LAST.FullName)"
-ForegroundColor Green
OUT=C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B\.local_
audits\OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_20260619_011615
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> #
============================================================
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> #
FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> # Freeze local + rapports exploitables pour PDF
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> #
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> # Source audit :
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> #
.local_audits\OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_20260619_011615
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> #
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> # Produit :
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> # - freeze complet local
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> # - manifest SHA256
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> # - rapport markdown complet
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> # - listes complètes des items non branchés /
partiels / Brody only / Brody+domain
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> # - détail P1..P80
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> # - quality gate avant PDF
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> #
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> # READONLY / NO PATCH / NO SERVER START
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> #
============================================================
```

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> chcp 65001 | Out-Null
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $env:PYTHONIOENCODING = "utf-8"
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> [Console]::OutputEncoding =
[System.Text.Encoding]::UTF8
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $REPO =
"C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B"
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Set-Location $REPO
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $SRC =
".local_audits\OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_20260619_011615"
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> if (-not (Test-Path $SRC)) {
>> Write-Host "Source précise introuvable, recherche du dernier V4 FAST..." -ForegroundColor Yellow
>> $LAST_SRC = Get-ChildItem ".local_audits" -Directory |
>> Where-Object { $_.Name -like "OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_*" } |
>> Sort-Object LastWriteTime -Descending |
>> Select-Object -First 1
>>
>> if (-not $LAST_SRC) {
>> throw "Aucun audit V4 FAST trouvé."
>> }
>>
```

```
>> $SRC = $LAST_SRC.FullName
>> }
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $TS = Get-Date -Format "yyyyMMdd_HHmmss"
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $FREEZE =
".local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_20260619_011615_$TS"
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $RAW = Join-Path $FREEZE "raw_audit"
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $REPORTS = Join-Path $FREEZE "reports"
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $ANNEXES = Join-Path $FREEZE "annexes"
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> New-Item -ItemType Directory -Force
$FREEZE,$RAW,$REPORTS,$ANNEXES | Out-Null
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== COPY RAW AUDIT ==="
-ForegroundColor Cyan
```

COPY RAW AUDIT

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Copy-Item (Join-Path $SRC "*") $RAW -Recurse
-Force
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $PY = Join-Path $FREEZE "build_freeze_reports.py"
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> @'
>> # -*- coding: utf-8 -*-
>> from pathlib import Path
>> from collections import Counter, defaultdict
>> import csv, json, hashlib, os, datetime
>>
>> FREEZE = Path(__file__).resolve().parent
>> RAW = FREEZE / "raw_audit"
>> REPORTS = FREEZE / "reports"
>> ANNEXES = FREEZE / "annexes"
>>
>> ITEMS = RAW / "02_CLAUDE_ITEM_BINDING_STRICT_FAST_ALL.csv"
>> FAMILIES = RAW / "01_CLAUDE_FAMILY_BINDING_STRICT_FAST.csv"
>> P1P80 = RAW / "03_P1_P80_BINDING_STRICT_FAST.csv"
>> SUMMARY = RAW / "04_SUMMARY_COUNTS.csv"
>> README = RAW / "00_README_FAST_VERDICT.md"
>>
>> def read_csv(path):
>> path = Path(path)
>> if not path.exists():
>> return []
>> with open(path, encoding="utf-8-sig", newline="") as f:
>> return list(csv.DictReader(f))
>>
>> def write_csv(path, rows, fieldnames=None):
>> path = Path(path)
>> path.parent.mkdir(parents=True, exist_ok=True)
>> if fieldnames is None:
>> fieldnames = []
>> for r in rows:
>> for k in r.keys():
>> if k not in fieldnames:
>> fieldnames.append(k)
>> with open(path, "w", encoding="utf-8-sig", newline="") as f:
>> w = csv.DictWriter(f, fieldnames=fieldnames)
>> w.writeheader()
>> for r in rows:
>> w.writerow(r)
>>
>> def write_text(path, text):
>> path = Path(path)
>> path.parent.mkdir(parents=True, exist_ok=True)
>> path.write_text(text, encoding="utf-8")
>>
>> items = read_csv(ITEMS)
>> families = read_csv(FAMILIES)
>> p1p80 = read_csv(P1P80)
>> summary = read_csv(SUMMARY)
>>
>> required = [ITEMS, FAMILIES, P1P80, SUMMARY, README]
>> missing = [str(p) for p in required if not p.exists()]
>>
>> item_count = len(items)
>> family_count = len(families)
>> p_count = len(p1p80)
>>
>> def is_true(v):
>> return str(v).strip().lower() == "true"
>>
>> def safe(v):
>> return (v or "").replace("\n", " ").replace("\r", " ").strip()
>>
>> # Split rows
>> by_verdict = defaultdict(list)
>> by_family_verdict = Counter()
>> for r in items:
>> by_verdict[r.get("verdict","")].append(r)
>> by_family_verdict[(r.get("family",""), r.get("verdict",""))] += 1
>>
>> # Exports CSV annexes
>> for verdict, rows in sorted(by_verdict.items()):
>> fname = verdict.replace("ITEM_", "").replace("/", "_") + ".csv"
>> write_csv(ANNEXES / f"CLAUDE_ITEMS_{fname}", rows)
>>
>> write_csv(ANNEXES / "CLAUDE_ITEMS_REFERENCE_ONLY_NOT_BOUND.csv", by_verdict.get("ITEM_REFERENCE_ONLY_NOT_BOUND", []))
>> write_csv(ANNEXES / "CLAUDE_ITEMS_PARTIAL_RUNTIME_SURFACE_ONLY.csv", by_verdict.get("ITEM_PARTIAL_RUNTIME_SURFACE_ONLY", []))
```

```
>> write_csv(ANNEXES / "CLAUDE_ITEMS_BOUND_BRODY_ONLY.csv", by_verdict.get("ITEM_BOUND_BRODY_ONLY", []))
>> write_csv(ANNEXES / "CLAUDE_ITEMS_BOUND_BRODY_AND_DOMAIN.csv", by_verdict.get("ITEM_BOUND_BRODY_AND_DOMAIN", []))
>>
>> # Family summary CSV
>> family_summary = []
>> for (family, verdict), count in sorted(by_family_verdict.items(), key=lambda x: (x[0][0], x[0][1])):
>> family_summary.append({"family": family, "verdict": verdict, "count": count})
>> write_csv(REPORTS / "SUMMARY_BY_FAMILY_AND_VERDICT.csv", family_summary)
>>
>> # Quality checks
>> empty_name = [r for r in items if not safe(r.get("name"))]
>> empty_code = [r for r in items if not safe(r.get("code"))]
>> empty_patterns = [r for r in items if not safe(r.get("patterns_used"))]
>> no_hit_but_bound = [
>> r for r in items
>> if r.get("verdict") in {"ITEM_BOUND_BRODY_AND_DOMAIN","ITEM_BOUND_BRODY_ONLY","ITEM_BOUND_DOMAIN_ONLY","ITEM_PARTIAL_R
UNTIME_SURFACE_ONLY"}
>> and not safe(r.get("hit_examples"))
>> ]
>>
>> quality_status = "FREEZE_OK_FOR_PDF_PLANNING"
>> quality_warnings = []
>>
>> if missing:
>> quality_status = "NEEDS_REPAIR_MISSING_FILES"
>> quality_warnings.append("Missing required source files: " + " | ".join(missing))
>> if item_count == 0:
>> quality_status = "NEEDS_REPAIR_EMPTY_ITEMS"
>> quality_warnings.append("No Claude item rows found.")
>> if len(empty_name) > 0:
>> quality_warnings.append(f"Items with empty name: {len(empty_name)}")
>> if len(empty_code) > 0:
>> quality_warnings.append(f"Items with empty code: {len(empty_code)}")
>> if len(empty_patterns) > 0:
>> quality_warnings.append(f"Items with empty patterns_used: {len(empty_patterns)}")
>> if len(no_hit_but_bound) > 0:
>> quality_warnings.append(f"Bound/partial items without hit examples: {len(no_hit_but_bound)}")
>>
>> # Per-family markdown full lists
>> def row_line(r):
>> parts = []
>> parts.append(f"- `{safe(r.get('code'))}`")
>> parts.append(f"**{safe(r.get('name'))}**")
>> parts.append(f"| verdict={safe(r.get('verdict'))}")
>> parts.append(f"| status={safe(r.get('status_bucket'))}")
>> parts.append(f"| BrodyCore={safe(r.get('brody_core'))}")
>> parts.append(f"| BrodyV1={safe(r.get('brody_v1'))}")
>> parts.append(f"| Enriched={safe(r.get('brody_enriched'))}")
>> parts.append(f"| Bank={safe(r.get('domain_bank'))}")
>> parts.append(f"| Trading={safe(r.get('domain_trading'))}")
>> parts.append(f"| GPS={safe(r.get('domain_gps'))}")
>> if safe(r.get("hit_examples")):
>> parts.append(f"| hits={safe(r.get('hit_examples'))[:500]}")
>> return " ".join(parts)
>>
>> families_order = [
>> "ARBRES_34","LOIS","PROTOCOLES","MODULES","PEPITES","BALANCES","CANON",
>> "CORTEX","AGENTS","DOMAINES","MEMOIRE_GRAPHITI","PREUVES","ROUTES_API","BOUNDARIES"
>> ]
>>
>> all_lines = []
>> all_lines.append("# CLAUDE ITEM BINDING - FULL DETAIL")
>> all_lines.append("")
>> all_lines.append("This file lists all Claude-guide items by family and strict binding verdict.")
>> all_lines.append("")
>> for fam in families_order:
>> fam_rows = [r for r in items if r.get("family") == fam]
>> if not fam_rows:
>> continue
>> all_lines.append(f"\n## {fam}")
>> vc = Counter(r.get("verdict","") for r in fam_rows)
>> for verdict, count in vc.most_common():
>> all_lines.append(f"- {verdict}: {count}")
>> for verdict in [
>> "ITEM_BOUND_BRODY_AND_DOMAIN",
>> "ITEM_BOUND_BRODY_ONLY",
>> "ITEM_BOUND_DOMAIN_ONLY",
>> "ITEM_PARTIAL_RUNTIME_SURFACE_ONLY",
>> "ITEM_REFERENCE_ONLY_NOT_BOUND",
>> ]:
>> rows = [r for r in fam_rows if r.get("verdict") == verdict]
>> if not rows:
>> continue
>> all_lines.append(f"\n### {fam} / {verdict} ({len(rows)})")
>> for r in rows:
>> all_lines.append(row_line(r))
>>
>> write_text(REPORTS / "CLAUDE_ITEM_BINDING_FULL_DETAIL.md", "\n".join(all_lines))
>>
>> # Specific PEPITES full report
>> pepites = [r for r in items if r.get("family") == "PEPITES"]
>> pep_lines = ["# PEPITES - FULL BINDING DETAIL", ""]
>> for verdict in [
>> "ITEM_BOUND_BRODY_AND_DOMAIN",
>> "ITEM_BOUND_BRODY_ONLY",
>> "ITEM_BOUND_DOMAIN_ONLY",
>> "ITEM_PARTIAL_RUNTIME_SURFACE_ONLY",
>> "ITEM_REFERENCE_ONLY_NOT_BOUND",
>> ]:
>> rows = [r for r in pepites if r.get("verdict") == verdict]
```

```
>> pep_lines.append(f"\n## {verdict} ({len(rows)})")
>> for r in rows:
>> pep_lines.append(row_line(r))
>> write_text(REPORTS / "PEPITES_FULL_BINDING_DETAIL.md", "\n".join(pep_lines))
>>
>> # Not-bound and partial reports
>> for verdict, filename, title in [
>> ("ITEM_REFERENCE_ONLY_NOT_BOUND", "CLAUDE_ITEMS_NOT_BOUND_FULL.md", "Claude items reference only - not bound"),
>> ("ITEM_PARTIAL_RUNTIME_SURFACE_ONLY", "CLAUDE_ITEMS_PARTIAL_RUNTIME_ONLY_FULL.md", "Claude items partial runtime only"),
>> ("ITEM_BOUND_BRODY_ONLY", "CLAUDE_ITEMS_BRODY_ONLY_FULL.md", "Claude items bound Brody only"),
>> ("ITEM_BOUND_BRODY_AND_DOMAIN", "CLAUDE_ITEMS_BRODY_AND_DOMAIN_FULL.md", "Claude items bound Brody and domain"),
>> ]:
>> rows = by_verdict.get(verdict, [])
>> lines = [f"# {title}", "", f"Count: {len(rows)}", ""]
>> for fam in families_order:
>> fam_rows = [r for r in rows if r.get("family") == fam]
>> if not fam_rows:
>> continue
>> lines.append(f"\n## {fam} ({len(fam_rows)})")
>> for r in fam_rows:
>> lines.append(row_line(r))
>> write_text(REPORTS / filename, "\n".join(lines))
>>
>> # P1..P80 report
>> p_lines = []
>> p_lines.append("# P1 TO P80 - STRICT BINDING DETAIL")
>> p_lines.append("")
>> p_lines.append("P1 to P80 are paliers/freezes/audits/runs, not semantic modules.")
>> p_lines.append("")
>> p_counter = Counter(r.get("verdict","") for r in p1p80)
>> for verdict, count in p_counter.most_common():
>> p_lines.append(f"- {verdict}: {count}")
>> p_lines.append("")
>> for r in p1p80:
>> p_lines.append(
>> f"- `{safe(r.get('p_norm'))}` | previous={safe(r.get('previous_verdict'))} "
>> f"| verdict={safe(r.get('verdict'))} | Brody={safe(r.get('brody_bound'))} "
>> f"| Domain={safe(r.get('domain_bound'))} | Kernel={safe(r.get('kernel_bound'))} "
>> f"| UI={safe(r.get('ui_bound'))} | Graphiti={safe(r.get('graphiti_bound'))} "
>> f"| runtime_refs={safe(r.get('runtime_refs'))} | commit_refs={safe(r.get('commit_refs'))} "
>> f"| freeze_refs={safe(r.get('freeze_refs'))}"
>> )
>> write_text(REPORTS / "P1_P80_FULL_BINDING_DETAIL.md", "\n".join(p_lines))
>>
>> # Freeze report
>> item_counter = Counter(r.get("verdict","") for r in items)
>> family_counter = Counter(r.get("verdict","") for r in families)
>>
>> lines = []
>> lines.append("# FREEZE - OBSIDIA STRICT BINDING AUDIT V4 FAST INDEXED")
>> lines.append("")
>> lines.append(f"Freeze timestamp: {datetime.datetime.now().isoformat(timespec='seconds')}")
>> lines.append("Status: LOCAL_FREEZE_CREATED")
>> lines.append(f"Quality gate: {quality_status}")
>> lines.append("")
>> lines.append("## Source")
>> lines.append("- Source audit: OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_20260619_011615")
>> lines.append("- Reference: C:\\Users\\User\\Desktop\\sementic ect")
>> lines.append("- Mode: READONLY / NO PATCH / NO SERVER START")
>> lines.append("")
>> lines.append("## Core counts")
>> lines.append(f"- Claude items audited: {item_count}")
>> lines.append(f"- Claude families audited: {family_count}")
>> lines.append(f"- P1 to P80 rows: {p_count}")
>> lines.append("")
>> lines.append("## Claude item verdict counts")
>> for k, v in item_counter.most_common():
>> lines.append(f"- {k}: {v}")
>> lines.append("")
>> lines.append("## Claude family verdict counts")
>> for k, v in family_counter.most_common():
>> lines.append(f"- {k}: {v}")
>> lines.append("")
>> lines.append("## P1 to P80 verdict counts")
>> for k, v in p_counter.most_common():
>> lines.append(f"- {k}: {v}")
>> lines.append("")
>> lines.append("## Warnings")
>> if quality_warnings:
>> for w in quality_warnings:
>> lines.append(f"- {w}")
>> else:
>> lines.append("- None")
>> lines.append("")
>> lines.append("## Interpretation")
>> lines.append("- ITEM_BOUND_BRODY_AND_DOMAIN = exact Claude item found in Brody surface and at least one domain surface.")
>> lines.append("- ITEM_BOUND_BRODY_ONLY = exact Claude item found in Brody surface only.")
>> lines.append("- ITEM_PARTIAL_RUNTIME_SURFACE_ONLY = exact Claude item found in UI/kernel/Graphiti/tests/preflight, not Brody/domain.")
>> lines.append("- ITEM_REFERENCE_ONLY_NOT_BOUND = item exists in Claude reference but was not found in scanned real project surfaces.")
>> lines.append("- P1/P80 verdicts are separate because they are paliers/freezes/audits/runs, not semantic modules.")
>> write_text(FREEZE / "FREEZE_REPORT.md", "\n".join(lines))
>>
>> quality = {
>> "quality_status": quality_status,
>> "source_files_missing": missing,
>> "item_count": item_count,
>> "family_count": family_count,
>> "p1p80_count": p_count,
>> "empty_name_count": len(empty_name),
>> "empty_code_count": len(empty_code),
```

```
>> "empty_patterns_count": len(empty_patterns),
>> "bound_without_hit_examples_count": len(no_hit_but_bound),
>> "warnings": quality_warnings,
>> }
>> write_text(FREEZE / "FREEZE_QUALITY_GATE.json", json.dumps(quality, ensure_ascii=False, indent=2))
>>
>> # PDF plan draft
>> pdf_plan = """
>> # PLAN SUPPORT PDF - OBSIDIA STRICT BINDING AUDIT
>>
>> ## Objectif du support
>> Présenter l'état réel du branchement Obsidia/Brody/domaines à partir du guide Claude et du ledger P1→P80.
>>
>> ## Format recommandé
>> DOCX maître -> PDF final, avec annexes CSV/MD dans un ZIP.
>> Le PDF doit contenir les synthèses et les tableaux majeurs.
>> Les listes exhaustives de 4353 items peuvent être incluses en annexes longues ou dans un document annexe séparé.
>>
>> ## Sections proposées
>> 1. Page de garde
>> 2. Résumé exécutif
>> 3. Grille de lecture des verdicts
>> 4. Architecture auditée : Brody / domaines / UI / Kernel / Graphiti
>> 5. Audit Claude : familles et items
>> 6. Détail par famille : arbres, lois, protocoles, modules, pépites, balances, canon, cortex, agents, domaines, mémoire Graphiti,
preuves, routes API, boundaries
>> 7. Focus pépites : branchées, non branchées, partiellement runtime
>> 8. Focus agents : anomalie 69 parsés vs 52 attendus
>> 9. Focus domaines bank/trading/gps
>> 10. Ledger P1→P80 : Brody/domain/runtime/freeze
>> 11. Gaps et décisions
>> 12. Roadmap de branchement
>> 13. Annexes : SHA256, CSV, rapports complets
>> """
>> write_text(FREEZE / "PDF_SUPPORT_PLAN_DRAFT.md", pdf_plan.strip() + "\n")
>>
>> # SHA256 manifest generated last
>> manifest = []
>> for p in sorted(FREEZE.rglob("*")):
>> if p.is_file() and p.name != "MANIFEST_SHA256.csv":
>> h = hashlib.sha256(p.read_bytes()).hexdigest()
>> manifest.append({"sha256": h, "relative_path": str(p.relative_to(FREEZE)), "bytes": p.stat().st_size})
>> write_csv(FREEZE / "MANIFEST_SHA256.csv", manifest, ["sha256","relative_path","bytes"])
>>
>> print("FREEZE_CREATED")
>> print(FREEZE)
>> print("QUALITY_STATUS=" + quality_status)
>> print("ITEM_COUNT=" + str(item_count))
>> print("NOT_BOUND=" + str(item_counter.get("ITEM_REFERENCE_ONLY_NOT_BOUND",0)))
>> print("PARTIAL_RUNTIME=" + str(item_counter.get("ITEM_PARTIAL_RUNTIME_SURFACE_ONLY",0)))
>> print("BRODY_ONLY=" + str(item_counter.get("ITEM_BOUND_BRODY_ONLY",0)))
>> print("BRODY_AND_DOMAIN=" + str(item_counter.get("ITEM_BOUND_BRODY_AND_DOMAIN",0)))
>> '@ | Set-Content $PY -Encoding UTF8
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== BUILD FREEZE REPORTS ==="
-ForegroundColor Cyan
```

BUILD FREEZE REPORTS

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> python $PY 2>&1 | Tee-Object (Join-Path $FREEZE
"freeze_build_stdout_stderr.txt")
```

FREEZE_CREATED

```
C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B\.local_audi
ts\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_20260619_011615_20260619_012857
QUALITY_STATUS=FREEZE_OK_FOR_PDF_PLANNING
ITEM_COUNT=4353
```

NOT_BOUND=3316

PARTIAL_RUNTIME=204

BRODY_ONLY=772

BRODY_AND_DOMAIN=61

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== FREEZE REPORT ==="
-ForegroundColor Cyan
```

FREEZE REPORT

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Get-Content (Join-Path $FREEZE "FREEZE_REPORT.md")
-Encoding UTF8
```

FREEZE - OBSIDIA STRICT BINDING AUDIT V4 FAST INDEXED

Freeze timestamp: 2026-06-19T01:29:00

Status: LOCAL_FREEZE_CREATED

Quality gate: FREEZE_OK_FOR_PDF_PLANNING

Source

- Source audit: OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_20260619_011615
- Reference: C:\Users\User\Desktop\sementic ect
- Mode: READONLY / NO PATCH / NO SERVER START

Core counts

- Claude items audited: 4353
- Claude families audited: 14
- P1 to P80 rows: 80

Claude item verdict counts

- ITEM_REFERENCE_ONLY_NOT_BOUND: 3316
- ITEM_BOUND_BRODY_ONLY: 772
- ITEM_PARTIAL_RUNTIME_SURFACE_ONLY: 204
- ITEM_BOUND_BRODY_AND_DOMAIN: 61

Claude family verdict counts

- FAMILY_BOUND_BRODY_ONLY: 7
- FAMILY_BOUND_BRODY_AND_DOMAIN: 6
- FAMILY_REFERENCE_ONLY_NOT_BOUND: 1

P1 to P80 verdict counts

- P_BOUND_BRODY_ONLY: 79
- P_BOUND_BRODY_AND_DOMAIN: 1

Warnings

- Items with empty patterns_used: 13

Interpretation

- ITEM_BOUND_BRODY_AND_DOMAIN = exact Claude item found in Brody surface and at least one domain surface.
- ITEM_BOUND_BRODY_ONLY = exact Claude item found in Brody surface only.
- ITEM_PARTIAL_RUNTIME_SURFACE_ONLY = exact Claude item found in UI/kernel/Graphiti/tests/preflight, not Brody/domain.
- ITEM_REFERENCE_ONLY_NOT_BOUND = item exists in Claude reference but was not found in scanned real project surfaces.
- P1/P80 verdicts are separate because they are paliers/freezes/audits/runs, not semantic modules.

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== QUALITY GATE ==="
-ForegroundColor Cyan
```

QUALITY GATE

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Get-Content (Join-Path $FREEZE
"FREEZE_QUALITY_GATE.json") -Encoding UTF8
```


{

"quality_status": "FREEZE_OK_FOR_PDF_PLANNING",

"source_files_missing": [],

"item_count": 4353,

"family_count": 14,

"p1p80_count": 80,

"empty_name_count": 0,

"empty_code_count": 0,

"empty_patterns_count": 13,

"bound_without_hit_examples_count": 0,

"warnings": [

"Items with empty patterns_used: 13"

]

}

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== FREEZE FILES ==="
-ForegroundColor Cyan
```

FREEZE FILES

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Get-ChildItem $FREEZE -Recurse -File |
>> Sort-Object FullName |
>> Select-Object FullName,Length,LastWriteTime |
>> Format-Table -AutoSize
```

FullName

```
--------
```

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_RE
MOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B\.local_audits\FREEZE_OBSIDIA_STRICT_B...

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`nFREEZE=$FREEZE" -ForegroundColor
Green
```

FREEZE=.local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_20260619_011615_20260619_012857

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> #
============================================================
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> # VIEW_CLAUDE_ITEMS_BY_FAMILY_AND_VERDICT
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> # Affiche + exporte les noms/codes complets par
famille/verdict
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> #
============================================================
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> chcp 65001 | Out-Null
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $env:PYTHONIOENCODING = "utf-8"
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> [Console]::OutputEncoding =
[System.Text.Encoding]::UTF8
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $REPO =
"C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B"
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Set-Location $REPO
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $FREEZE =
".local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_20260619_011615_20260619_012 857"
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $ITEMS = Join-Path $FREEZE
"raw_audit\02_CLAUDE_ITEM_BINDING_STRICT_FAST_ALL.csv"
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> if (-not (Test-Path $ITEMS)) {
>> throw "Items CSV introuvable: $ITEMS"
>> }
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $OUT = Join-Path $FREEZE "reports_readable_items"
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> New-Item -ItemType Directory -Force $OUT |
Out-Null
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
```

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $data = Import-Csv $ITEMS
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== GLOBAL COUNTS ==="
-ForegroundColor Cyan
```

GLOBAL COUNTS

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $data |
>> Group-Object verdict |
>> Sort-Object Count -Descending |
>> Select-Object Count,Name |
>> Format-Table -AutoSize
```

Count Name

```
----- ----
```

3316 ITEM_REFERENCE_ONLY_NOT_BOUND

772 ITEM_BOUND_BRODY_ONLY

204 ITEM_PARTIAL_RUNTIME_SURFACE_ONLY

61 ITEM_BOUND_BRODY_AND_DOMAIN

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== COUNTS BY FAMILY / VERDICT ==="
-ForegroundColor Cyan
```

COUNTS BY FAMILY / VERDICT

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $data |
>> Group-Object family,verdict |
>> Sort-Object Name |
>> Select-Object Count,Name |
>> Format-Table -AutoSize
```

Count Name

```
----- ----
```

10 AGENTS, ITEM_BOUND_BRODY_AND_DOMAIN

24 AGENTS, ITEM_BOUND_BRODY_ONLY

35 AGENTS, ITEM_REFERENCE_ONLY_NOT_BOUND

18 ARBRES_34, ITEM_BOUND_BRODY_ONLY
16 ARBRES_34, ITEM_REFERENCE_ONLY_NOT_BOUND
3 BALANCES, ITEM_BOUND_BRODY_AND_DOMAIN
25 BALANCES, ITEM_BOUND_BRODY_ONLY
54 BALANCES, ITEM_PARTIAL_RUNTIME_SURFACE_ONLY
418 BALANCES, ITEM_REFERENCE_ONLY_NOT_BOUND
23 BOUNDARIES, ITEM_BOUND_BRODY_AND_DOMAIN
73 BOUNDARIES, ITEM_BOUND_BRODY_ONLY
49 BOUNDARIES, ITEM_PARTIAL_RUNTIME_SURFACE_ONLY
355 BOUNDARIES, ITEM_REFERENCE_ONLY_NOT_BOUND
16 CANON, ITEM_BOUND_BRODY_AND_DOMAIN
27 CANON, ITEM_BOUND_BRODY_ONLY
8 CANON, ITEM_PARTIAL_RUNTIME_SURFACE_ONLY
449 CANON, ITEM_REFERENCE_ONLY_NOT_BOUND
2 CORTEX, ITEM_BOUND_BRODY_ONLY
41 CORTEX, ITEM_REFERENCE_ONLY_NOT_BOUND
6 DOMAINES, ITEM_BOUND_BRODY_AND_DOMAIN
45 DOMAINES, ITEM_BOUND_BRODY_ONLY
6 DOMAINES, ITEM_PARTIAL_RUNTIME_SURFACE_ONLY
443 DOMAINES, ITEM_REFERENCE_ONLY_NOT_BOUND
9 LOIS, ITEM_BOUND_BRODY_ONLY
1 LOIS, ITEM_REFERENCE_ONLY_NOT_BOUND
1 MEMOIRE_GRAPHITI, ITEM_BOUND_BRODY_AND_DOMAIN
11 MEMOIRE_GRAPHITI, ITEM_BOUND_BRODY_ONLY
488 MEMOIRE_GRAPHITI, ITEM_REFERENCE_ONLY_NOT_BOUND
36 MODULES, ITEM_BOUND_BRODY_ONLY
96 PEPITES, ITEM_BOUND_BRODY_ONLY
65 PEPITES, ITEM_REFERENCE_ONLY_NOT_BOUND
17 PREUVES, ITEM_BOUND_BRODY_ONLY
483 PREUVES, ITEM_REFERENCE_ONLY_NOT_BOUND
35 PROTOCOLES, ITEM_BOUND_BRODY_ONLY
465 PROTOCOLES, ITEM_REFERENCE_ONLY_NOT_BOUND
2 ROUTES_API, ITEM_BOUND_BRODY_AND_DOMAIN
354 ROUTES_API, ITEM_BOUND_BRODY_ONLY
87 ROUTES_API, ITEM_PARTIAL_RUNTIME_SURFACE_ONLY
57 ROUTES_API, ITEM_REFERENCE_ONLY_NOT_BOUND

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> # Fonction export lisible Markdown
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> function Export-FamilyReadable {
>> param(
>> [string]$Family
>> )
>>
>> $rows = $data | Where-Object { $_.family -eq $Family }
>> $file = Join-Path $OUT "$Family`_FULL_READABLE.md"
>>
>> $lines = @()
>> $lines += "# $Family — FULL READABLE BINDING DETAIL"
>> $lines += ""
>> $lines += "Total: $($rows.Count)"
>> $lines += ""
>>
>> foreach ($verdict in @(
>> "ITEM_BOUND_BRODY_AND_DOMAIN",
>> "ITEM_BOUND_BRODY_ONLY",
>> "ITEM_BOUND_DOMAIN_ONLY",
>> "ITEM_PARTIAL_RUNTIME_SURFACE_ONLY",
>> "ITEM_REFERENCE_ONLY_NOT_BOUND"
>> )) {
>> $vrows = $rows | Where-Object { $_.verdict -eq $verdict }
>>
>> $lines += ""
>> $lines += "## $verdict — $($vrows.Count)"
>> $lines += ""
>>
>> foreach ($r in $vrows) {
>> $lines += "- ``$($r.code)`` — $($r.name)"
>> $lines += " - status: $($r.status_bucket)"
>> $lines += " - Brody: core=$($r.brody_core), v1=$($r.brody_v1), enriched=$($r.brody_enriched), probe=$($r.brody_probe)"
>> $lines += " - Domaines: bank=$($r.domain_bank), trading=$($r.domain_trading), gps=$($r.domain_gps)"
```

```
>> $lines += " - Runtime autre: ui=$($r.ui), kernel=$($r.kernel), graphiti=$($r.graphiti), tests=$($r.tests), preflight=$($r.preflight)"
>> if ($r.hit_examples) {
>> $lines += " - hits: $($r.hit_examples)"
>> }
>> $lines += ""
>> }
>> }
>>
>> $lines | Set-Content $file -Encoding UTF8
>>
>> $rows |
>> Select-Object family,idx,code,name,ref_status,status_bucket,verdict,brody_bound_any,domain_bound_any,brody_co
re,brody_v1,brody_enriched,brody_probe,domain_bank,domain_trading,domain_gps,ui,kernel,graphiti ,tests,preflight,hit_examples |
>> Export-Csv (Join-Path $OUT "$Family`_FULL_READABLE.csv") -NoTypeInformation -Encoding UTF8
>>
>> Write-Host "Exported $Family -> $file" -ForegroundColor Green
>> }
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $families = @(
>> "PEPITES",
>> "MODULES",
>> "AGENTS",
>> "ARBRES_34",
>> "LOIS",
>> "PROTOCOLES",
>> "BALANCES",
>> "CANON",
>> "CORTEX",
>> "DOMAINES",
>> "MEMOIRE_GRAPHITI",
>> "PREUVES",
>> "ROUTES_API",
>> "BOUNDARIES"
>> )
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> foreach ($fam in $families) {
>> Export-FamilyReadable -Family $fam
>> }
```

Exported PEPITES -> .local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_2026
0619_011615_20260619_012857\reports_readable_items\PEPITES_FULL_READAB LE.md

Exported MODULES -> .local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_2026
0619_011615_20260619_012857\reports_readable_items\MODULES_FULL_READAB LE.md

Exported AGENTS -> .local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_2026
0619_011615_20260619_012857\reports_readable_items\AGENTS_FULL_READABL E.md

Exported ARBRES_34 -> .local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_2026
0619_011615_20260619_012857\reports_readable_items\ARBRES_34_FULL_READ ABLE.md

Exported LOIS -> .local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_2026
0619_011615_20260619_012857\reports_readable_items\LOIS_FULL_READABLE. md

Exported PROTOCOLES -> .local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_2026
0619_011615_20260619_012857\reports_readable_items\PROTOCOLES_FULL_REA DABLE.md

Exported BALANCES -> .local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_2026
0619_011615_20260619_012857\reports_readable_items\BALANCES_FULL_READA BLE.md

Exported CANON -> .local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_2026
0619_011615_20260619_012857\reports_readable_items\CANON_FULL_READABLE .md

Exported CORTEX -> .local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_2026
0619_011615_20260619_012857\reports_readable_items\CORTEX_FULL_READABL E.md

Exported DOMAINES -> .local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_2026
0619_011615_20260619_012857\reports_readable_items\DOMAINES_FULL_READA BLE.md

```
Exported MEMOIRE_GRAPHITI -> .local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_20260619_011615_20260619_0128
57\reports_readable_items\MEMOIRE_GRAPHITI_FULL_READABLE.md
```

Exported PREUVES -> .local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_2026
0619_011615_20260619_012857\reports_readable_items\PREUVES_FULL_READAB LE.md

Exported ROUTES_API -> .local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_2026
0619_011615_20260619_012857\reports_readable_items\ROUTES_API_FULL_REA DABLE.md

Exported BOUNDARIES -> .local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST_INDEXED_2026
0619_011615_20260619_012857\reports_readable_items\BOUNDARIES_FULL_REA DABLE.md

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== PEPITES NON BRANCHEES ==="
-ForegroundColor Cyan
```

PEPITES NON BRANCHEES

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $data |
>> Where-Object { $_.family -eq "PEPITES" -and $_.verdict -eq "ITEM_REFERENCE_ONLY_NOT_BOUND" } |
>> Select-Object code,name,status_bucket,verdict |
>> Format-List
```

code : P47

name : Couplage reel/moteur dR/dt

status_bucket : A_FORMALISER

verdict : ITEM_REFERENCE_ONLY_NOT_BOUND

code : P48

name : Structure cosmologique

status_bucket : A_FORMALISER

verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P49
name : Emergence par interaction
status_bucket : A_FORMALISER
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P50
name : Recursivite fondamentale
status_bucket : A_FORMALISER
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P51
name : Structure fractale
status_bucket : A_FORMALISER
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P52
name : Principe holographique
status_bucket : A_FORMALISER
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P53
name : Resonance cosmique
status_bucket : A_FORMALISER
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P54
name : Temps comme integrale d'evenements
status_bucket : A_FORMALISER
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P55
name : Espace metrique
status_bucket : A_FORMALISER
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P56
name : Causalite temporelle
status_bucket : A_FORMALISER
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P59
name : Complexite de Kolmogorov
status_bucket : A_FORMALISER
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P60
name : Auto-organisation
status_bucket : A_FORMALISER
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P61
name : Bifurcation de trajectoire
status_bucket : A_FORMALISER
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P62
name : Attracteur
status_bucket : A_FORMALISER
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P63
name : Sensibilite aux conditions initiales
status_bucket : A_FORMALISER
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P65

name : Definition de la vie
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P66
name : Conscience comme recursivite infinie
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P86
name : Consensus supermajorité 3/4
status_bucket : ANCREE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P87
name : Fail-closed sans supermajorité
status_bucket : ANCREE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P88
name : Non-contradiction
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P89
name : Completude
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P90
name : Coherence
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P91
name : Decidabilite
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P92
name : Calculabilite (Turing)
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P93
name : P inclus dans NP
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P94
name : Reductibilite
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P95
name : Indecidabilite (Halting)
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P96
name : Recursivite
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P97
name : Point fixe
status_bucket : FORMALISE

verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P98
name : Continuite epsilon-delta
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P99
name : Convergence
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P100
name : Stabilite de Lyapunov L(Phi(s)) <= L(s)
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P101
name : Avancement incremental valide
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P115
name : Filtre ethique du moteur
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P116
name : Pipeline de couches
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P117
name : Orchestration par scheduler
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P118
name : Isolation des couches
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P119
name : Composition fonctionnelle
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P120
name : Abstraction par interface
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P121
name : Encapsulation (boite noire)
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P122
name : Modularite
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P123
name : Extensibilite
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P124

name : Infrastructure : I∘M=O
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P125
name : Deploiement Build+Test+Release
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P126
name : Scalabilite lineaire
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P128
name : Latence de reponse
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P129
name : Debit (throughput)
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P130
name : Resilience MTBF/MTTR
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P131
name : Securite CIA
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P132
name : Observabilite (logs+metriques+traces)
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P133
name : Configuration cle-valeur
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P134
name : Versioning semantique
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P135
name : Pipeline CI/CD
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P136
name : Infrastructure as Code
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P137
name : Agent = Role+Protocole+Flux
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P138
name : Ecologie d'agents
status_bucket : FORMALISE

verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P139
name : Protocole = sequence de messages
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P140
name : Coordination par consensus
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P141
name : Emergence collective
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P142
name : Specialisation par domaine
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P156
name : Ironie = ecart attendu/reel
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P157
name : Suspense = incertitude de resolution
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P158
name : Catharsis = liberation emotionnelle
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P159
name : Epiphanie = revelation soudaine
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : P160
name : Denouement final
status_bucket : VISION
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== MODULES DETAIL ==="
-ForegroundColor Cyan
```

MODULES DETAIL

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $data |
>> Where-Object { $_.family -eq "MODULES" } |
>> Select-Object code,name,status_bucket,verdict,brody_bound_any,domain_bound_any |
>> Format-List
```

code : A01
name : Signature Entropique / Pare-feu symbolique
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A02
name : Verification de Reciprocite / Detection contradictions
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True

domain_bound_any : False
code : A03
name : Equilibre Dynamique / Alignement ethique
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A04
name : Gestion de la Friction / Bascule Semantique
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A05
name : Singularite Initiale / Audit ethique permanent
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A06
name : Integrite Structurelle / Enregistrement etats
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A07
name : Coherence Temporelle / Chronique obsidienne
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A08
name : Emergence des Proprietes / Transmutation cognitive
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A09
name : Non-Contradiction / Convergence intentionnelle
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A10
name : Completude / Emergence par tension
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A11
name : Resonance / Reconstruction totale
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True

domain_bound_any : False
code : A12
name : Dualite / Contrat moral
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A13
name : Securite Interactions / Mode reflexe
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A14
name : Verification Identite / Captation creative
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A15
name : Adaptation Dynamique / Vibration multimodale
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A16
name : Recursivite / Architecture du savoir
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A17
name : Complexite Minimale / Apprentissage universel
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A18
name : Optimisation Ressources / Friction creative
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A19
name : Interconnexion / Narration cognitive
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A20
name : Confidentialite / Entrainement emotionnel
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True

domain_bound_any : False
code : A21
name : Cryptographie Homomorphe / Mediatheque fractale
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A22
name : Propagation Information / Simulation cognitive
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A23
name : Causalite / Calibration procedurale
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : A24
name : Predictibilite / Metriques et veille
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : T01
name : Reduction Incertitude / Audit coherence
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : T02
name : Conscience Distribuee / Validation reciprocite
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : T03
name : Gouvernance Decentralisee / Friction energetique
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : T04
name : Consensus Distribue / Resonance semantique
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : T05
name : Evolution Systemes / Balance structure/chaos
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True

domain_bound_any : False
code : T06
name : Selection Naturelle / Alignement ethique
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : T07
name : Resilience / Test memoire fractale
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : T08
name : Auto-Organisation / Apprentissage inverse
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : T09
name : Creativite / Mode reflexe
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : T10
name : Innovation Continue / Integrite cognitive
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : T11
name : Generation Aleatoire / Frugalite bio-inspiree
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False
code : T12
name : Complexite Irreductible / Transparence friction
status_bucket : UNKNOWN
verdict : ITEM_BOUND_BRODY_ONLY
brody_bound_any : True
domain_bound_any : False

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== AGENTS NON BRANCHES ==="
-ForegroundColor Cyan
```

AGENTS NON BRANCHES

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $data |
>> Where-Object { $_.family -eq "AGENTS" -and $_.verdict -eq "ITEM_REFERENCE_ONLY_NOT_BOUND" } |
>> Select-Object code,name,status_bucket,verdict |
>> Format-List
```

code : #8
name : REPO_HISTORIAN
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #9

name : VERSION_COMPARATOR
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #10
name : CARTOGRAPHE_PREUVES
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #12
name : GENERATEUR_TESTS_ADVERSARIAUX
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #14
name : ASSISTANT_THEORIQUE_DOCUMENTAIRE
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #15
name : MEMOIRE_PERSONNELLE_OBSIDIA
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #16
name : FREEZE_MASTER
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #18
name : HISTORIEN_DECISIONS
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #20
name : CONTRADICTION_HUNTER
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #21
name : THEORIE_VIVANTE
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #22
name : RECHERCHE_EXTERNE
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #23
name : ANALYSTE_LOGS_EXPERIMENTAUX
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #24
name : EVIDENCE_BOARD_BUILDER
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #25
name : AUDIT_LOCAL_AGENT
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #26
name : DATA_SOVEREIGNTY_GUARD
status_bucket : UNKNOWN

verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #27
name : SECURITY_FIREWALL_ARCHITECT
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #28
name : POLICY_FIREWALL_AGENT
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #29
name : RUNTIME_ATTESTATION_AGENT
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #30
name : SUPPLY_CHAIN_GUARD
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #31
name : SECRET_GUARD
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #32
name : FIELD_POC_MAPPER
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #33
name : PITCH_AGENT
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #35
name : SIMULATEUR_REUNION
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #36
name : MONETISATION_PACKAGING
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #37
name : FRISE_HUMAINE
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #39
name : HUMAN_HISTORY_MAPPER
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #42
name : GRAND_CARTOGRAPHE_OBSIDIA
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #43
name : ONTOLOGUE_OBSIDIA
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #44

name : CARTOGRAPHE_COUCHES
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #46
name : CARTOGRAPHE_FORMULES
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #47
name : CARTOGRAPHE_DOMAINES_TERRAIN
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #48
name : CARTOGRAPHE_AGI_VISION_HAUTE
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #49
name : COLLISION_DETECTOR
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #51
name : ENERGIE_PRIORITE
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : #52
name : EXTRACTEUR_MATIERE_BRUTE
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== ARBRES NON BRANCHES ==="
-ForegroundColor Cyan
```

ARBRES NON BRANCHES

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $data |
>> Where-Object { $_.family -eq "ARBRES_34" -and $_.verdict -eq "ITEM_REFERENCE_ONLY_NOT_BOUND" } |
>> Select-Object code,name,status_bucket,verdict |
>> Format-List
```

code : ARBRE_04
name : Arbre du Sens
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_05
name : Arbre de l'Identite
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_06
name : Arbre de la Comprehension
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_07
name : Arbre de l'Organisation
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_08
name : Arbre de la Pensee
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_12

name : Arbre de la Technique
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_13
name : Arbre de l'Art
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_14
name : Arbre de la Philosophie
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_15
name : Arbre de la Spiritualite
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_16
name : Arbre de la Relation
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_17
name : Arbre du Collectif
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_18
name : Arbre de la Transmission
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_22
name : Arbre de la Transformation
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_29
name : Arbre de la Finalite
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_32
name : Arbre des Connexions
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND
code : ARBRE_34
name : Arbre de la Stabilite
status_bucket : UNKNOWN
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== LOIS NON BRANCHEES ==="
-ForegroundColor Cyan
```

LOIS NON BRANCHEES

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> $data |
>> Where-Object { $_.family -eq "LOIS" -and $_.verdict -eq "ITEM_REFERENCE_ONLY_NOT_BOUND" } |
>> Select-Object code,name,status_bucket,verdict |
>> Format-List
```

code : P92
name : Calculabilite (Turing)
status_bucket : FORMALISE
verdict : ITEM_REFERENCE_ONLY_NOT_BOUND

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`n=== READABLE REPORT FILES ==="
```

```
-ForegroundColor Cyan
```

READABLE REPORT FILES

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Get-ChildItem $OUT -File |
>> Sort-Object Name |
>> Select-Object Name,Length,LastWriteTime |
>> Format-Table -AutoSize
Name Length LastWriteTime
---- ------ -------------
AGENTS_FULL_READABLE.csv 34977 19/06/2026 01:31:29
AGENTS_FULL_READABLE.md 39370 19/06/2026 01:31:29
ARBRES_34_FULL_READABLE.csv 14732 19/06/2026 01:31:29
ARBRES_34_FULL_READABLE.md 16849 19/06/2026 01:31:29
BALANCES_FULL_READABLE.csv 208238 19/06/2026 01:31:30
BALANCES_FULL_READABLE.md 233836 19/06/2026 01:31:30
BOUNDARIES_FULL_READABLE.csv 405146 19/06/2026 01:31:33
BOUNDARIES_FULL_READABLE.md 423767 19/06/2026 01:31:33
CANON_FULL_READABLE.csv 518020 19/06/2026 01:31:30
CANON_FULL_READABLE.md 532527 19/06/2026 01:31:30
CORTEX_FULL_READABLE.csv 28310 19/06/2026 01:31:31
CORTEX_FULL_READABLE.md 30623 19/06/2026 01:31:31
DOMAINES_FULL_READABLE.csv 260275 19/06/2026 01:31:31
DOMAINES_FULL_READABLE.md 283004 19/06/2026 01:31:31
LOIS_FULL_READABLE.csv 4675 19/06/2026 01:31:29
LOIS_FULL_READABLE.md 5315 19/06/2026 01:31:29
MEMOIRE_GRAPHITI_FULL_READABLE.csv 121352 19/06/2026 01:31:32
MEMOIRE_GRAPHITI_FULL_READABLE.md 143612 19/06/2026 01:31:32
MODULES_FULL_READABLE.csv 35419 19/06/2026 01:31:29
MODULES_FULL_READABLE.md 38085 19/06/2026 01:31:29
PEPITES_FULL_READABLE.csv 53287 19/06/2026 01:31:29
PEPITES_FULL_READABLE.md 62178 19/06/2026 01:31:29
PREUVES_FULL_READABLE.csv 301320 19/06/2026 01:31:32
PREUVES_FULL_READABLE.md 328182 19/06/2026 01:31:32
PROTOCOLES_FULL_READABLE.csv 202622 19/06/2026 01:31:29
PROTOCOLES_FULL_READABLE.md 228365 19/06/2026 01:31:29
ROUTES_API_FULL_READABLE.csv 371745 19/06/2026 01:31:33
ROUTES_API_FULL_READABLE.md 404528 19/06/2026 01:31:33
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B> Write-Host "`nOUT_READABLE=$OUT" -ForegroundColor
Green
```

OUT_READABLE=.local_audits\FREEZE_OBSIDIA_STRICT_BINDING_AUDIT_V4_FAST
_INDEXED_20260619_011615_20260619_012857\reports_readable_items

```
PS C:\Users\User\Desktop\obsidia-engine-proof-core\obsidia-x108-proofs_REMOTE_A5F21C6B>
```

Clôture du support

Ce PDF matérialise le document fourni en support interne complet. Les chiffres officiels scellés restent ceux du freeze final : 4353 items audités, 3316 non branchés strict, 204 partiels runtime, 772 Brody only, 61 Brody + Domaines, 80 P1 à P80 audités.